Children, Household Specialization and Relationship Quality*

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Abstract

We study how having children affects the quality of couples' relationships. Relationship quality captures the non-material gains from being in a relationship, which are central to models of family formation and dissolution. Using rich longitudinal data from the United Kingdom, we construct and validate a novel measure of relationship quality (RQ). Exploiting variation in the timing of first births in a dynamic difference-in-differences framework, we find a sharp and persistent RQ decline of half a standard deviation, sustained for seven years. This effect is similar for mothers and fathers, in contrast to the large gender asymmetries documented in the parenthood literature. We establish gender-based household specialization as a key mechanism.

Keywords: Fertility, Marital decisions, Time allocation, Household specialization

JEL Codes: J12, J13, J22, D13

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1 Introduction

Having children changes people's lives. With more responsibilities at home, including housework and childcare, couples need to reorganize their time. Mothers substantially reduce their time in the labour market (Goldin, 2021) and take on more home production (Aguilar-Gomez et al., 2026; Siminski and Yetsenga, 2022), while both parents experience declines in leisure and rest (Aguiar and Hurst, 2007; Costa-Font and Flèche, 2020). Despite these significant adjustments, the deeper consequences that parenthood and the associated reallocation of time may have on couples have received limited attention.

This paper studies the impact of having children on the quality of couples' relationships. Relationship quality captures the non-material benefits that individuals derive from their relationships. Together with material factors, such as risk sharing and economies of scale, it shapes the value of being in a relationship, directly entering individuals' utility (Chiappori et al., 2018). Theoretical models of couple formation and dissolution treat relationship quality as a central driver of couple decisions. In these models, it determines couple formation, dissolution, and fertility decisions (e.g. Browning et al., 2014; Chiappori and Mazzocco, 2017; Chiappori, 2020), governs intra-household cooperation and parental investment (Chiappori and Weiss, 2007), and affects child outcomes (Edwards and Roff, 2016; Goldberg and Carlson, 2014).

Direct empirical measures of relationship quality, however, remain scarce, largely due to measurement difficulties. Existing studies have relied on indirect proxies, such as partner characteristics, residual measures of subjective well-being, or separate questions on happiness and conflict, typically in small cross-sectional samples (Weiss and Willis, 1997; Bertrand et al., 2015; Chiappori et al., 2018). Yet relationship quality has likely become increasingly important over time. Rising female labour force participation and the expansion of welfare states have reduced financial dependence within couples, increasing the role of non-material considerations in relationship formation and stability (Lundberg and Pollak, 2007).

The first contribution of this paper is developing a novel measure of relationship quality (RQ). We use a questionnaire from Understanding Society, a longitudinal household survey in the United Kingdom, which collects detailed information on respondents' relationships with their cohabiting partners. The questionnaire covers two broad dimensions of relationship quality: subjective assessments of the relationship, such as the degree of happiness with the couple or the frequency with which they consider splitting, and couple time use, like how often they work together on a project or kiss. This rich information allows us to gain a nuanced understanding of the non-material benefits individuals derive from their relationship. To capture these multiple dimensions parsimoniously, we combine responses to the questionnaire using factor analysis and construct our measure of RQ.

We conduct a series of tests to verify that our measure provides economically meaningful

information about couples' relationships. We evaluate two key theoretical properties: informativeness and interpersonal comparability (Ferrer-i Carbonell and Frijters, 2004). First, we show that RQ strongly predicts subsequent couple decisions. Periods before separation are characterised by significantly lower RQ, while it is higher for individuals who will end up getting married or having children. Second, we find a high correlation in reported RQ across partners, with partner RQ being the most influential predictor of an individual's own RQ.

The second contribution of this paper is to quantify the effect of parenthood on relationship quality. The transition to parenthood is one of the most consequential events in couples' lives. A large literature has documented its effects on multiple dimensions, including labour market outcomes, time allocation within the household, and individual well-being (e.g. Cetre et al., 2015; Goldin, 2021; Siminski and Yetsenga, 2022), and has also documented negative effects on couple stability (Lillard and Waite, 1993; Svarer and Verner, 2008). Yet focusing on separation provides an incomplete picture of how relationships respond to parenthood, since it captures only the extensive margin of relationship adjustment. Our measure of relationship quality allows us to study how parenthood affects relationships within existing couples, providing a more complete picture of relationship adjustment following the arrival of children.

We exploit variation in the timing of first births and adopt a dynamic difference-in-differences approach within the sample of individuals who become parents, following the literature on transition to parenthood (e.g., Kleven et al., 2019b). This approach compares parents with not-yet-parents, addressing the endogeneity of fertility decisions and exploiting instead sharp within-individual deviations from pre-birth trajectories (Callaway and Sant'Anna, 2021). Because identification relies only on individuals who eventually have children, our estimates should be interpreted as an average treatment effect on the treated (ATT), capturing the effect of childbirth on those who become parents.

We find a sharp and persistent decrease in RQ following the birth of the first child. Within five years after birth, RQ falls by approximately half a standard deviation relative to pre-birth levels and remains low throughout the seven-year post-birth window we study. Quantitatively, this effect is large: half a standard deviation corresponds to the difference between the 75th percentile and the median of the RQ distribution, and is associated with roughly a 20% higher probability of couple dissolution.

The magnitude and persistence of this effect stand out compared to the effect of parenthood on other outcomes and relative to other shocks. We find that the effect on general happiness is positive, although less pronounced and short-lived, lasting only one period. Moreover, we show that the impact of childbirth on RQ is larger than other major life events, such as becoming unemployed, which has virtually no impact on RQ.

The causal interpretation of these results rests on two assumptions: no anticipation and parallel trends. The no-anticipation assumption requires that changes in RQ do not influence the timing of childbirth. It would be violated if births were timed to periods of particularly high

or low RQ. Consistent with this assumption, we find no evidence of differential pre-trends and show that pre-birth changes in RQ do not predict subsequent childbirth. The parallel trends assumption requires that, absent childbirth, RQ would have evolved similarly across individuals regardless of first birth timing. Beyond the absence of pre-trends, we provide two additional pieces of evidence supporting this assumption. First, parents experiencing fertility delays exhibit similar pre-birth levels and trajectories of RQ relative to other parents. Second, our estimates remain unchanged when restricting comparisons to individuals with similar pre-birth RQ.

To address concerns about selection into parenthood, we consider two alternative control groups. First, we compare new parents to individuals who attempted but failed to have children. These individuals resemble future parents, including a similar pre-birth distribution of RQ, but do not experience a live birth. Exploiting the timing of their first unsuccessful pregnancy yields estimates that closely mirror our baseline results. Second, we compare parents to individuals who are never observed to enter or attempt parenthood. Using a placebo-birth design for this group again produces estimates similar to our main findings.

Importantly, while RQ is a couple-level object, it is measured at the individual level, allowing us to separately identify how each partner is affected, unlike commonly used outcomes such as dissolution (Lillard and Waite, 1993; Svarer and Verner, 2008). In contrast to much of the literature studying the consequences of parenthood, which documents large gender asymmetries (e.g., Kleven et al., 2019a; Goldin, 2021), we find that RQ declines substantially for both partners. While the drop is slightly larger for mothers, fathers also experience a sizeable decline, challenging the notion that fathers are largely unaffected by parenthood.

We substantiate the robustness of our findings addressing four potential concerns. First, we show that the results are not driven solely by reduced time spent together by studying the different components of RQ separately. Second, we rule out that subsequent births drive the persistent decline of RQ by estimating effects separately by completed fertility. Third, we address potential selection bias from sample attrition due to separation by showing that such attrition is limited and that the results remain virtually unchanged when restricting the sample to couples who remain together. Lastly, we verify that parental age and relationship tenure do not influence our results.

Finally, we explore household specialization as a mechanism behind the decline in RQ after the first birth. This is motivated by evidence that a more equal division of household tasks is associated with higher relationship satisfaction (e.g., Blom et al., 2017; Carlson et al., 2018). We also find that changes in household specialization after childbirth are larger than changes in other potential channels, such as gender norms, household income, and the within-couple earnings gap. Specifically, mothers increase their weekly housework time (excluding childcare) by five to seven hours and reduce their labour market hours by about eleven hours per week. In contrast, fathers' housework and labour market time remain largely unchanged, implying that

the reallocation of roles after birth is driven primarily by changes in mothers' behaviour.

To assess the role of specialization, we classify couples into four types based on their pre-birth division of paid and unpaid work. We show that all types transition toward gender-based household specialization after birth, regardless of their pre-birth task division. However, the magnitude of this reallocation differs across types. We exploit this variation to estimate the effect of childbirth on RQ separately by type. We find that couples experiencing the largest shifts in intra-household specialization also experience the largest declines in RQ, while those whose division of labour remains largely unchanged show no significant change in RQ.

Our findings suggest that household specialization driven by the arrival of children has adverse consequences for couples' relationship quality. This has implications for the design of policies aimed at promoting fertility and supporting families after childbirth. Policies that encourage more equitable divisions of responsibilities after birth may mitigate the decline in RQ that follows the transition to parenthood (Doepke and Kindermann, 2019). Such policies have been shown to improve female labour market outcomes (Olivetti and Petrongolo, 2017), increase fathers' contributions to housework (Farré and González, 2019; González and Zoabi, 2021) and foster more egalitarian gender norms among children exposed to them (Farré et al., 2022). At the same time, Avdic and Karimi (2018) suggests that they may also affect the timing of couple separations, potentially by making differences in preferences over household task allocation more salient. Therefore, implementing such policies could yield both immediate benefits, by improving parents' RQ, and long-term advantages, by perpetuating more equitable household arrangements that persist across generations.

Related literature. This paper connects to the extensive literature on the economics of family formation and dissolution. Standard models studying family formation and dissolution commonly integrate the theoretical concept of relationship or match quality into the decision-making process (Weiss and Willis, 1997; Browning et al., 2014; Chiappori and Mazzocco, 2017; Chiappori, 2020). However, the empirical determinants of this variable are little understood.[1] Related work uses structural matching models and aggregate marriage data to estimate marital surplus (Choo and Siow, 2006; Galichon and Salanié, 2022). While these estimates capture the total gains from matching, they do not separately identify the non-material component of these gains. Empirical efforts to measure this component are limited, proposing general characteristics of partners (Weiss and Willis, 1997; Lundberg, 2012; Eckstein et al., 2019; Low,

[1]The models of couple formation and dissolution incorporate match quality in two ways: through a match quality parameter and through partner quality. A set of papers have modelled match quality as a parameter that is known in the current period but that receives stochastic shocks in the future (Chiappori and Weiss, 2006, 2007; Gemici and Laufer, 2011; Bruze et al., 2015; Voena, 2015; Goussé et al., 2017; Greenwood et al., 2017; Low et al., 2018). Others consider it to be deterministic but unobserved, where couple members only get a noisy signal each period about true quality (Brien et al., 2006; Blasutto et al., 2020; Antler et al., 2022; Blasutto, 2024). A second strand of models consider match quality to be contingent on partner quality in terms of different socio-economic characteristics (Greenwood et al., 2017; Eckstein et al., 2019; Low, 2024).

2024), a residual of subjective well-being (Chiappori et al., 2018) or separate questions related to happiness and conflict (Bertrand et al., 2015) as proxies for match quality. We develop and validate a novel measure that explicitly integrates relationship-specific information. Compared with existing approaches, it minimizes measurement error and consolidates multiple dimensions into a single, interpretable metric, allowing us to track relationship quality over the life cycle and examine how it responds to key life events.

Previous studies in psychology have proposed and tested various measures of marital satisfaction and conflict (e.g., Norton, 1983; Busby et al., 1995; Joel et al., 2020). These variables have previously been studied across social sciences in relation to different outcomes. For instance, Carlson and VanOrman (2017) study the relationship quality trajectories of married and unmarried parents, only after birth. Our measure and analysis present a number of advantages over prior research. First, RQ integrates multiple aspects of relationships, for which only separate measures have been available, into a single, parsimonious measure of match quality (Busby et al., 1995). Second, the combination of a larger sample and longitudinal data allow us to employ causal identification methods that were previously unfeasible, overcoming limitations acknowledged in earlier studies (Amato and Booth, 2001; Hassebrauck and Fehr, 2002; Amato and Patterson, 2017).

This paper adds to our understanding of the consequences of having children. Seminal papers in gender economics have documented the differential effect of fertility on women relative to men on several labour market outcomes (Blau and Kahn, 2017; Bertrand, 2020; Goldin, 2021). A large body of work has focused on accounting for the share of the gender wage gap explained by fertility and on studying the channels through which it operates (Goldin, 2014; Adda et al., 2017; Kleven et al., 2019b; Cortés and Pan, 2023; Cavapozzi et al., 2021; Kleven, 2022; Albanesi et al., 2023), in different countries (Kleven et al., 2019a, 2025) and across different cohorts (Goldin, 2021). Other work has discussed the deeper implications of having children on outcomes like identity (Akerlof and Kranton, 2000, 2010; Bertrand et al., 2015; Andresen and Nix, 2022), well-being and happiness (Dolan et al., 2008; Clark et al., 2008; Blanchflower, 2009; Ferrer-i Carbonell, 2013; Cetre et al., 2015), welfare (Guo and Xie, 2024) and mental health (Ahammer et al., 2023).

The findings across these studies consistently reveal a stark asymmetry: fathers experience little to no impact, while mothers face substantial consequences. In contrast, this paper identifies a consequence of childbirth shared by both parents, offering novel insights into the experiences of fathers. Importantly, this differs from studies of couple-level outcomes like relationship tenure or divorce, which inherently involve both partners (Lillard and Waite, 1993; Svarer and Verner, 2008). By focusing on an individual-level outcome like RQ, we can test whether childbirth uniquely affects each parent. Moreover, the implications of RQ extend beyond parents to future outcomes of their children, who are affected by exposure to a poor-quality parental relationships (Piketty, 2003; Björklund and Sundström, 2006) and by parental

divorce (Gruber, 2004; Björklund et al., 2007).

Finally, this paper provides new evidence on household time allocation, which is a long-standing issue (Lundberg and Rose, 2000; Lundberg, 2005). This topic has gained even greater relevance during and after the COVID-19 pandemic, which introduced unprecedented disruptions to housework and childcare (Sevilla and Smith, 2020; Hupkau and Petrongolo, 2020; Alon et al., 2020; Farré et al., 2021; Del Boca et al., 2020). The dynamic impact of the birth of the first child on paid and unpaid work has previously been studied by Aguilar-Gomez et al. (2026) in Mexico and Siminski and Yetsenga (2022) in Australia. We replicate their findings for the United Kingdom and extend them to uncover heterogeneity in this impact based on household time arrangements before birth, providing the novel finding that gender-based specialization occurs after birth, irrespective of pre-birth arrangements. Furthermore, we provide evidence of the non-material costs of household specialization, highlighting its impact on relationship quality.

Roadmap. The rest of the paper is organized as follows. Section 2 describes the data and introduces our measure of relationship quality. Section 3 describes the empirical strategy and discusses identification. Section 4 presents the main results and considers their robustness. Section 5 explores changes to household specialization as the potential mechanism at play. Section 6 concludes.

2 Data and Measures

The analysis is based on data from Understanding Society, a longitudinal household survey in the United Kingdom representative of the country's population (University of Essex and Research, 2024). Information on the primary outcome spans from 2009 to 2024 and is based on data collected about cohabiting couples in waves 1, 5, 7, 9, 11, 13, and 15. We source additional data from the British Household Panel Survey (BHPS), which predates Understanding Society and covers the period from 1991 to 2008, to construct individuals' relationship histories. In particular, the BHPS allows us to identify both past and current partners, along with detailed characteristics of the relationships, including tenure.

2.1 Measure of Relationship Quality

Partner questionnaire. Understanding Society conducts a 10-question survey every other data collection wave. These questions revolve around individuals' relationships with their cohabiting partners, including questions like "How often do you and your partner quarrel?". Respondents rate these questions on a scale from "All of the time" to "Never" on a six-point Likert scale. The survey also includes questions regarding relationship happiness and shared

interests. Table 1 contains the full set of questions.[2] These questions are asked individually to all respondents living with a partner, regardless of marital status. This data is available in alternating waves spanning from 2009 to 2024.[3]

Table 1: Questions in the Understanding Society Partner module.

(a) Subjective assessment		(b) Couple time use	
Question	Loading	Question	Loading
How often do you... ?		*How often do you... ?*	
discuss or consider splitting	0.652	work together on a project	0.626
regret that you married or lived together	0.706	have stimulating exchanges of ideas	0.645
quarrel	0.620	calmly discuss something	0.701
get on each other's nerves	0.675	kiss your partner	0.531
What is the... ?		*Do you and your partner... ?*	
degree of happiness of your relationship	0.676	engage in outside interests together	0.512

Notes: This table lists all the questions used to construct the relationship quality measure (RQ), drawn from the Understanding Society Partner module, and their corresponding factor loadings. This self-completion questionnaire is administered in alternating survey collection waves, spanning the period 2009-2024. We divide the items in (a) subjective assessments of the relationship and (b) couple time use, based on the information they convey. The first four questions in each block are responded in terms of frequency in a six-point Likert scale; the happiness question is answered in terms of degree in a seven-point scale; the number of outside interests is given in a five-point scale. Appendix A.1 reports the exact phrasing of the questions and response options. All items are recoded prior to the factor analysis so that higher values correspond to better quality matches. We perform a factor analysis on all available responses and retain the first factor as the RQ measure. The reported factor loadings correspond to this first factor, which has eigenvalue 4.06 and explains 40.64% of the variation in the data.

We categorize the items in Table 1 in two blocks based on the information they convey: (a) subjective assessments of the relationship and (b) couple time use items. The items in the first block contain information related to the degree of happiness and conflict in the relationship, which are individual perceptions that may not be shared by both partners. In the second block, the items refer to joint activities and how couples spend their time together. Including both types of items provides a nuanced characterization of the relationship.

The response distributions in Table A.1 illustrate the information captured by these two blocks. Respondents generally report positive relationship experiences, with frequent constructive interactions, high relationship happiness, and few respondents regularly considering separation. At the same time, conflict is also present, with respondents reporting quarrelling and getting on each other's nerves. The correlations across items, reported in Table A.2, further show that items within each block tend to be more strongly correlated than items across blocks. This pattern suggests that subjective assessments and couple time use capture related

[2]This questionnaire, excluding the questions on happiness and kissing, was originally used in psychology research to construct the cohesion and satisfaction sub-scales of the Revised Dyadic Adjustment Scale (Busby et al., 1995). Appendix A.1 provides the exact phrasing of the questions and the response options of the questionnaire.

[3]An implementation error in data collection wave 3 (starting on year 2011) makes the item "Do you and your partner engage in outside interests together?" incompatible with the rest of the waves. Hence, we exclude wave 3 and construct the measure using the remaining waves.

but distinct dimensions of relationships, motivating their combination into a single measure of relationship quality.

Construction of the measure. We construct the measure of relationship quality (RQ) using all the available responses to the ten items in Table 1.[4] We use factor analysis to extract the common component of responses across these items. The underlying idea is that relationship quality is a latent characteristic that cannot be directly observed but manifests itself in different aspects of couples' relationships. Factor analysis identifies the variation shared across items, distinguishing it from variation specific to individual questions. Before estimation, we transform the responses such that lower values correspond to *poorer* couple attitudes, such as never working together on a project or considering separation all of the time. This transformation ensures that all items are increasing in the quality of the relationship.

We then conduct a factor analysis on the transformed items and retain the first factor as the RQ measure. The decision to retain a single factor is based on the eigenvalues plotted in Figure A.1. The first factor has an eigenvalue of 4.06, which is more than twice as large as that of the next factor. The eigenvalue of the rest of the latent variables remains below one. The first factor accounts for 40.64% of the total variation across items. Additionally, as shown in Table 1, all items have positive factor loadings that are larger than 0.5, indicating strong associations between RQ and each item, rather than the measure being driven by a particular subset of questions. Further details on the estimation and factor loadings are provided in Appendix A.2. The resulting variable is standardized with a mean of zero and a standard deviation of one. In this context, higher values of RQ indicate higher quality relationships.

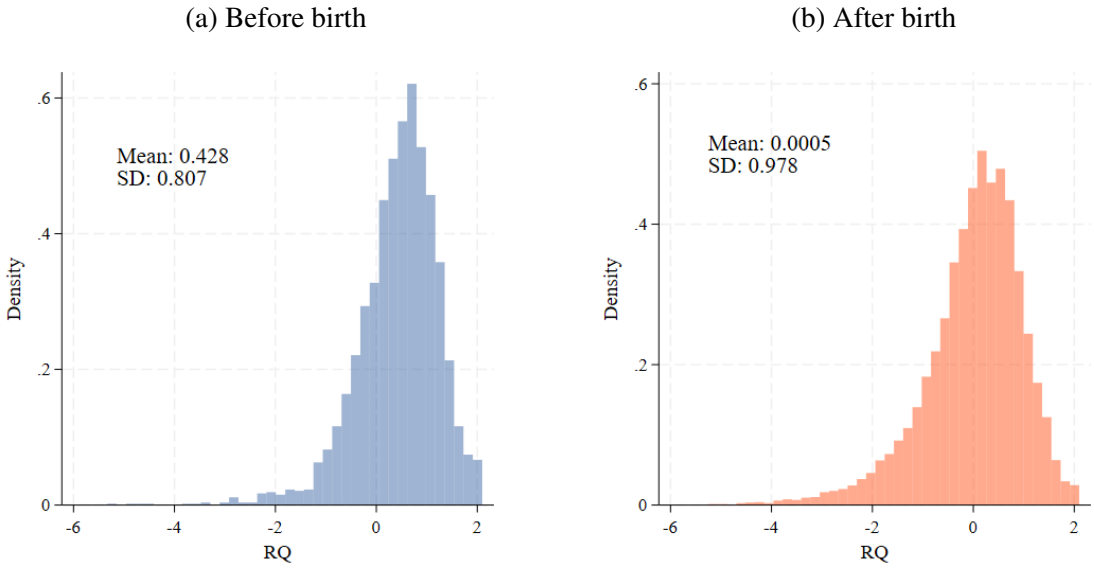
Figure 1 shows the distribution of RQ for individuals who become parents, separately for periods (a) before and (b) after the birth of their first child. In both cases, the distribution is left-skewed, indicating a higher frequency of high-quality relationships. Before birth, average RQ is higher than in the full sample and less dispersed. After birth, instead, mean RQ drops to the full-data average. This is preliminary evidence of a decrease in RQ after having a child. This pattern is consistent with the evidence in Figure A.3, which shows a generalized downward shift in RQ, with most within-individual changes being negative, while largely preserving individuals' relative positions in the distribution.

2.1.1 Measure Validation

Given the novelty of the measure, in Appendix A.3 we conduct a series of validation exercises to verify that RQ provides meaningful information about match quality. First, we evaluate

[4] We repeat the factor analysis separately for pre- and post-birth observations in our analysis sample of first time parents. The resulting factor loadings, presented in Figure A.2, are consistent with the full-sample benchmark, and our main results remain unchanged when using the RQ measure constructed exclusively from post-birth observations in Figure D.2.

Figure 1: Distribution of RQ in the sample.



Notes: This figure plots the distribution of RQ in the main analysis sample of parents, separately for the periods (a) before and (b) after the birth of their first child. The sample consists of individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18-45 (women) or 18-50 (men) at the time of birth, and were not cohabiting with children from previous relationships. Panel (a) includes observations up to four years before birth. Panel (b) spans from the year of birth to seven years after. These time frames align with the main analysis period. In the full data, RQ has a mean of 0 and a standard deviation of 1.

whether the measure exhibits a set of desirable properties. Following the life satisfaction literature, we assess two key theoretical properties: informativeness and interpersonal comparability (Ferrer-i Carbonell and Frijters, 2004). We also verify that RQ evolves smoothly with age and relationship tenure, as large random fluctuations could confound the estimated effects in the main analysis. Second, we benchmark RQ against existing empirical attempts. We relate it to observable characteristics commonly used as proxies for match quality (e.g., Weiss and Willis, 1997) and to the limited number of explicit empirical measures of match quality in the economics literature (Busby et al., 1995; Bertrand et al., 2015; Chiappori et al., 2018). These validation exercises use all the available information on RQ rather than restricting the sample to parents, as is done in the main analysis.

Theoretical properties. We begin by assessing whether RQ is informative about major couple decisions. We compare the distribution of RQ before marital transitions (marriage and separation) to its distribution in the full data, and the distribution prior to first birth to that in the sample of first-time parents. Figure A.4 plots the empirical cumulative distribution functions. RQ is shifted to the right prior to marriage and childbirth, consistent with marriage acting as a commitment mechanism that raises separation costs (Brien et al., 2006) and with fertility being more likely in higher-quality relationships. In contrast, RQ is substantially lower prior to separation, indicating that couples tend to dissolve only after substantial deterioration. Kol-

mogorov–Smirnov tests in Table A.3 confirm that all pre-event distributions differ significantly from their benchmarks. Consistent with this, the regression results in Table A.4 show that a one-standard-deviation higher RQ is associated with a 0.8 percentage point lower probability of separating in the subsequent period.

We next examine the extent to which RQ captures a concept shared between partners. In Table A.5 we regress women's RQ on their male partners' RQ and find a correlation of 0.6. The rank-rank plot in Figure A.5 reveals non-linearities in this association: the slope is steepest at the top and bottom of the distribution and about half as large in the middle, suggesting that extreme assessments of the quality of the relationship are more strongly shared than moderate ones. Table A.6 further shows positive within couple correlations across all items in Table 1, with the highest agreement in questions related to kissing and quarrelling and the lowest for stimulating exchanges of ideas and calm discussions.

Finally, we verify that RQ evolves smoothly with age and relationship tenure. We estimate a two-way fixed effects model including age and relationship tenure non-parametrically, allowing us to recover their profiles in the full data. Both profiles, shown in Figure A.6, are smooth and display no sharp discontinuities. The age profile is increasing and nearly linear, while the tenure profile declines steeply in the early years before flattening out. This pattern mirrors the evolution of divorce hazards over marital tenure, which are highest in the first decade of marriage and decline thereafter (Stevenson and Wolfers, 2007). This similarity is consistent with declining RQ contributing to marital dissolution, although selective dissolution may also shape the observed tenure profile.

Comparison to existing measures. The most common approach in the literature to proxy match quality is using observable individual and couple characteristics (e.g., Weiss and Willis, 1997; Eckstein et al., 2019; Low, 2024). We follow this approach and examine how RQ relates to a set of individual and couple characteristics commonly used as proxies: sex, college education, employment status, and log monthly income for both partners, and marital status, and the presence of children. Table A.7 reports results for the full data and for the sample of first-time parents. Coefficient magnitudes and signs are stable across samples. Consistent with the literature, which often emphasizes education as a key proxy for match quality, RQ is positively associated with both own and partner's education. Marriage is associated with higher RQ, whereas the presence of children is linked to lower RQ. When including individual fixed effects, most individual characteristics lose significance, suggesting that the cross-sectional correlations are largely driven by compositional differences. In contrast, the coefficients on marriage and children remain significant.

Beyond these approaches based on observable characteristics, explicit empirical measures of match quality in economics are scarce. We assess the value of RQ by comparing it to these existing measures. These include a residualised subjective well-being measure, which aims to

isolate the relationship-specific component of life satisfaction (Chiappori et al., 2018), and survey questions on couple happiness and whether partners have discussed separation (Bertrand et al., 2015). We also consider the cohesion and satisfaction sub-scales of the revised Dyadic Adjustment Scale (DAS) from the psychology literature (Busby et al., 1995), which can be constructed from a subset of the items in Table 1. Table A.8 reports their correlations with RQ. In all cases, the correlation with RQ exceeds the correlation among the alternative measures, suggesting that RQ summarizes their information more parsimoniously. The residualised subjective well-being measure exhibits the lowest correlation with RQ, while the DAS sub-scales display the highest correlations.

2.2 Household Specialization

Household specialization refers to the division of labour between partners in terms of home and market production. Understanding Society provides data on the number of hours each individual works in the labour market and spends on routine housework per week.[5] We use this information to compute the shares of home and market production out of the couple total carried out by women as our proxies for household specialization. We refer to these as the *female share* of unpaid housework and paid labour market hours. A 50% share in both types of work indicates no specialization, larger shares imply female specialization and smaller ones male specialization.

Figure E.1 plots the distribution of the female share of paid and unpaid hours (a) before and (b) after the birth of the first child. Before birth, household specialization varies considerably. The distribution of the female share of paid hours is centred around 50%, while the distribution of unpaid work is more evenly spread, primarily above the 50% mark. After birth, both distributions experience noticeable changes: the mass of paid hours moves below the 50% threshold, while the share of unpaid hours becomes more concentrated above this point.

We classify couples based on the female share of paid and unpaid hours *before* the birth of their first child, distinguishing four types: (i) traditional couples, where women perform more than 50% of housework and men take a larger share of paid work; (ii) unbalanced couples, where women contribute more than half of both types of work; (iii) egalitarian couples, where men and women share both types of work roughly equally, with female shares between 40% and 60%; and (iv) counter-traditional couples, where men perform more than 50% of housework.[6]

[5]Specifically, the questions asked are: "About how many hours do you spend on housework in an average week, such as time spent cooking, cleaning and doing the laundry?" and "Thinking about your (main) job, how many hours, excluding overtime and meal breaks, are you expected to work in a normal week?". The responses to these questions are available for approximately half of the sample of parents. Following Borra et al. (2021), we do not consider childcare to be part of routine housework. Time use information on childcare is unavailable.

[6]Figure E.2 illustrates this classification based on the female shares of paid and unpaid hours. Each couple belongs to a single category, with no overlap between classes. Couples with female shares between 40% and 60% in both paid and unpaid work are always classified as egalitarian.

2.3 Sample Restrictions and Descriptives

The population of interest consists of individuals in cohabiting relationships, married or not, who become parents.[7] Our analysis focuses on the subset of individuals who become *biological* parents for the first time within the observation period. To ensure the robustness of our sample, we impose several criteria. First, we restrict our analysis to individuals who were aged 18 to 45 (women) or 50 (men) when they first became parents. Second, we exclude couples cohabiting with children from previous relationships and retain only observations corresponding to the couple in which the first birth occurs. Third, we use only heterosexual couples. The resulting sample is an unbalanced panel of 2,015 individuals observed up to 7 times, capturing 97% of the fertility events in the data.

Table 2 presents summary statistics for the analysis sample in the period preceding childbirth. Panel (a) reports individual characteristics separately for fathers and mothers. On average, fathers are 32 years old, while mothers are three years younger. A smaller share of mothers than fathers are active in the labour market or employed. Mothers work three hours less per week in the labour market, earn lower monthly labour income, and spend three more hours on housework in comparison to fathers. Both report above-average RQ compared to the full sample. Panel (b) summarizes couple characteristics. On average, respondents cohabit for four years before having their first child. Consistent with the time-use patterns, mothers contribute slightly less than 50% of paid labour market hours and take on more than 60% of unpaid housework.

Table E.1 summarizes the same set of characteristics separately for each couple type based on pre-birth household specialization.[8] Importantly, there are no significant differences across couple types in the timing of childbirth in terms of age and relationship tenure, or in marital status. The most pronounced differences in labour market outcomes emerge between traditional couples and the other groups. Traditional couples have, on average, lower educational attainment and lower household income. They also exhibit the largest within-household differences in paid and unpaid work hours (differing by 10 and 7 hours, respectively), and in monthly labour earnings. RQ is lowest among unbalanced couples, aligning with the full data average, and highest among egalitarian couples. Egalitarian couples also have the highest household income.

[7] Some studies also examine infertile individuals who attempt to have children but are unable to conceive, offering insights into the counterfactual scenario where no child is born (Bögl et al., 2024). While our main analysis focuses on individuals who become parents, since we argue this is the policy-relevant population, we conduct robustness checks using infertile individuals as the control group in Appendix B.3.

[8] The reduced sample sizes result from the limited availability of responses to time use questions and the unbalanced nature of the sample, where not all individuals are observed in the period immediately before birth.

Table 2: Summary statistics for the period before the birth of the first child.

	(1) Fathers	(2) Mothers		(3) Couples
(a) Individual characteristics			(b) Couple characteristics	
Age	31.77	28.92	Tenure	4.412
	(6.037)	(5.294)		(4.048)
College educated (%)	39.04	49.81	Married (%)	52.44
	(48.78)	(50.00)		(49.56)
Active in labour mkt (%)	73.43	71.92	Monthly household income (£)	4356.3
	(44.16)	(44.90)		(2617.1)
Employed (%)	70.44	68.37	Female share of paid work	0.495
	(45.60)	(46.46)		(0.257)
Monthly labour income (£)	2469.7	1918.3	Female share of housework	0.615
	(1664.0)	(1332.7)		(0.206)
Weekly work hours	26.22	23.76		
	(19.14)	(17.61)		
Weekly housework hours	5.473	8.631		
	(4.027)	(5.942)		
RQ	0.386	0.423		
	(0.877)	(0.859)		
Observations	2608	2940	Observations	3387

Notes: This table presents summary statistics for the main analysis sample of parents in the period before the birth of their first child. The sample consists of individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18-45 (women) or 18-50 (men) at the time of birth, and were not cohabiting with children from previous relationships. Panel (a) reports the mean values of individual characteristics separately for women and men, and panel (b) summarizes couple characteristics. Couple characteristics can be retrieved even when only one partner is observed. Female shares of paid and unpaid work represent the share of total weekly household paid labour market hours and unpaid housework hours performed by women. Income measures are presented in gross terms. In the full data, RQ has a mean of 0 and a standard deviation of 1. Standard deviations are reported in parentheses.

3 Empirical Strategy

Having a child is a deliberate decision, and individuals who become parents differ from those who remain childless in several aspects (e.g., Cetre et al., 2015). In particular, as documented in Table B.2, parents tend to have higher levels of RQ before childbirth than individuals who never have children, reflecting the fact that individuals in higher quality couples are more likely to choose to have children. Comparing parents with never parents may therefore confound the effect of childbirth with pre-existing differences in RQ and subsequent mean reversion.

Our objective is to estimate the average treatment effect on the treated (ATT), namely the effect of first childbirth on individuals who become parents, rather than the average effect in the broader population. We exploit variation in the *timing* of first births and adopt a dynamic difference-in-differences (DiD) approach restricted to the sample of individuals who eventually become parents, following the approach used in related work to study the impact of the

transition to parenthood (e.g., Kleven et al., 2019b; Siminski and Yetsenga, 2022; Ahammer et al., 2023). We compare individuals who have already had their first child with individuals who have not yet had a child but will do so later. Thus, by restricting the comparison group to eventual parents, our design avoids relying on comparisons between parents and individuals who never have children.

Formally, consider the usual two-way fixed effects (TWFE) regression model:

$$y_{i,t} = \alpha_i + \mu_t + \sum_j \mathbb{1}\{j = t - G_i\}\delta_j + u_{i,t} \tag{1}$$

where $y_{i,t}$ denotes the RQ of individual i in period t. We denote as G_i the year in which individual i has a first child, so $t - G_i$ represents event-time, the number of periods since birth. Equation 1 includes the full set of event-time dummies, as well as individual fixed effects, α_i, and period dummies, μ_t. By including individual fixed effects, identification comes from deviations of each individual's RQ from their own pre-birth trajectory, rather than from cross-sectional comparisons. The coefficients of interest are δ_j, which capture changes in RQ around first child birth.

Recent econometric literature has highlighted biases derived from estimating TWFE models via OLS in settings with staggered treatment adoption (De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; Gardner, 2022; Roth et al., 2023). To address these concerns, we use the estimator proposed by Callaway and Sant'Anna (2021).[9] We aggregate the resulting group-time treatment effects by event-time to trace the dynamic effect of childbirth on RQ.

3.1 Identification Assumptions

The causal interpretation of the event-time coefficients δ_j as the ATT relies on two identification assumptions: no anticipation and parallel trends. While these assumptions cannot be directly tested, we assess their plausibility in Appendix B.2.

No anticipation – This assumption requires that changes in RQ do not systematically predict when individuals have their first child.[10] Crucially, it concerns *changes* in RQ, not levels. The cross-sectional finding that parents have higher average RQ before birth is fully absorbed by individual fixed effects and does not threaten identification.

To assess its plausibility, we first verify that the evolution of RQ is flat during the periods preceding the birth of the first child. However, flat pre-trends could arise from averaging out couples with varying circumstances: some may decide to have a child in response to a tempo-

[9] A detailed summary of this methodology is provided in Appendix B.1. We implement it using the `csdid` package on Stata.

[10] This assumption can be relaxed to limited anticipation if the anticipation horizon can be specified. We relax this assumption in complementary analyses to allow for outcomes to react during pregnancy.

rary decline in RQ, while others may do so when their relationship is going particularly well. In Figure B.1, we provide evidence that this is not the case, as variation in RQ is not larger during the pre-birth periods, and there are no significant deviations from individual-specific means before birth. Second, we perform "long" comparisons of pre-birth RQ values relative to the baseline period in Figure B.2, following Roth (2024), and find that the coefficients are not statistically different from zero. Third, we examine in Table B.1 whether the timing of childbirth is predictable. We regress an indicator for birth in the subsequent period on pre-birth RQ, economic characteristics, and couple characteristics, controlling for individual and period fixed effects. We confirm that changes in RQ before birth do not predict timing. The only significant predictor is marriage.

Parallel trends – This assumption states that, in the absence of childbirth, RQ would have evolved similarly across individuals, regardless of when they had their first child. The flat pre-trends documented above also support this assumption. As additional evidence, in Figure B.3 (a) we show that parents who experienced delays in the birth of their first child, namely those who have used fertility treatments or faced involuntary pregnancy interruptions, do not exhibit significantly different pre-birth RQ trends from those who did not. In addition, our results hold when restricting comparisons to individuals with similar RQ levels before birth in Figure B.3 (b), a restriction that makes parallel trends more plausible by construction.

Beyond these two assumptions, the aggregated event-study estimates additionally require *homogeneity of treatment effects across cohorts*. We find no evidence of such heterogeneity, both from estimating effects separately by cohort in Figure B.4 and from the pre-birth coefficient in our main specification, which provides a direct test for baseline differences across cohorts and is statistically zero.[11]

3.2 Selection into Parenthood

Although our identification strategy focuses on the ATT among eventual parents, we further examine whether selection into parenthood may affect our results. We consider two potential sources of selection. First, our results may partly reflect characteristics shared by individuals who attempt parenthood. Second, couples whose RQ deteriorates may be more likely to separate before having a child, so that our sample of eventual parents excludes couples whose relationships end before a birth occurs.

Attempting parenthood. We assess whether our results reflect selection into attempting parenthood rather than the transition to parenthood itself. In Appendix B.3 we replicate the main

[11] Recall that the questions used to construct RQ are asked every other Understanding Society wave, so each individual contributes to every second event time. Consequently, individuals contribute observations at every second event time, meaning that some treatment cohorts contribute to odd event-times and others to even event-times.

analysis using two alternative comparison groups: individuals who tried but failed to have children (infertile individuals) and individuals who neither wanted nor had children (never-parents).

The infertile group is particularly informative in this respect, as these couples share the same positive selection into attempted parenthood as the treatment group, including a similar pre-birth RQ distribution, yet did not experience a live birth (Bögl et al., 2024). We identify them as individuals who become pregnant but never experience a live birth.[12] Using this group as the control and exploiting the timing of their first failed pregnancy, we find that the estimated effect of childbirth on RQ remains close in magnitude to the main result as shown in Figure B.6, providing evidence against the interpretation that our estimates primarily reflect selection associated with the decision to attempt parenthood.

We also consider individuals who neither want nor have children, identified as those who have never been pregnant and remain childless by the end of their fertility cycle. Following Kleven et al. (2019b), we employ a design that assigns placebo births to never parents. While this group was initially excluded due to concerns about comparability to future parents and about the plausibility of the identification assumptions, the results in Figure B.6 are similar to our main findings, with improved precision.

Survival until first birth. We also examine whether our results are sensitive to selection arising from couples who separate before having a child. Because RQ is observed only while couples remain together, our sample excludes couples whose relationships dissolve before a first birth. If couples whose RQ deteriorates are more likely to separate before having a child, this may affect the comparability of earlier- and later-treated parents. The flat pre-trends among never-parents documented above provide some reassurance that the absence of pre-birth changes is not specific to our sample of eventual parents. However, this comparison does not directly address selection through separation before parenthood.

To assess the potential importance of this selection for our baseline results, we conduct a sensitivity analysis that adjusts the comparison group at each event-time for predicted separations among childless couples. We vary the assumed share of separating couples who would otherwise have become parents to assess how sensitive our findings are to this source of selection. As shown in Figure B.7, the estimated decline in RQ remains negative and statistically significant under moderate assumptions about selection.

[12] The data do not allow us to distinguish involuntary pregnancy losses, such as miscarriages or stillbirths, from induced abortions, so this group may include individuals who were not actively attempting to have a child. Reassuringly, their pre-pregnancy RQ distribution is similar to that of parents.

4 Impact of Children on Relationship Quality

Figure 2 presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The reported coefficients represent the impact of childbirth on RQ at each period relative to birth. Since RQ is standardized, the estimates are interpreted in standard deviations. The available data allow us to examine up to four periods before and seven periods after the birth of the first child. We also plot 95% confidence intervals.

The coefficients corresponding to periods before birth are not significantly different from zero.[13] These flat pre-trends are consistent with the identification assumptions of no anticipation and parallel trends discussed in Subsection 3.1. The coefficient corresponding to the period of birth is also statistically indistinguishable from zero. This means that RQ does not immediately react to the birth of the first child.

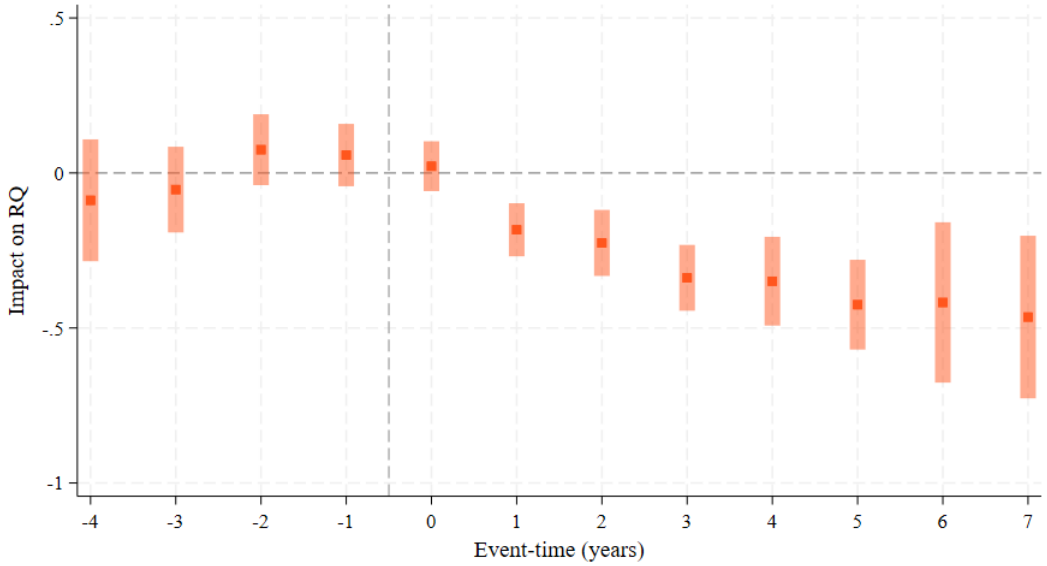
There is a significant decrease in RQ during the first five years after birth. By this point, the decrease stabilizes at approximately half a standard deviation. As an illustration, individuals ranked in the 75th percentile of the RQ distribution before the birth of their first child are pushed down to median RQ over this time period. This negative impact stabilizes after five years and until seven years after child birth. This suggests that, although having a child shifts RQ down, it does not change how this variable evolves over time.

Impact on mothers and fathers. Most of the literature on the impact of parenthood documents strongly asymmetric effects of first child birth on mothers and fathers. Studies on labour market outcomes find that women reduce their participation in the labour force, work fewer hours, and experience declines in wages and earnings following childbirth, while men exhibit little to no change. These differences contribute to within-household inequality and widen the gender wage gap (Adda et al., 2017; Blau and Kahn, 2017; Kleven et al., 2019b; Cortés and Pan, 2023; Goldin, 2021; Albanesi et al., 2023). Beyond labour market outcomes, gender differences in the impact of childbirth also emerge in other domains, including time spent on housework (Aguilar-Gomez et al., 2026; Siminski and Yetsenga, 2022), leisure (Aguiar and Hurst, 2007), sleep (Costa-Font and Flèche, 2020) and even mental health (Ahammer et al., 2023).

Motivated by this evidence, we study whether similar gender differences emerge in RQ around first child birth. Even though RQ is a shared concept, as discussed in Subsection 2.1, individual perceptions may differ. However, as shown in Table 2, mothers and fathers report

[13] Unlike in conventional event-study plots, the coefficient at event-time (-1) is not normalised to zero. Following Callaway and Sant'Anna (2021), pre-birth coefficients capture changes relative to the preceding period, while post-birth coefficients are measured relative to the last observed pre-birth period. Thus, the coefficient at (-1) tests whether the treated-control difference changes between the preceding period and event-time (-1). We find no statistically significant change.

Figure 2: Dynamic effect of first child birth on RQ.



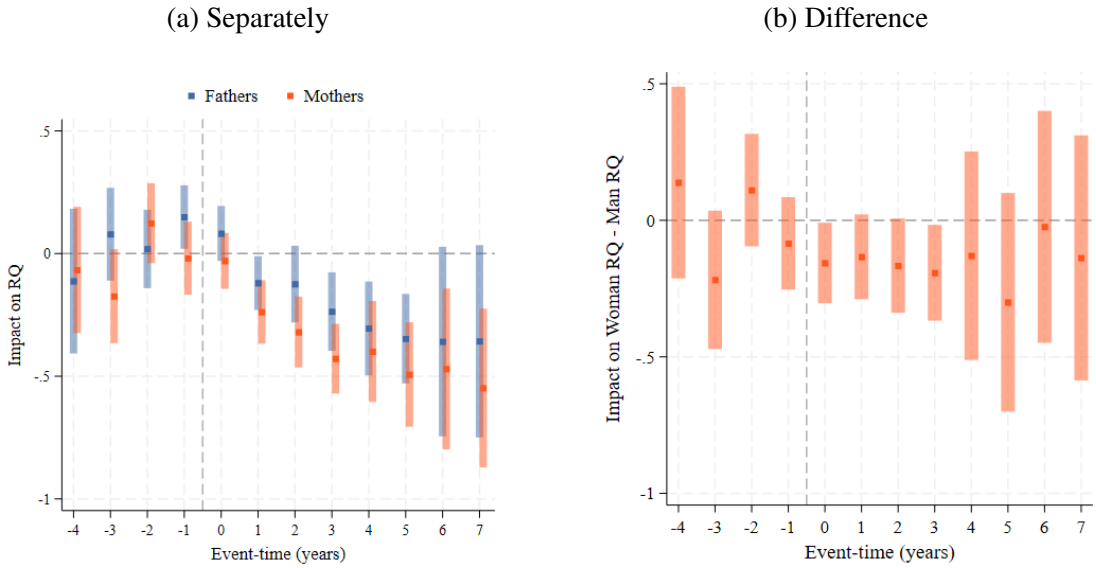
Notes: This figure presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. We also plot 95% confidence intervals.

very similar levels of RQ before childbirth. In Figure 3 (a), we repeat the main analysis separately for men and women. We find a comparable decline in RQ for both parents over the first three years after birth. The decline is somewhat steeper for mothers, but it stabilizes thereafter for both parents, at a slightly lower level for women. Figure 3 (b) tests whether the gender differences are statistically significant by comparing RQ responses between both partners. Although the difference in RQ between women and men becomes negative after childbirth, it remains small and not statistically distinguishable from zero in most periods, with only marginal significance in the first and third years after birth.

The similar declines for mothers and fathers distinguish our results from much of the prior literature on the effects of childbirth, which documents particularly pronounced changes for mothers across many outcomes. RQ deteriorates for fathers as well, even though they typically do not experience significant changes in many of the outcomes traditionally studied. This finding highlights an important dimension of the transition to parenthood that affects both partners and provides valuable insight into fathers' experiences, contrasting with the notion that fathers are unaffected by parenthood.

Implications for couple dissolution. To better understand the relevance of these results, we compute a back-of-the-envelope calculation of the implications for couple dissolution. In Section 2 we provide evidence of the capacity of RQ to predict marital transitions, finding lower values of RQ preceding couple separation. The regression analysis in Table A.4 indicates that,

Figure 3: Impact of first child birth on RQ for mothers and fathers.



Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). Panel (a) reports separate estimates for mothers and fathers in the main sample of parents. Panel (b) uses the difference between women's RQ and their male partners' RQ as the outcome. The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. We plot 95% confidence intervals.

in the main analysis sample of parents, a one standard deviation higher RQ is associated with a 0.8 percentage point lower probability of separation in the subsequent period.

The magnitude of this coefficient highlights the relevance of RQ in dictating couple dissolution. Based on this correlation, the estimated half standard deviation decline in RQ following childbirth would be associated with a 0.4 percentage point higher probability of separation. Relative to the baseline separation rate in our sample of around 2%, this would correspond to roughly a 20% higher separation probability.[14] Although this calculation is purely correlational, it illustrates that the magnitude of the effect on RQ is economically meaningful in the context of couple stability, and is consistent with prior evidence on the impact of children on couple dissolution (Lillard and Waite, 1993; Svarer and Verner, 2008).

As argued in Browning et al. (2014), RQ is just one aspect of how the presence of children influences couple dissolution decisions. Having a child tends to raise separation costs for parents while also increasing the value of maintaining the relationship through economies of scale within the couple, both of which work against dissolution and offset the effect of the RQ decline on stability. While exploring these factors jointly is crucial given the detrimental effects that separation can have on children, studying them separately may be equally relevant,

[14] Figure C.1 plots the annual share of couples separating among all couples observed in each year. Approximately 2% of couples separate each year. For comparison, Office for National Statistics (2022) reports that an average of 1.44% of married couples aged 20-45 in England and Wales divorced annually between 2009 and 2021. Equivalent data for cohabiting couples is not available.

especially considering the adverse effects of being exposed to a low parental RQ.

General happiness and RQ. We next study the effect of the birth of the first child on general happiness to obtain a more comprehensive picture of the impact of children on individual utility. RQ can be interpreted as part of the utility exclusively attached to the couple, distinct from the utility derived from having a child. Therefore, the impact of childbirth on happiness and RQ would not necessarily align in the same direction.

In Figure C.2 (a) we repeat the main analysis using standardized general happiness as the outcome.[15] We find that the birth of the first child induces an immediate increase in general happiness during the pregnancy and birth periods, with the increase being slightly larger for mothers than for fathers. However, the absolute magnitude of this impact is smaller than the one documented for RQ. Moreover, the impact on general happiness only lasts one period, not mirroring the persistence observed in the impact on RQ.[16]

The depicted patterns align with the hypothesis that happiness encompasses a component attached to the couple (RQ) and another attached to the child.[17] The main finding of this paper indicates that the couple component decreases after birth. Hence, the fact that the sum of both components - couple and child - does not decrease suggests that the child component is positive. Taken together, while the event of childbirth may indirectly reduce happiness, entailing negative consequences for RQ, this evidence suggests that the direct contribution of children to happiness is not negative.

Other measures and shocks. We compare the magnitude and persistence of our main finding with the effect of this event on other subjective outcomes at the individual level. Figure C.2 (b) displays the results for subjective well-being.[18] In contrast to our main finding, we do not document an effect of first child birth on subjective well-being beyond the first period post-birth. Ahammer et al. (2023) use a similar approach to study the impact on mental health and find a significant increase in antidepressant prescriptions for the sub-population of individuals that visit the psychiatrist. They find a greater effect on mothers, which they attribute to the

[15]The question used is "Have you recently been feeling reasonably happy, all things considered?". We take two periods before birth as the baseline to allow for anticipation during the pregnancy period.

[16]There is a decreasing trend in general happiness after birth. Blanchflower and Oswald (2004) find that general happiness is U-shaped in age, reaching the minimum at ages 37-41 in the United Kingdom. In our sample, the average age at birth is 30, as seen in Table 2 (32 for fathers and 29 for mothers). Thus, the documented decline corresponds to a life-cycle trend that our main methodology does not allow accounting for.

[17]In Table C.1, we quantify the association between happiness and RQ for individuals who have not become parents by age 50, the end of the fertility cycle. We find that a one standard deviation higher RQ is associated with a 0.2 standard deviation higher happiness measure, on average.

[18]The subjective well-being measure combines the responses to the 12 General Health Questionnaire (GHQ-12), related to physical and mental health. Some of the questions relate to sleep and being under strain, which are mechanically affected by having a child. Previous literature has discussed the association between children and well-being, although not in a causal way (Dolan et al., 2008; Clark et al., 2008; Blanchflower, 2009; Ferrer-i Carbonell, 2013).

higher childcare burden borne by them.

Finally, we examine the distinctiveness of the birth of the first child as a shock to RQ by comparing it to another major life event: becoming unemployed. Previous studies have found a positive correlation between male unemployment and divorce probabilities (Jensen and Smith, 1990; Charles and Stephens, 2004; Eliason, 2012; Doiron and Mendolia, 2012). Figure C.3 displays the results obtained from computing our main estimation around unemployment events on RQ.[19] Unlike childbirth, unemployment is not significantly associated with changes in RQ, which contrasts with the findings on divorce.

These insights can inform models of family formation and dissolution in multiple ways. As mentioned earlier, RQ is a measure of the non-material benefits of being in a relationship. The difference between the association of unemployment with RQ and with divorce may indicate that, while the job loss of a partner reduces the material gains from being in a relationship with them, it may not have such impact on non-material gains. Furthermore, our main finding evidences the smooth and path-dependent nature of RQ, which could guide the modelling of the distribution of this factor within family formation and dissolution models.

4.1 Robustness

We address four potential concerns that could affect the interpretation of our results. First, we explore whether the impact on RQ is driven by specific items within the measure. This could result from an uneven response of items across different blocks, or from a change in how the items are valued after the birth of a child. Second, we consider whether the persistent impact on RQ is solely due to subsequent children or if it represents a lasting effect of the first child's arrival. Third, we take into account the limitation that RQ cannot be observed after a couple dissolves, which may introduce some selection bias that could distort our results. Fourth, we examine whether the timing of child birth, in terms of the age of the parents and the duration of the relationship, has any influence on our findings.

4.1.1 Time Invariance of RQ

It is a concern whether the decrease in RQ might be solely attributed to specific items within the measure, particularly those related to couple time use. This concern arises from the fact that, after the birth of the first child, the available time for couples to spend together decreases significantly due to the new responsibilities that occupy a substantial portion of their time (Bianchi, 2011). Given that individuals highly value spending leisure time with their partners (Georges-Kot et al., 2024), this decrease could directly reduce RQ, potentially being the only driver of

[19] We focus on the first instance of unemployment for an individual within our observation period and exclusively consider individuals who become unemployed at some point ("not yet unemployed") as our control group. We do not make any causal claims about these results.

the documented decrease.

To explore this, we repeat the factor analysis excluding the time use items and using solely the items containing subjective assessments of the relationship. Figure D.1 (a) plots the results of conducting our main analysis on the resulting measure. The documented decrease is steeper than what we observed on RQ. For completeness, we repeat this procedure using only the time use items in Figure D.1 (b). In this case, the impact is less pronounced and not statistically different from zero after five years. This suggests that our primary finding is not only driven by reduced time together, but is mainly influenced by changes in how individuals assess their relationship.[20]

However, the possibility remains that individuals change how they value different items within the RQ measure after becoming parents. To address this, we construct a new RQ measure using data solely from individuals who are already parents. We use those observations to obtain the factor loadings in the factor analysis and construct the measure for the entire sample. As shown in Figure A.2, subjective assessment items become more important relative to the main measure, with the regret item exhibiting the highest loading. Figure D.2 displays the results of our main analysis using this measure, showing that our primary result remains consistent.

4.1.2 Subsequent Fertility

Our analysis focuses exclusively on the birth of the first child. It is therefore unclear whether the persisting impact on RQ represents a lasting effect of the first child's arrival or if it is merely a consequence of subsequent children. In Figure D.5 we split the analysis sample in two groups: (a) individuals with only one child or (b) those with subsequent children. About 38% of the individuals in our sample are observed having an only child. The decline in RQ emerges somewhat more gradually for individuals who have additional children, but the five-year effect is similar across both groups. Thus, we cannot attribute the documented persistence on RQ to parents of more than one child.

However, this analysis masks censoring in the sample, as many individuals have not reached the end of their fertility cycle at the last observation. We restrict the analysis to individuals observed at age 40 or older, which better approximates their lifetime fertility. This restriction remarkably reduces the sample and does not allow for a dynamic analysis. Table D.1 displays the static DiD results obtained through the usual estimation method. Note that the obtained ATT is an average of the impact across all periods after, where periods with more observations get higher weights (Callaway and Sant'Anna, 2021). The results suggest that subsequent children do not have an additional negative impact on parental RQ.

[20] Figure D.3 and Figure D.4 display the impact of first child birth on each subjective assessment and couple time use item separately. The item responses are standardized for comparability. The degree of happiness with the couple and the frequency with which partners kiss experience the largest impact.

It is important to note, however, that this could be a selected sample, as couples who choose to have a second child may have had a different experience compared to those who did not. We focus exclusively on second-time parents to examine whether the birth of a second child has any impact on RQ. Figure D.6 shows the results of the main analysis around the birth of the second child. Although we observe a pre-trend, likely reflecting the decline in RQ following first child birth, there is no significant effect associated to the birth of the second child.

4.1.3 Attrition due to Post-birth Couple Dissolution

Due to the nature of our measure, RQ can only be observed while couples remain together. Hence, couple dissolution leads to attrition from our outcome measure, potentially introducing selection bias that could distort our results. In our sample, 12.5% of couples dissolve during the observation window. We first assess whether our results are driven by these couples by repeating the main analysis excluding all couples who separate during this period. The results in Figure D.7 (a) closely mirror our main findings, with point estimates that are statistically indistinguishable from those in our baseline specification. This suggests that our results are not driven by couples who subsequently separate.

To further assess the potential for selection bias, we conduct a sensitivity analysis based on extreme imputations, following the logic of worst-case bounds for missing outcomes (e.g. Manski, 1989; Horowitz and Manski, 2000). Specifically, for couples who separate, we impute the unobserved RQ values in subsequent periods using two extreme values: the 1st and 99th percentiles of the RQ distribution within our sample. This represents a deliberately conservative exercise, assigning very low or very high RQ values to observations lost following couple dissolution.

Figure D.7 (b) presents the results. The point estimates under both imputations remain negative throughout the observation window, reinforcing our main finding that RQ declines following childbirth. However, for the 99th-percentile imputation the confidence intervals include zero at event-times beyond year five, when the share of unobserved outcomes is larger. Overall, the results remain qualitatively similar under these extreme imputations, providing further reassurance that our main findings are not driven by attrition following couple dissolution.

4.1.4 Age and Tenure at Birth

A first concern is that treated and control groups may differ in relationship tenure, particularly at longer event-times. For example, to serve as a control at event-time seven, a couple must have remained together for at least seven years without having a first child. These couples may differ from those who become parents earlier. To assess the importance of these differences, we first examine the distribution of first births across relationship tenures. As shown in Figure D.8 (a), 65% of first births in our sample occur between tenures three and seven but some births occur

even 10 years after that.

We then restrict the analysis to couples whose first child is born within this most common tenure window, making the comparison groups more similar in terms of tenure. This restriction substantially reduces the event-time window we can estimate. Even for couples who have a child at tenure three, the earliest tenure in our restricted window, we observe at most four post-birth years. Consequently, we can estimate effects only up to two years after childbirth, as shown in Figure D.8 (b). The obtained estimates closely resemble our main results, although the shorter event window prevents us from assessing whether the pattern persists in the longer run.

A second concern is that, in addition to the birth of the first child, the timing of this event may significantly affect RQ (Kleven et al., 2019b). We explore this by separately examining individuals who had children at different ages and relationship tenures. Figure D.9 presents the average RQ at each event-time period by (a) age bin and (b) tenure bin, normalizing RQ to zero in the period before birth. While baseline RQ levels differ for individuals who had children at different ages, the post-birth evolution of RQ remains consistent across groups.

Baseline levels also differ across tenure groups. In this case, individuals who had their first child at the earliest stages of their relationship experience a slightly greater decrease in RQ . However, this more substantial decrease could be partially attributed to the general trend of RQ decreasing during the early stages of a relationship, as seen in Figure A.6.

To ensure the robustness of our main findings, we include these variables as controls, as presented in Figure D.10. Note that Callaway and Sant'Anna (2021) does not allow us to study the dynamic impact of age and tenure on RQ, since it only allows to control for pre-birth levels of covariates. The differences for all sets of controls remain minimal until six years after the birth of the first child. At that point the coefficients of the specification including tenure become smaller.

To account for potential dynamic effects across age and tenure, we follow the event-study approach popularized by Kleven et al. (2019b). Specifically, we estimate the effects using OLS while replacing individual fixed effects with the full set of age and relationship tenure dummies. This analysis relies on the assumption that treatment timing is independent of unobserved factors once age, tenure and calendar time are conditioned on. The results displayed in Figure D.11 confirm that our main finding holds when accounting for the dynamic evolution of RQ over the life cycle and relationship duration.

5 Mechanism: Changes in Household Specialization

The birth of a child introduces a significant and unprecedented shift in how couples allocate their time. New responsibilities related to childcare emerge and the demand for routine house-

work increases, requiring couples to adjust how they divide paid work in the labour market and unpaid work at home. This reallocation of responsibilities shapes the roles that each partner assumes within the household, such as primary earner or main caregiver.

Research shows that men's labour market outcomes remain largely unaffected by the transition to parenthood, while women experience substantial declines in labour force participation and working hours (e.g., Kleven et al., 2019b; Goldin, 2021) and take on longer home production times (Aguilar-Gomez et al., 2026; Siminski and Yetsenga, 2022). Kuziemko et al. (2018) show that the magnitude of these changes is often unforeseen. Women, particularly high educated ones, underestimate the impact of having a child on their labour market outcomes and do not envision themselves as full-time caretakers after motherhood. Moreover, a more equal division of household tasks has been associated with higher relationship satisfaction (e.g., Blom et al., 2017; Carlson et al., 2018).

We therefore argue that the unanticipated shift towards traditional gender-based household specialization, where women assume primary responsibility for home production and men for market work, is an important mechanism underlying the decline in RQ following first birth.[21]

5.1 Impact of Children on Household Specialization

We start by documenting how mothers and fathers adjust their time use around the birth of their first child. Specifically, we re-estimate our main specification using weekly hours in paid employment and unpaid housework, as defined in Section 2, separately for mothers and fathers.[22] Figure 4 (a) plots the findings on paid work and Figure 4 (b) does it for unpaid work.

As seen in Table 2, before birth, fathers spend on average 26 hours in paid work per week, while mothers work roughly three hours less. Figure 4 (a) indicates that, after childbirth, mothers reduce their labour market hours sharply and persistently, stabilizing at about 11 fewer hours within two years. In contrast, fathers display only a slight decline later in the child's life, which is consistent with life-cycle trends. In terms of housework, in Figure 4 (b), mothers increase their weekly unpaid work time by about five hours over the five years immediately after birth, relative to a pre-birth baseline of about nine hours. This increase reaches around seven hours above the baseline seven years after birth. Fathers, by contrast, show no statistically significant increase in housework time.

Taken together, these results show that, following first birth, women give up the equivalent of more than one day of full-time work and assume nearly a full extra day of unpaid work

[21] In Appendix F, we examine other potential channels, including gender norms, household income, and the within-couple earnings gap. These factors also change around first birth, but their changes are smaller than those in household specialization.

[22] To account for life-cycle trends in employment and home production, we control for age in Equation 1. The identifying assumptions remain those in Section 3, with the nuance that the parallel trends assumption needed is conditional on age. Since the Callaway and Sant'Anna (2021) method can only account for baseline levels of control variables, overlooking non-linear trends, we may still observe patterns attributable to the life-cycle.

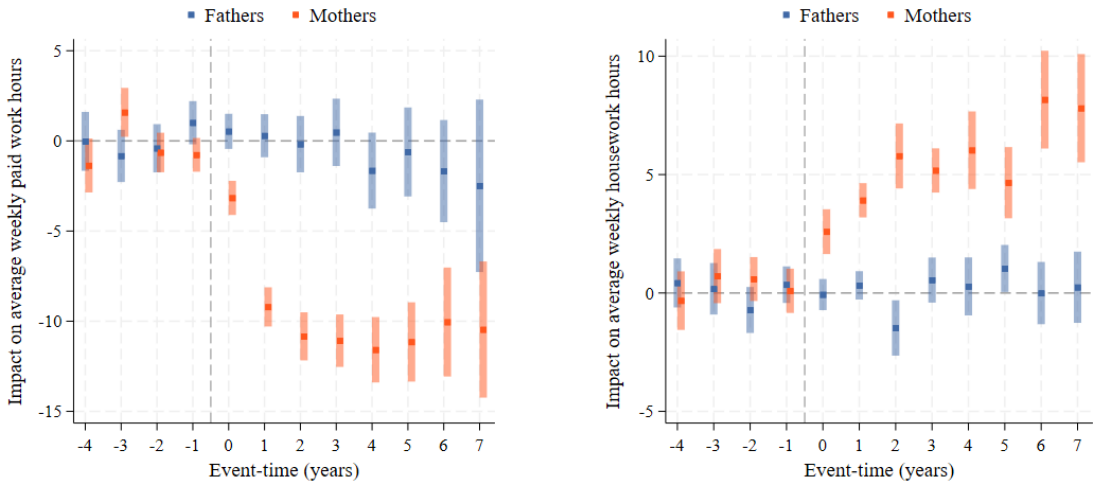
each week. Conversely, men maintain both their housework time and labour market hours. These findings are consistent with prior evidence on the impact of having children on labour market outcomes (Kleven, 2022) and home production (Aguilar-Gomez et al., 2026; Siminski and Yetsenga, 2022). They reveal asymmetries in how couples absorb the additional workload from children, with women assuming the bulk of new household responsibilities.[23]

Despite these asymmetric adjustments, both partners report similar declines in RQ. We cannot empirically disentangle whether this pattern reflects dissatisfaction arising first among mothers and then spilling over to fathers, or whether the reorganization of tasks strains the relationship as a whole from the outset. In either case, the evidence indicates that the unequal adjustment to parenthood erodes RQ at the couple level, rather than generating discontent for only one partner.

Figure 4: Impact of first child birth on paid and unpaid hours.

(a) Weekly labour market hours

(b) Weekly housework hours



Notes: This figure presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on time use variables, using the method proposed by Callaway and Sant'Anna (2021). The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. Panel (a) displays the results using average weekly paid labour market hours as an outcome, and panel (b) presents the results for average weekly housework hours (cooking, cleaning, doing laundry, etc.). On the period before first birth, fathers work 26 hours in the labour market and 5 hours at home, while mothers spend 24 hours in paid work and 9 in housework. We control for age at first birth to account for potential cohort-age differences. We also plot 95% confidence intervals.

To study how roles and household specialization evolve, we track changes in women's relative contribution to market and home production. To that end, we calculate the female share of total paid and unpaid work hours within the couple and classify couples into four categories based on their pre-birth values, as outlined in Section 2: traditional, unbalanced, egalitarian

[23] Andrew et al. (2021) document that mothers also spend more time on childcare than fathers do. This may account for the time freed by the reduction in labour market work that is not explained by the increase in housework time among mothers. However, information on time spent on childcare is not available.

and counter-traditional. These pre-birth values of the female shares provide information about the roles in place before parenthood and the existing degree of household specialization, which may reflect underlying comparative advantages within the couple.

We investigate the association between first child birth and these shares by couple type. Figure 5 (a) plots the average female share of paid work at each period relative to birth. The contribution of women to paid work decreases after birth across all types, accounting for less than half of the household total. The largest drop occurs in unbalanced couples, where the female share falls by 20 percentage points. In egalitarian and counter-traditional couples women's share stabilizes around 40%, representing a decrease of approximately 10 percentage points. Traditional couples are the least affected in this dimension, although the female contribution remains low compared to the other groups.

Figure 5 (b) displays the same shares for unpaid work. Regardless of the initial allocation, women in all couple types assume a larger share of unpaid work than men do after birth. Counter-traditional and egalitarian couples experience the largest changes, where female shares rise by about 25 and 15 percentage points, respectively, accounting for over 60% of the total. In contrast, traditional and unbalanced couples, where women already did over three quarters of unpaid work before birth, do not experience significant changes.[24]

This evidence suggests that couples across all types adjust their degree of household specialization after childbirth, transitioning towards more traditional, gender-based roles. However, the changes they undergo to establish these new roles vary considerably. Traditional couples experience the smallest adjustments, maintaining men's predominant role in the labour market and women's in home production. Among unbalanced couples, the division of housework remains unchanged, but the primary responsibility for paid work shifts from mothers to fathers. Notably, women in these couples are the only ones whose total time share declines after birth. Egalitarian couples experience substantial changes in the division of both types of work, abandoning the equal distribution of tasks. After birth, women adopt a dominant role at home whereas men specialize in the labour market. Counter-traditional couples experience the most pronounced shifts in the division of housework. Men abandon their predominant role in this domain, which is assumed by women, while preserving or strengthening their role in the labour market.

In traditional models of the household, paid and unpaid work are assumed to be allocated among household members based on the degree of substitutability between their labour market and domestic inputs (Becker, 1991). Our findings challenge the plausibility of this assumption. If the pre-birth distribution of tasks reflects potential comparative advantages within the couple, then the post-birth exchange of responsibilities cannot be fully explained by differences in skill substitutability. Instead, it points to the presence of other forces, such as frictions in the labour

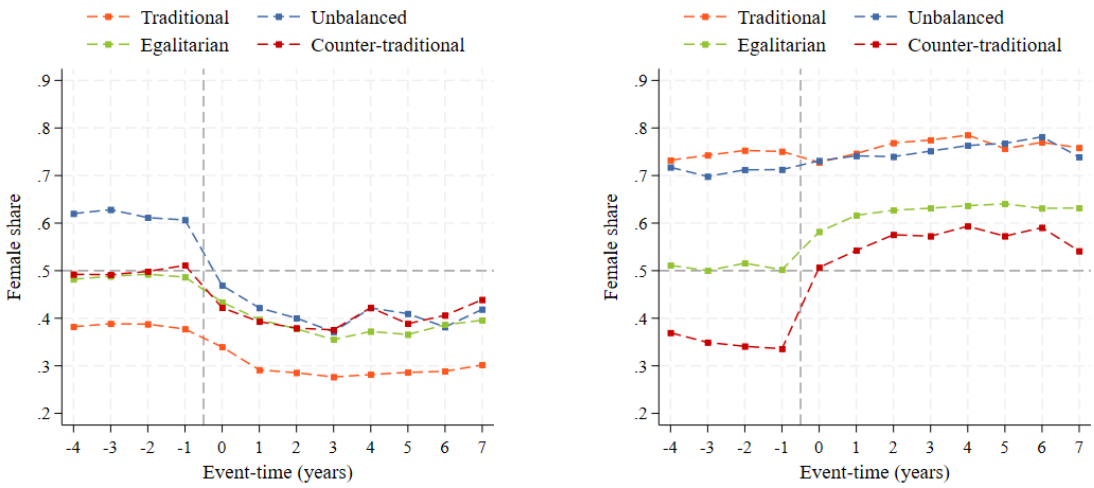
[24] In Figure E.3, we examine changes in labour market work and housework hours by couple type and gender, confirming that the shifts in female shares are driven by changes in women's time use.

market, for instance in the shape of discrimination against mothers (Petit, 2007; Becker, 2010), or identity considerations (Akerlof and Kranton, 2000; Ichino et al., 2019; Farré et al., 2021; Andresen and Nix, 2022). This evidence suggests that such factors become prevalent with parenthood, consistent with the view that individuals may face difficulty anticipating changes in roles and in household specialization (Kuziemko et al., 2018).

Figure 5: Household specialization around first child birth by couple type.

(a) Paid work share

(b) Unpaid work share



Notes: This figure displays the average female shares of (a) paid and (b) unpaid work — defined as the share of total weekly household paid labour market hours and unpaid housework hours performed by women — at each time around the birth of the first child. These averages are plotted separately by couple type, defined based on the distribution of tasks before first child birth. The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships.

5.2 Household Specialization and Relationship Quality

If the unanticipated shift toward gender-based specialization is the mechanism driving the decline in RQ, then the decline should be largest for couples whose pre-birth arrangements are most incompatible with gender-based roles, and smallest for couples who were already gender-specialized before birth. Leveraging the heterogeneity in task reallocation documented in Figure 5, we study how changes in household specialization after first birth may be associated with the effects on RQ. To do so, we repeat the main analysis on the impact of child birth on RQ separately for each couple type. Because the reduced sample size within each group does not allow us to carry out a dynamic analysis as in Equation 1, we adopt a static approach and estimate the following regression for each couple type:

$$y_{i,t} = \alpha_i + \mu_t + \delta D_{i,t} + u_{i,t} \tag{2}$$

where $D_{i,t}$ is a binary variable equal to one if individual i already has a child at time t, and α_i and μ_t denote the full set of individual and time dummies.

We estimate this regression using the estimator proposed by Callaway and Sant'Anna (2021). In a static setting, the coefficient δ captures an average treatment effect on the treated across post-birth periods.[25] The results in Figure 2 indicate heterogeneity in treatment effects with time relative to childbirth, suggesting that the interpretation of δ as a single, time-invariant effect is not appropriate in our context. Due to this and since we cannot disentangle the impact of first birth from unobserved characteristics of each couple type, we do not interpret these estimates as causal.

Table E.1 reports the average pre-birth RQ levels for each couple type. All types report RQ values above the average in the full dataset. Egalitarian couples exhibit the highest levels, followed by counter-traditional ones. Unbalanced couples display the lowest levels, although they are similar to those of traditional couples. The largest gender gap in RQ is observed in unbalanced couples, with men reporting lower values.[26]

Table 3 reports the results.[27] The estimates are broadly consistent with the prediction above.[28] Traditional couples were already organized along gender-based lines before the first birth and face no role shift after birth. Their RQ is correspondingly unaffected: the coefficient is slightly positive, but close to zero and not statistically significant. At the other extreme, egalitarian couples experience the largest decline in RQ. These couples face the greatest departure from their pre-birth arrangements: after the first birth, women substantially increase their share of housework while men specialize in paid work, transitioning from an equal division tasks to one resembling the traditional pattern.

The results for unbalanced and counter-traditional couples are more nuanced, but remain consistent with the mechanism. For unbalanced couples, the estimated effect is negative but close to zero. Although these couples experience the largest drop in the female share of paid work, their division of housework remains largely unchanged, and crucially, the overall allocation of total hours between partners becomes more equal after birth: women in these couples are the only ones whose total time share declines. Counter-traditional couples display a sizeable negative coefficient, although it is imprecisely estimated, likely reflecting the smaller sample

[25] As explained in Appendix B, the static aggregation constitutes a weighted average of the treatment effects for different treatment cohorts at different times relative to treatment. These weights are assigned based on the number of observations in each time bin, with more populous bins receiving higher weights. Hence, without this assumption, the estimates would be skewed towards the event-times with a larger number of observations.

[26] Equation 2 assumes that the impact of first child birth on RQ is linear for all types of couples, and hence it does not depend on the pre-birth values of RQ.

[27] Note that the event-time periods with a larger number of observations are those closest to the time of birth. As seen in Figure 2, the dynamic impact of children on RQ is increasing with time since birth. Thus, these estimates should be considered a lower bound to the full impact.

[28] Although the point estimates differ across couple types, pairwise tests cannot reject equality of the estimated effects across groups, likely reflecting the imprecision of the estimates. The differences in magnitudes should therefore be interpreted cautiously. See Figure E.4 for the p-values from these comparisons.

size for this group. Importantly, the direction is consistent with the mechanism: these couples experience the most pronounced shift in housework, with women assuming a domain previously led by men, and transitioning toward a gender-based allocation.

Table 3: Impact of first child birth on RQ by couple type.

	(1) Traditional	(2) Unbalanced	(3) Egalitarian	(4) Counter-traditional
ATT	0.0248	-0.00539	-0.189**	-0.137
	(0.096)	(0.099)	(0.072)	(0.091)
Observations	618	397	641	543

Notes: This table presents the results from a static difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. The estimation is computed separately by couple type, as defined in Figure E.2. We include observations from four years before first child birth to seven years after, in line with the period considered in the main analysis. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

We examine whether the heterogeneity across couple types differs by gender. Given the limited sample sizes, we pool traditional with unbalanced couples and egalitarian with counter-traditional couples, and estimate the effects separately for fathers and mothers. Figure E.5 reports the results. The estimates are more negative for women in both groups. For traditional and unbalanced couples, neither estimate is statistically significant. For egalitarian and counter-traditional couples, the estimates are negative for both genders but statistically significant only for women. These patterns broadly align with Table 3 and suggest that the decline in RQ may be more pronounced for women, although the estimates do not establish statistically significant gender differences.

Two factors therefore appear to jointly shape the heterogeneity across couple types documented in Table 3. First, RQ might be particularly sensitive to unequal divisions of total household production time. The attenuation observed for unbalanced couples suggests that the equalization in total hours could partially offset the tensions arising from role rearrangement. Second, the redistribution of housework may be the key factor triggering the deterioration in RQ, as changes in housework may be less foreseeable than changes in labour market work. While labour interruptions and protections for mothers are a central concern of many fertility policies, changes in home production are arguably less salient. Thus, the unforeseen and unequal distribution of housework may play an important role in the decline in RQ.

6 Concluding Remarks

This paper investigates the causal relation between first child birth and RQ, a proxy for the non-material gains from being in a relationship. The arrival of children triggers a series of decisions related to time allocation in both the labour market and at home. These decisions have implications not only for individual well-being, but also for the quality of the couple's relationship. While having children can lead to increased overall happiness, it does not necessarily guarantee happiness in every aspect of one's life.

We find that having a child significantly reduces RQ, and that this effect endures over time. In other words, the level of RQ is shifted downward after the birth of a child, maintaining the pre-existing life-cycle trend. This impact on RQ is substantial and persistent compared to other life events, such as unemployment, and other well-being indicators, like general happiness.

The paper posits that the arrival of children increases the demands on couples, including childcare responsibilities and additional housework. Consistent with existing literature, we find that women largely bear the increased housework at the expense of their participation in the labour market, a consequence not typically foreseen by mothers before childbirth. We exploit heterogeneity in changes to household specialization to explore the extent to which these changes mediate the impact of child birth on RQ. The results indicate that couples undergoing the most significant changes in household specialization also experience the greatest declines in RQ.

These findings open avenues for further research on the consequences of having children. From a policy perspective, several measures addressing the impact of children on various outcomes have been explored. Policies like paternity leave, shared parental leaves, and childcare provisions have been shown to enhance female labour market outcomes, potentially by redistributing household tasks and responsibilities more equitably among couple members. Our findings suggest that such policies may mitigate the negative effects of children on RQ, an outcome closely tied to couple dissolution and the decisions surrounding investments in children, which ultimately influence their future well-being. The 2015 introduction of Shared Parental Leave in the UK is a natural candidate for testing this claim. However, take-up among eligible fathers remained extremely low, preventing us from drawing causal conclusions about this specific policy.

Moreover, this paper introduces a novel factor for consideration in the study of decisions to have children. As noted earlier, the impact on relationship quality is, to some extent, unforeseen when deciding to have the first child. Nevertheless, the observed decline in RQ and the associated utility losses, despite potential utility gains from the child itself, may become a significant factor in deciding to have subsequent children. This factor could contribute to explaining the decline in fertility observed on the intensive margin.

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A Measure of Relationship Quality

A.1 Self-Completion Partner Module

Survey text. *For each of the following questions, please indicate which best describes your relationship with your partner at the moment. Please select only one answer per question.*

How often do you have a stimulating exchange of ideas?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you calmly discuss something?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you work together on a project?

1	Never
2	Less than once a month
3	Once or twice a month
4	Once or twice a week
5	Once a day
6	More often

How often do you discuss or consider divorce, separation or terminating your relationship?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you ever regret that you married or lived together?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

How often do you and your partner quarrel?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

How often do you and your partner "get on each other's nerves"?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you kiss your partner?

1	All of the time
2	Most of the time
3	More often than not
4	Occasionally
5	Rarely
6	Never

Do you and your partner engage in outside interests together?

1	All of them
2	Most of them
3	Some of them
4	Very few of them
5	None of them

The responses below represent different degrees of happiness in your relationship. The middle point, "happy", represents the degree of happiness of most relationships. Please select the number which best describes the degree of happiness, all things considered, of your relationship.

1	Extremely unhappy
2	Fairly unhappy
3	A little unhappy
4	Happy
5	Very happy
6	Extremely happy
7	Perfect

Responding to the module. Due to the personal nature of the questionnaire, preserving the privacy of respondents is a priority of Understanding Society. The questionnaire is part of a self-completion module, meaning that it is not administered by an interviewer but completed individually, either on paper or a computer, regardless of the interview mode.[29]

Although the questionnaire is answered individually, respondents may still be concerned about others being present during the interview. Individuals are given the option to refuse to answer the self-completion part of the survey, and if they do so due to the presence of another person, this reason is recorded separately. However, this is rarely reported: in the most recent wave, no one mentioned it, and in the previous wave, only three out of 363 refusals were due to another person being present.[30] Additionally, when an interviewer is present, they record whether another person is present and whether that person influences the interview. In the majority of cases, no one else is present during the interview.

Description of the responses. Table A.1 reports the distribution of responses across categories for each item in the questionnaire. Most respondents report frequent positive interactions: 72% exchange ideas at least weekly, 83% discuss matters calmly at least weekly and 72% kiss their partner most or all of the time. Most couples also share at least some outside interests (76%). Working together on a project is less frequent, with 44% reporting to do so at most once or twice a month. Regarding subjective assessments, around 90% rarely or never consider splitting up or regret getting married. Conflict is more common, with around 83% reporting quarrelling and getting on each other's nerves occasionally or rarely. Finally, relationship happiness is high, with 63% reporting being very or extremely happy, or that their relationship is perfect.

The correlations between items in Table A.2 further illustrate how these dimensions relate to one another. All correlations are statistically significant and have the expected signs, indicating that the items capture related dimensions of relationships. The strongest correlations arise between subjective assessment items, particularly considering splitting and regretting marriage (0.638), and between quarrelling and getting on each other's nerves (0.658). The time-use items are also strongly correlated, especially exchanging ideas and discussing matters calmly (0.672). By contrast, correlations between subjective assessments and couple time-use items are generally weaker, suggesting that the items capture complementary dimensions of relationships. This supports combining them into a single measure that encompasses both dimensions.

[29] Understanding Society conducts interviews in three modes: face-to-face, by phone, and online. Until wave six, all interviews were conducted either face-to-face (98%) or by phone. Web mode was introduced in wave seven, and its use has steadily increased, reaching 87% in the most recent wave.

[30] Other reasons for refusal include disliking the computer format, a child needing attention, confidentiality concerns, lack of motivation, or time constraints.

Table A.1: Percentage of responses per item.

(a) Time use items						
	never	less than once a month	once or twice a month	once or twice a week	once a day	more often
Stimulating exchange of ideas	3.33	8.78	15.90	36.29	21.95	13.76
Calmly discuss something	1.84	4.91	10.54	30.55	29.98	22.19
Work together on a project	6.77	15.67	21.81	27.64	12.65	15.46
(b) Subjective assessment items						
	all of the time	most of the time	more often than not	occasionally	rarely	never
Consider splitting	0.62	0.81	1.36	5.81	16.32	75.08
Regret getting married	0.57	0.81	1.22	7.50	16.43	73.49
Quarrel	0.55	1.58	4.34	38.97	44.24	10.32
Get on each other's nerves	1.02	2.66	5.93	47.49	34.19	8.72
(c) How often do you kiss your partner						
	never	rarely	occasionally	more often than not	most of the time	all of the time
Kiss	2.61	6.33	18.81	16.06	25.82	30.36
(d) Do you and your partner engage in outside interests together						
	none of them	very few of them	some of them	most of them	all of them	
Shared interests	6.49	17.24	42.96	26.29	7.02	
(e) What is the degree of happiness with your relationship						
	extremely unhappy	fairly unhappy	a little unhappy	happy	very happy	extremely happy	perfect
Degree of happiness	2.31	2.75	6.11	26.03	28.36	27.10	7.34

Notes: This table reports the percentage of responses in each response category for each item in the Self-Completion Partner Module of Understanding Society. The sample includes all available responses to these items. This yields 141,129 responses from 46,674 individuals observed up to 7 times between 2009 and 2024. The panels correspond to the different question and answer formulations listed above.

Table A.2: Correlation between item responses.

	(1) Consider splitting	(2) Regret marriage	(3) Quarrel	(4) Get on nerves	(5) Degree of happiness	(6) Joint project	(7) Exchange of ideas	(8) Calm discussions	(9) Kiss
(1) Consider splitting	1								
(2) Regret marriage	0.638	1							
(3) Quarrel	0.427	0.418	1						
(4) Get on nerves	0.427	0.454	0.658	1					
(5) Degree of happiness	0.393	0.455	0.344	0.389	1				
(6) Joint project	0.215	0.261	0.211	0.271	0.299	1			
(7) Exchange of ideas	0.241	0.280	0.191	0.250	0.329	0.563	1		
(8) Calmly discussions	0.302	0.336	0.272	0.304	0.360	0.569	0.672	1	
(9) Kiss	0.232	0.317	0.197	0.236	0.400	0.247	0.270	0.304	1
(10) Shared interests	0.198	0.247	0.194	0.254	0.301	0.361	0.286	0.280	0.307

Notes: This table reports pairwise correlations between items in the Self-Completion Partner Module of Understanding Society. Items (1)–(5) capture subjective assessments of the relationship, while items (6)–(10) capture couple time use and shared activities. The sample includes all available responses to these items.

A.2 Factor Analysis

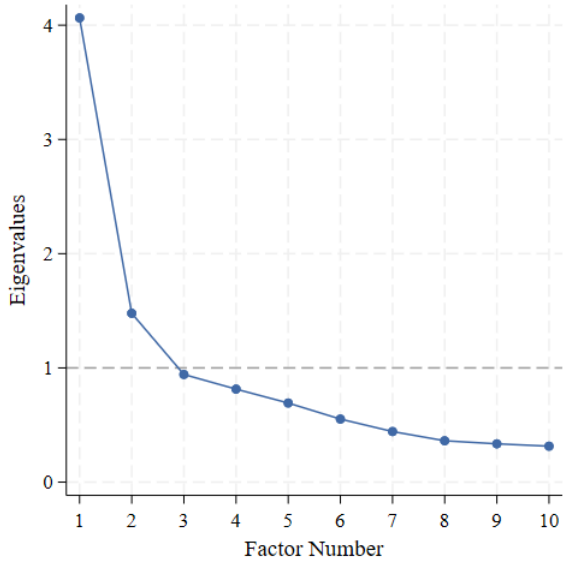
Factor analysis is a statistical dimension reduction method that seeks to disentangle latent associations between different items. The goal is to identify a set of latent variables, or factors, that explain the shared variance among a set of observed items. The variance of each item is assumed to be influenced by these common factors, as well as by a unique error term specific to each item. The factors, which are unobservable, are inferred from their effects on the items. They are identified by solving a system of equations that produces a set of weights, called factor loadings, which indicate how strongly each item is associated with each factor.

Crucially, factor analysis differs from principal component analysis (PCA), another common data reduction method, in both assumptions and interpretation. PCA aims to reduce dimensionality by finding the linear combinations of items that capture the largest possible variance, driven purely by the data and without assuming an underlying structure. In contrast, factor analysis assumes that observed variables are influenced by latent factors and explicitly models the unique variance of each variable. Its goal is to uncover the latent constructs rather than simply reducing the number of dimensions. In our case, the responses to the questionnaire items in Table 1 are the observed variables. Instead of treating relationship quality as a linear combination of the items, as PCA would, we consider that this variable is a latent factor influencing respondents' answers to each item of the questionnaire.

Before conducting the factor analysis, we recode the responses to the items in Appendix A.1 so that zero represents the least favourable couple behaviours and habits, like constantly quarrelling or never kissing one's partner, to ensure that all item responses are increasing in the quality of the relationship. We then perform factor analysis on all available responses to the ten items, and retain the first factor as the RQ measure. The decision to retain a single factor is based on the eigenvalues plotted in Figure A.1. The first factor has an eigenvalue of 4.06, more than twice that of the second factor, indicating a substantial drop in the explained variation. All factors from the third onward have eigenvalues below 1. This first factor accounts for 40.64% of the variation in the items.

Figure A.2 presents the factor loadings corresponding to RQ. All loadings are positive and greater than 0.5, indicating a strong association between RQ and each questionnaire item. The highest correlations with RQ are found in items related to having calm discussions and regretting getting married, while the lowest correlations are with items related to the frequency of kisses and to the number of joint outside interests. We also plot the factor loadings derived from conducting the factor analysis separately for men and women. The factor loadings for men are generally lower than those of the overall RQ measure, indicating weaker correlations between the extracted factor and the individual items. Similarly, the eigenvalue for the male sample (3.84) is lower than that of RQ in the full sample. In contrast, the factor loadings for women are generally higher than those of RQ, with a correspondingly larger eigenvalue (4.24).

Figure A.1: Eigenvalues from factor analysis.



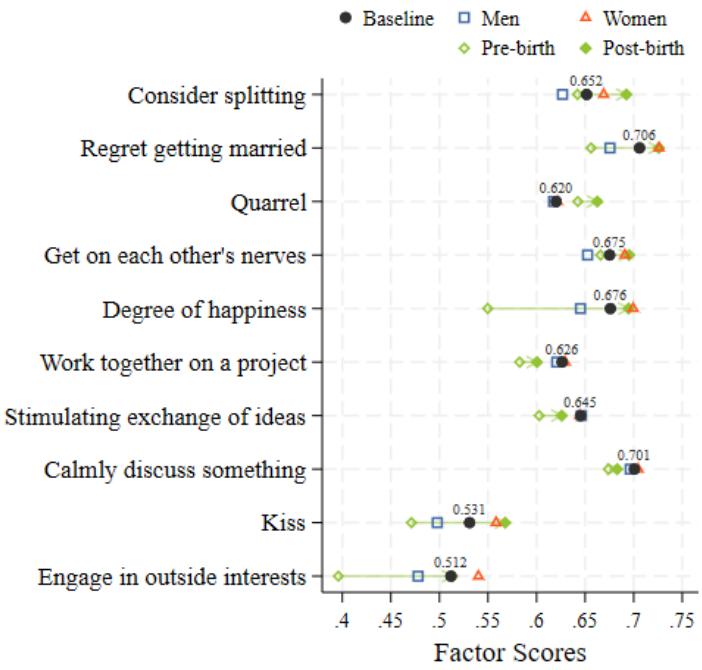
Notes: This graph presents the eigenvalues of all factors derived from performing a factor analysis on the responses to the ten questions in the Understanding Society Partner module estimated on all available Understanding Society waves (2009–2024). The eigenvalues indicate the amount of variation in the sample explained by each factor. The first factor from this analysis is the RQ measure.

Despite these differences, the ranking of items remains largely consistent across genders, with women assigning higher loadings and men consistently lower ones. The only exception is the item related to the frequency of stimulating exchanges of ideas.

We also repeat the factor analysis within our analysis sample of first time parents, separately for observations before and after childbirth. The post-birth factor loadings are more similar to those obtained in the full sample, suggesting that childbirth does not substantially alter the relative valuation of individual items. This is reassuring, as we might otherwise be concerned that the transition to parenthood changes how different aspects of a relationship map into RQ in comparison to the full data benchmark. In contrast, the pre-birth factor loadings are smaller than the benchmark, with the loading on shared outside interests being the lowest at just below 0.4. Consistent with this pattern, the first factor has a higher eigenvalue in the post-birth sample than in the pre-birth sample, with 4.22 and 3.54, respectively.

Figure A.3 provides a descriptive assessment of changes in the distribution of relationship quality (RQ) around the birth of the first child. Panel (a) plots transitions across deciles of the RQ distribution before and after birth. Most transitions occur within nearby deciles, indicating that individuals tend to maintain their relative position in the distribution. In particular, remaining in the highest and lowest deciles are the most common outcomes. At the same time, the distribution shifts downward after birth, suggesting that these rank-preserving transitions are accompanied by an overall decline in RQ. Panel (b) presents the distribution of within-individual changes in RQ. While a subset of individuals experience increases in RQ,

Figure A.2: Factor loadings of RQ measures.

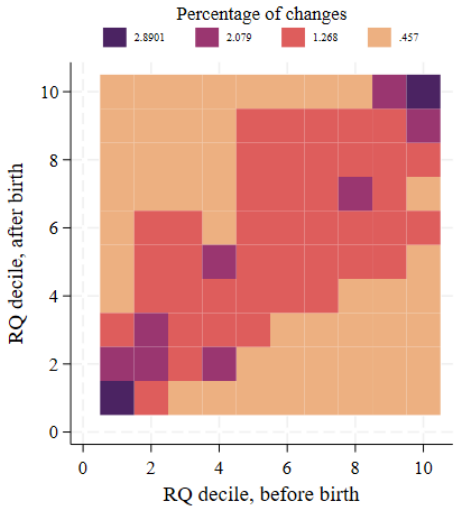


Notes: This graph displays the factor loadings for the RQ measure, along with the factor loadings from factor analyses conducted separately for men and women, and the factor loadings from factor analyses conducted in the sample of parents before and after first child birth. In all cases, the factor analysis is based on responses to the ten questions in the Understanding Society Partner module. The black circles represent the factor loadings of RQ, obtained from the full sample, with an eigenvalue of 4.06. The blue squares correspond to the factor loadings estimated using only male observations, yielding an eigenvalue of 3.84. The orange triangles represent factor loadings from the female-only sample, with an eigenvalue of 4.24. The hollow green diamonds correspond to factor loadings from the pre-birth parent sample, with an eigenvalue of 3.54, and the solid green diamonds correspond to post-birth observations, with an eigenvalue of 4.22.

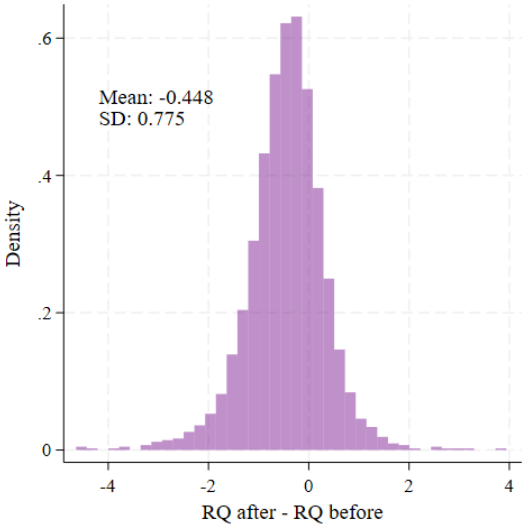
the distribution is clearly shifted toward negative values, indicating declines of varying magnitudes. The average change is consistent with the magnitude of the effect estimated in the main analysis.

Figure A.3: RQ distribution before and after first child birth.

(a) Decile changes



(b) Distribution of differences



Notes: This figure examines differences in the distribution of RQ before and after the birth of the first child. Panel (a) plots transitions across deciles of the RQ distribution before and after childbirth, where the horizontal axis shows the pre-birth decile, the vertical axis shows the post-birth decile, and darker cells indicate a higher share of observations. Panel (b) presents the distribution of within-individual changes in RQ, defined as the difference between average RQ in the post-birth periods (up to seven periods after birth) and average RQ in the pre-birth periods (up to four periods before birth), consistent with the main analysis.

A.3 Validity of the Measure

To verify that the constructed measure provides sensible information about cohabiting partners' relationships, we conduct a number of validity tests. First, we follow the life satisfaction literature (Ferrer-i Carbonell and Frijters, 2004), which highlights the importance of two fundamental theoretical properties that empirical measures of satisfaction should plausibly satisfy: informativeness and interpersonal comparability. We additionally verify that RQ evolves smoothly over time and does not experience abrupt jumps which could confound the estimates obtained in the main analysis of the paper. Second, we compare RQ to existing empirical attempts to measure match quality. To do so, we correlate the measure to individual and couple characteristics previously used to proxy match quality (e.g., Weiss and Willis, 1997; Eckstein et al., 2019) and to the few explicit measures of match quality employed by the economics literature (Bertrand et al., 2015; Chiappori et al., 2018).

A.3.1 Theoretical Properties

Informativeness. First, we verify that RQ provides meaningful information about the state of a relationship by examining its ability to predict major couple decisions: marriage, separation and fertility decisions. Marriage raises separation costs, serving as a commitment mechanism (Brien et al., 2006). Hence, couples transitioning into marriage are expected to report relatively

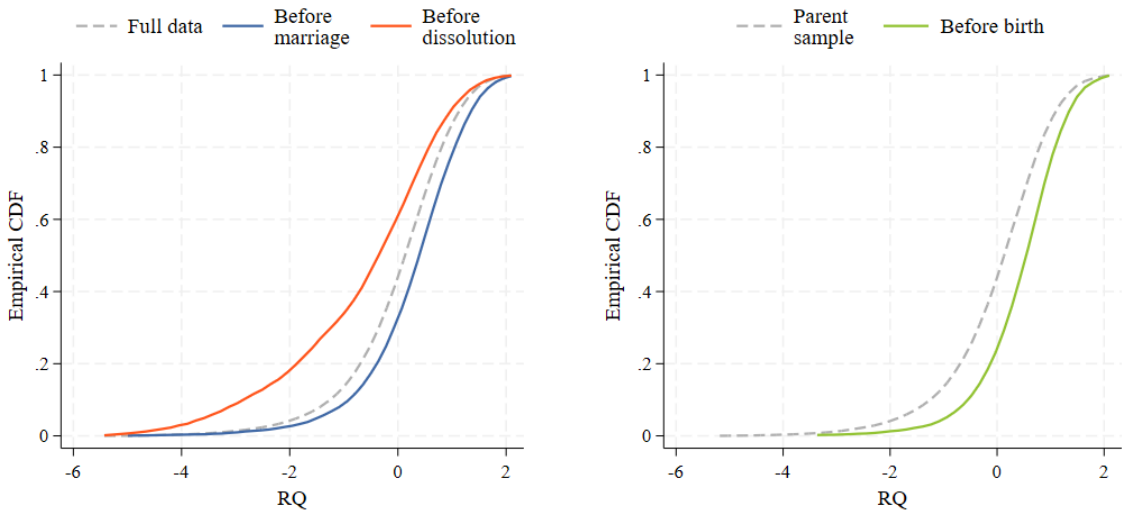
higher RQ. In contrast, separations typically result from poor-quality relationships. Thus, we expect lower than average RQ levels on couples at the brink of dissolution. Finally, we hypothesise that couples deciding to have a child exhibit higher RQ levels.

In Figure A.4, we compare the distribution of RQ the period before these events with its overall distribution, presents the empirical cumulative distribution function (CDF) across these samples. Panel (a) compares the overall distribution of RQ with that observed one period before marriage and dissolution. The distribution before marriage is shifted to the right, indicating higher RQ among couples about to marry. In contrast, the distribution before separation is clearly shifted to the left. The shift before separation is notably larger, suggesting that couples often delay dissolution until RQ has deteriorated markedly, whereas smaller improvements are sufficient to trigger marriage. Panel (b) shows that among new parents, the RQ distribution is shifted upward before the birth of a first child, consistent with fertility being more likely in higher quality relationships.

Figure A.4: Informativeness: behaviour prediction.

(a) Marital transitions

(b) First child birth



Notes: This figure displays the empirical cumulative distribution function (CDF) of RQ for different samples. Panel (a) compares the overall distribution of RQ in the full data (dashed gray line) with its distribution one period before marriage (blue) and one period before couple dissolution (red). Panel (b) plots the overall distribution of RQ in the analysis sample of parents (dashed gray line) and the distribution one period before first child birth (green).

To formally test for differences between these distributions, we conduct a two-sample Kolmogorov-Smirnov equality-of-distributions test. This test assesses whether two samples are drawn from the same population by computing the distance between their empirical CDFs. Table A.3 reports the D-statistics and p-values for the samples considered, where the null hypothesis is that both samples are drawn from the same distribution. We find that the pre-divorce and pre-marital samples contain significantly lower and higher RQ values than the full data, respectively. Similarly, the pre-child sample exhibits significantly higher values than the com-

plete sample of first-time parents. Overall, the test results confirm that all three samples differ significantly from their respective benchmarks.

Table A.3: Two-sample Kolmogorov-Smirnov tests on RQ preceding marital decisions.

	d_0 = Full data		d_0 = Parent sample
	d_1 = Before separation	d_1 = Before marriage	d_1 = Before first child
$d_0 > d_1$	0.0007	0.1333	0.2521
	(0.998)	(0.000)	(0.000)
$d_0 < d_1$	-0.2107	-0.0004	-0.0003
	(0.000)	(0.999)	(1.000)
Combined	0.2107	0.1333	0.2521
	(0.000)	(0.000)	(0.000)

Notes: This table displays the results of two-sample Kolmogorov-Smirnov tests on different samples. The first two columns compare the distribution of RQ in the full data with the distribution the period before couple dissolution and marriage. The third column compares the overall distribution of the sample of parents with the distribution before first child birth within this sample. The null hypothesis is that the two samples compared are drawn from the same distribution. The reported coefficients are the resulting D-statistics and p-values, in parentheses.

We further examine the correlation between RQ and couple dissolution through the regression analysis in Table A.4. The outcome variable is a binary indicator for the period preceding dissolution. We perform this analysis separately in the full data and in the main analysis sample of first-time parents. Controls include age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, and period. A one–standard deviation increase in RQ is associated with a 0.8 percentage point lower probability of separation, a substantial association given the average annual dissolution rate of 2% reported in Figure C.1.

Table A.4: Correlation between RQ and couple dissolution.

	Full data		Parent sample	
	(1)	(2)	(3)	(4)
RQ	-0.0112***	-0.0106***	-0.00975***	-0.00841***
	(0.001)	(0.001)	(0.002)	(0.003)
Controls	✓	✓	✓	✓
Individual FE		✓		✓
R-squared	0.032	0.046	0.027	0.041
Observations	72567	72567	15961	15961

Notes: This table presents the results from regressing a binary variable indicating whether the period precedes couple dissolution on RQ. The coefficients in columns (1) and (2) are estimated using the full data, while columns (3) and (4) present estimation results from the analysis sample of first-time parents. Controls include age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, and period. Standard errors, clustered at the couple level, are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The periods preceding marital transitions are characterized by significant deviations from

the average RQ. We conclude that RQ provides valuable information about couple behaviour, which is consistent with the theoretical property of informativeness.

Interpersonal comparability. A valid measure of RQ should capture a concept that is at least partly shared between partners. We assess this by examining the correlation in RQ across couple members. We use a rank-rank plot to explore the non-linear relation between partners' RQ. Table A.5 reports the results from regressing women's RQ on their male partners' RQ. Men's RQ is a highly significant predictor of women's RQ and has the largest coefficient in magnitude, nearly five times larger than the second-largest predictor, the marriage indicator.

Table A.5: Regression of woman RQ on man RQ.

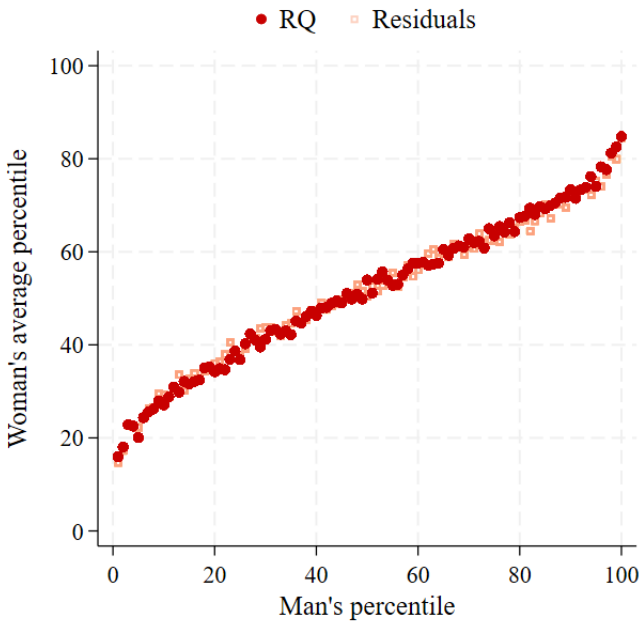
	Woman RQ		
	(1)	(2)	(3)
Man RQ	0.619***	0.610***	0.586***
	(0.007)	(0.007)	(0.010)
Age, tenure and wave dummies		✓	✓
Controls			✓
R-squared	0.324	0.333	0.339
Observations	54454	51394	22162

Notes: This table presents the results from regressing women's RQ on their (male) partners' RQ. Column (2) includes age, relationship tenure and survey wave dummies. Column (3) additionally controls for age, sex, college education, employment status, log monthly income, presence of children, relationship tenure, marital status, and period. Standard errors, clustered at the couple level, are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

To explore non-linearities in this association, Figure A.5 displays women's average RQ percentile by their partners' percentile rank. While the correlation is not perfect, which would produce a 45-degree line, there is a clear positive association between partners' RQ. The slope is steepest at the top and bottom of the distribution, approaching one, but flattens by nearly half in the middle ranks. This pattern suggests that extreme assessments of the relationship are more strongly shared between partners than intermediate ones. This pattern is sustained when controlling for the usual set of individual and couple characteristics.

We also examine whether the individual items underlying RQ are similarly correlated across partners. Table A.6 reports the results from regressing women's responses to each item in Table 1 on their partners' responses to the same question, controlling for standard individual and couple characteristics. All items display a positive and statistically significant association. The strongest correlation arises for kissing, while the weakest occurs among other couple time-use items, despite these being plausibly more jointly determined. Among subjective items, happiness with the relationship is the least shared, whereas the frequency of quarrels is the most strongly aligned across partners.

Figure A.5: Rank-rank correlation of RQ between couple members.



Notes: This figure presents the rank-rank correlation of RQ between couple members in the full data. We plot the average RQ percentile rank of women against their male partners' RQ percentile rank. Each dot represents a percentile. Red markers denote RQ averages. Light markers correspond to residuals obtained regressing RQ on individual and couple characteristics: age, survey collection wave, educational attainment, marital status, and tenure.

Smooth evolution. We next verify that the measure evolves smoothly over time and does not experience abrupt jumps. If systematic jumps occurred at specific ages or relationship tenures, the effects estimated in the main analysis could be driven by the passage of time rather than childbirth. We study how RQ evolves over time computing its life-cycle and relationship profiles by estimating the following regression model:

$$y_{i,t} = \alpha_i + \mu_t + \sum_a \mathbb{1}\{\text{age}_{i,t} = a\}\gamma_a + \sum_d \mathbb{1}\{\text{tenure}_{i,t} = d\}\lambda_d + u_{i,t} \tag{3}$$

where $y_{i,t}$ denotes RQ of individual i at wave t. The terms $\mathbb{1}\{\text{age}_{i,t} = a\}$ and $\mathbb{1}\{\text{tenure}_{i,t} = d\}$ denote the full set age and tenure dummies, respectively, and provide the life-cycle and relationship profiles of RQ. The model also includes individual and period fixed effects, α_i and μ_t, to isolate the age and tenure profiles of RQ by removing individual unobserved heterogeneity.

Figure A.6 (a) presents the age profile of RQ, plotting the estimates of γ_a for ages ranging from 25 to 70, taking 25 as the baseline. RQ increases almost linearly with age, without noticeable jumps.[31] A 25-year difference in age is associated with approximately a one standard deviation higher RQ. Similarly, (b) shows the evolution of RQ with relationship tenure,

[31] Note that Equation 3 is a non-parametric function of age and tenure, allowing full flexibility in the relation between RQ and age, which makes the linear pattern more remarkable.

Table A.6: Couple correlation by item.

	(a) Subjective assessment				
	(1) Consider splitting	(2) Regret marriage	(3) Quarrel	(4) Get on nerves	(5) Degree of happiness
Man's response	0.4601*** (0.0127)	0.3173*** (0.0125)	0.4649*** (0.0086)	0.3937*** (0.0087)	0.3047*** (0.0086)
Controls	✓	✓	✓	✓	✓
R-squared	0.212	0.114	0.234	0.178	0.130
Observations	27107	27087	27078	27079	26687
	(b) Couple time use				
	(1) Work together	(2) Exchange ideas	(3) Calmly discuss	(4) Kiss	(5) Shared interests
Man's response	0.2916*** (0.0078)	0.2574*** (0.0079)	0.2721*** (0.0080)	0.5259*** (0.0078)	0.3997*** (0.0083)
Controls	✓	✓	✓	✓	✓
R-squared	0.135	0.120	0.111	0.345	0.206
Observations	27059	27047	27080	27029	23020

Notes: This table presents the results from regressing women's response to the items in Table 1 on their male partners' response to the same item. Panel (a) corresponds to subjective assessment items, and panel (b) reports correlations for couple time use items. Controls include age, survey collection wave, educational attainment, marital status, and tenure. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

using one year relationships as the baseline. Unlike the age profile, RQ displays a decreasing tendency with tenure, with a slightly convex shape where the decline becomes less steep for longer relationships. The decrease in RQ over tenure is remarkably lower than the increase with age. These findings indicate that age and tenure have opposing effects on RQ. Older individuals report higher RQ levels regardless of relationship tenure, while RQ generally decreases as relationship tenure increases.

A.3.2 Comparison to Existing Measures

Observable characteristics. We study the correlation between our RQ measure and alternative measures of match quality proposed in the literature. Several empirical attempts have proxied for match quality using observable characteristics of individuals and couples (Weiss and Willis, 1997; Eckstein et al., 2019; Low, 2024). We examine the association between RQ and a set of individual and couple characteristics in a linear regression. Individual characteristics include sex, college education, employment status, and log personal monthly income of both the individual and their partner. Couple characteristics include relationship marital status, presence of children, and the female shares of both labour market work and housework, as defined in Subsection 2.2. Standard errors are clustered at the couple level.

Figure A.6: Age and tenure profiles of RQ.



Notes: This figure plots the (a) age and (b) relationship tenure profiles obtained from estimating Equation 3 on the full data. Panel (a) plots the estimates of γ_a for ages a ranging from 25 to 70, using 25 year-olds as the baseline, and (b) plots the estimates of λ_d for relationship tenures d from 1 to 35, using one year long relationships as the baseline.

Table A.7 presents the results for the full data and the parent sample, respectively. The magnitudes and signs of the coefficients remain stable across samples. Women consistently report lower RQ than men. College education, both own and partner's, is positively associated with RQ. Notably, couple characteristics show the strongest associations. Marriage is linked to higher RQ, whereas having children is associated with lower RQ.

While these estimates capture level differences in RQ across groups, unobserved individual heterogeneity may drive some of these correlations. To address this, we re-estimate the model including individual fixed effects, which remove time-invariant unobserved heterogeneity and provide within-individual estimates. The results are shown in Table A.7 columns (2) and (4). Individual characteristics lose significance, suggesting that compositional differences drive the initial correlations. However, the coefficients of marriage and children remain significant, implying that RQ tends to increase with marriage and decline with the arrival of children.

Explicit measures of match quality. Other than using observable characteristics as proxies, attempts to directly empirically measure match quality in economics are more scarce.

First, we consider the measure proposed by Chiappori et al. (2018), which is based on subjective well-being. General well-being, measured using a life satisfaction scale, is assumed to be the sum of marital quality, economic conditions and health. Their approach involves regressing overall life satisfaction on partner's life satisfaction, personal satisfaction with eco-

Table A.7: Regression of RQ on individual, couple and partner characteristics.

	Full data		Parent sample	
	(1)	(2)	(3)	(4)
(a) Individual characteristics				
Women	-0.081***		-0.096***	
	(0.012)		(0.025)	
College	0.059***	0.023	0.092**	0.041
	(0.016)	(0.054)	(0.031)	(0.090)
Employed	-0.014	0.023	0.048	-0.054
	(0.040)	(0.033)	(0.089)	(0.070)
Log monthly income	0.028**	0.002	0.012	0.031*
	(0.009)	(0.007)	(0.017)	(0.014)
(b) Couple characteristics				
Married	0.293***	0.055	0.323***	0.120**
	(0.030)	(0.030)	(0.053)	(0.046)
Children	-0.295***	-0.121***	-0.401***	-0.184***
	(0.022)	(0.020)	(0.038)	(0.036)
Female share of paid work	0.015	-0.020	0.004	-0.103
	(0.047)	(0.041)	(0.095)	(0.085)
Female share of housework	-0.032	0.010	0.003	-0.030
	(0.040)	(0.034)	(0.074)	(0.064)
(c) Partner characteristics				
Employed	-0.022	0.035	0.047	0.043
	(0.042)	(0.031)	(0.098)	(0.069)
College	0.079***	0.028	0.108***	0.085
	(0.016)	(0.054)	(0.031)	(0.090)
Log monthly income	0.014	0.001	0.020	0.005
	(0.008)	(0.007)	(0.017)	(0.011)
Age, tenure and wave	✓	✓	✓	✓
Individual FE		✓		✓
R-squared	0.055	0.076	0.098	0.145
Observations	34079	34079	9134	9134

Notes: This table presents the results from regressing RQ on a set of individual, couple and partner characteristics. Columns (1) and (2) show the results for the full data, and columns (3) and (4) display the results in the main analysis sample of parents. Individual and partner characteristics include sex, college education, employment and log monthly personal income. Couple characteristics include marital status, presence of children, and the female shares of labour market work and in housework, as defined in Subsection 2.2. We control for age, relationship tenure and survey collection wave non-parametrically. Columns (1) and (3) report pooled OLS estimates, while columns (2) and (4) present fixed effects results. Standard errors, clustered at the couple level, are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

nomic conditions and health status, own and partner's wages, age and time. This analysis is conducted separately for men and women, but accounts for interdependence by estimating the equations simultaneously. The residual from this regression is taken as the measure of relation-

ship quality. We use Understanding Society data to derive an equivalent measure.[32]

Second, we examine the questions used by Bertrand et al. (2015) to measure marital stability. The first question asks respondents to describe their relationship on a seven-point happiness scale. We proxy this using the happiness item in Table 1, even if the response scales differ slightly. The other two questions are binary, asking whether the respondent felt their relationship was in trouble or discussed separation with their partner during the past year.[33] While we cannot proxy the trouble question, we can transform the item in Table 1 about the frequency of divorce discussions into a binary variable. This variable equals one when respondents report thinking about divorce "more often than not" or more frequently.

Third, we examine two sub-scales of the revised DAS proposed by Busby et al. (1995): cohesion and satisfaction. These sub-scales can be directly constructed using a subset of the questions listed in Table 1.[34] The Cohesion sub-scale sums responses related to the frequency of working together on a project, having stimulating exchanges of ideas, calmly discussing something and the number of shared outside interests. The Satisfaction sub-scale adds responses concerning the frequency of discussing divorce, regretting marriage, quarrelling and getting on each other's nerves.

We standardize all measures and compute their correlations with RQ. Table A.8 presents the resulting correlation coefficients. The residualised subjective well-being measure has the lowest correlation with RQ. This measure relies on the assumption that all sources of life satisfaction other than marital satisfaction are accounted for in the residuals of overall well-being.[35] However, if other factors that could influence life satisfaction, such as having children or social life, are not fully accounted for, this could introduce measurement error. In contrast, the RQ measure specifically targets relationship quality with a partner, potentially reducing measurement error.

The questions used by Bertrand et al. (2015) are presented in (3) and (4). As expected, the correlation between RQ and couple happiness matches the factor loading of this item in the RQ measure. The question related to discussing separation is the only one negatively associated with RQ, which is reasonable given that discussions of separation indicate poorer relationship quality. This question is also part of the RQ measure, but assessed on a more granular scale. The advantage of RQ is that it integrates several aspects of couple relationships, like happiness, conflict or quality time together, into a unified measure, providing a comprehensive assessment

[32] Although the survey questions we use are designed to capture the same concept as those used by Chiappori et al. (2018), they are not identical. While their satisfaction scales range from 1 (least satisfied) to 5 (most satisfied), our scales are more granular, ranging from 1 to 7. Additionally, our estimation approach differs: we estimate this in a single regression and account for correlation in the residuals by clustering at the couple level.

[33] Other than Bertrand et al. (2015), various studies use couple happiness and conflict as measures of marital satisfaction and stability (e.g., Halliday Hardie and Lucas, 2010; Perelli-Harris and Blom, 2022).

[34] There is a third sub-scale within the DAS, consensus, which we cannot construct. While the full DAS consists of the sum of the three sub-scales, each has also been studied independently.

[35] It also assumes additivity in how each of these factors contribute to well-being, excluding potential interaction effects.

of match quality.

The last two columns and rows of Table A.8 correspond to the DAS sub-scales. Both sub-scales show a high correlation with RQ, being larger for satisfaction. This is expected since these sub-scales are constructed using a subset of the items used to create RQ. Additionally, the correlation between RQ and each sub-scale is higher than the correlation between the sub-scales themselves. Therefore, RQ effectively summarizes the information contained in the two sub-scales in a more parsimonious manner.

Table A.8: Correlation among alternative match quality measures.

	(1) RQ	(2) Subjective well-being	(3) Couple happiness	(4) Discuss separation	(5) DAS Cohesion	(6) DAS Satisfaction
(1) RQ	1					
(2) Subjective well-being	0.238	1				
(3) Couple happiness	0.676	0.205	1			
(4) Discuss separation	-0.336	-0.0659	-0.161	1		
(5) DAS Cohesion	0.803	0.170	0.413	-0.143	1	
(6) DAS Satisfaction	0.837	0.188	0.479	-0.427	0.406	1

Notes: This table reports the correlation coefficients among RQ and various match quality measures in the literature. Column and row (1) correspond to RQ. Column and row (2) present the residualised subjective well-being measure as proposed by Chiappori et al. (2018). Columns and rows (3) and (4) contain two of the marital stability measures used in Bertrand et al. (2015). Columns and rows (5) and (6) present the cohesion and satisfaction sub-scales of the revised DAS (Busby et al., 1995).

B Empirical Strategy

We take a dynamic difference-in-differences (DiD) approach to study the causal impact of children on RQ. This methodology exploits sharp changes in individual outcomes after the birth of their first child and allows for treatment heterogeneity with time relative to birth. In essence, it allows us to estimate the impact of having a child at each period relative to birth. Furthermore, it exploits differences in the timing of child birth to dynamically construct a control group.

In the main analysis, we use a sample of individuals who become parents to study the impact of the first birth (treatment) on RQ (outcome). This has two important implications. First, the treatment is staggered because different individuals have their first child in different periods. This divides the sample into different treatment adoption cohorts, depending on the calendar year when individuals become parents. Second, having a sample of new parents implies that everyone in the sample is treated at some point. This means that, in each period, we compare individuals who just had a child with individuals who have not become parents yet and with individuals who have been parents for more than one period.

The usual approach is to estimate the dynamic two-way fixed effects (TWFE) regression in Equation 1. The econometrics literature has pointed out several problems derived from estimating such specifications through OLS in contexts of staggered treatment adoption (see Goodman-Bacon, 2021, for a review). There are two main issues that threaten the causal interpretation of the obtained estimates. First, it performs forbidden comparisons between individuals changing status from control to treated and cohorts that have already been treated for more than one period. Using already-treated as controls mechanically generates negative or non-convex weights. Second, although such weights exist even under homogeneous treatment effects, heterogeneity across treatment cohorts makes them bias-inducing, as the estimator combines effects from different periods in a non-representative way. As we explain below, we take an alternative approach in our estimation.

B.1 Callaway and Sant'Anna (2021) method

The method proposed by Callaway and Sant'Anna (2021) overcomes the issues related to TWFE estimation by clearly separating identification, estimation and aggregation. This method is applicable in settings where a panel dataset is available and the binary and absorbing treatment is adopted in a staggered manner. We feel comfortable assuming that having a child is a binary (having a child or not) and absorbing treatment (having a child forever). The only situation where this treatment would not be absorbing is in the case of child death, which is a very rare occurrence in the data. Additionally, the treatment is adopted in a staggered manner, since individuals have children in different periods, i.e., there are different treatment adoption

cohorts.

Identification. The building blocks of this method are the average treatment effects on the treated (ATT) for each treatment adoption cohort g and at each period t, denoted as ATT(g,t). Identifying these requires two assumptions:

A1. Limited anticipation. If a unit is untreated in period t, its outcome in that period does not depend on when it will be treated in the future. In our context, we need to assume that changes in RQ before the birth of the first child do not predict when individuals have their first child. Note that "limited" anticipation implies that the method allows for some preceding reaction, but an assumption needs to be made on the amount of anticipation periods before treatment.

A2. (Conditional) parallel trends based on "not-yet treated". All treatment cohorts would have evolved in parallel in absence of treatment. In our context, individuals' RQ would have evolved in parallel regardless of the period when they had their first child. Importantly, the method allows for the looser assumption of parallel trends conditional on covariates.

Those two assumptions allow to construct the counterfactual for each ATT(g,t) using (i) the period before treatment as the baseline period, and (ii) all treatment cohorts that have not been treated by t as controls. Therefore, the control group at each t for the same cohort g varies because at each subsequent period new cohorts enter treatment status.

Estimation. Each ATT(g,t) is estimated as a 2×2 difference-in-differences coefficient using the baseline period and the control group described. The estimation can be done in three ways, which use information from different parts of the data generating process: using outcome regression, inverse probability weighting, or doubly robust estimands. Refer to Callaway and Sant'Anna (2021) for detailed information on the estimation methods. We use the last method, since it constitutes the combination of the other two.

Aggregation. The ATT(g,t) estimates are the building blocks used to summarize the treatment effects across treatment adoption cohorts, periods or time relative to treatment. In each case, aggregation involves carefully chosen and estimable weights for the different ATT(g,t). Our main aggregation involves obtaining the ATT at each event-time. Crucially, because each event-time estimate is a weighted average of the cohorts present at that event-time, it mechanically reflects the cohort composition prevailing in that period. This may prevent the causal interpretation of the aggregated coefficients, unless homogeneity across treatment adoption cohorts is assumed. In this scenario, we need to assume that the impact of first birth is the same regardless of the year of birth.

B.2 Identification

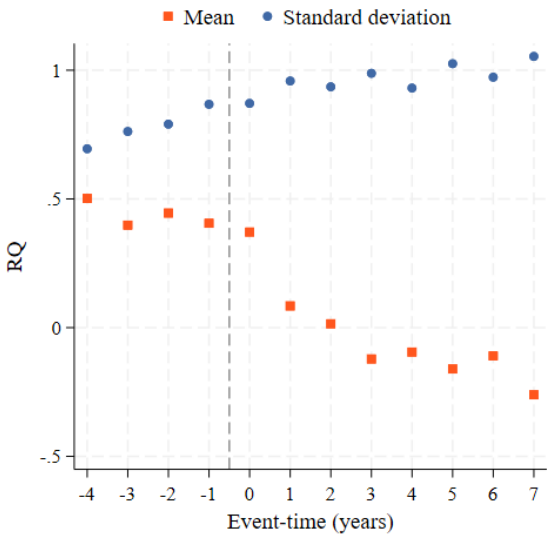
Although we cannot directly test whether the identification assumptions above are satisfied, we can provide evidence in favour of their plausibility.

The **limited anticipation** assumption would be violated if individuals decided to have children in response to either negative or positive shocks to RQ. Figure 2 shows no evidence that the birth of a first child may be preceded by deviations in RQ, on average, given that all the point estimates before the event are not statistically different from zero. However, this could be masking heterogeneity in the evolution of RQ before first child birth, averaging individuals that decide to have children after a positive or a negative RQ deviation. If that were the case, the standard deviation of RQ would be larger over the periods before the event.

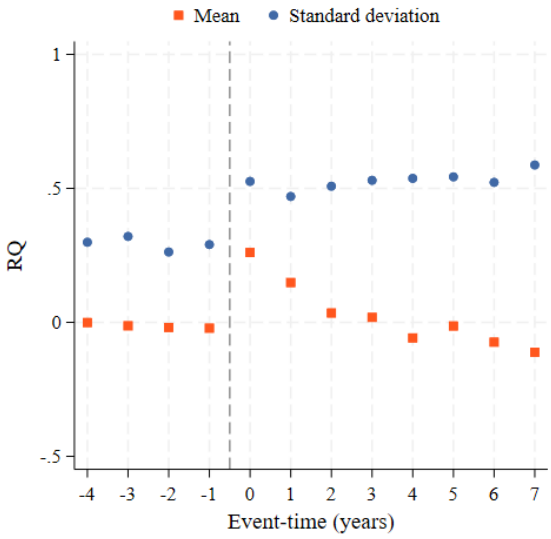
Figure B.1 (a) plots the mean and standard deviation of RQ at each event time. Consistent with Figure 2, average RQ remains stable before childbirth and declines sharply afterward. The standard deviation is lowest during the periods before childbirth, supporting the absence of anticipation, and steadily increases post-birth. Additionally, there are no significant deviations from individual RQ trajectories before birth that could predict its timing. Figure B.1 (b) shows the mean and the standard deviation of deviations from individual-specific RQ means, computed separately for pre- and post-birth periods. Before childbirth, deviations are nearly zero on average, indicating no systematic pre-birth RQ changes. The low standard deviation further suggests minimal heterogeneity in pre-birth RQ changes.

Figure B.1: Evidence on no anticipation assumption.

(a) RQ levels



(b) Deviations from individual means



Notes: This figure presents evidence supporting the plausibility of the no anticipation assumption. Panel (a) displays the mean and standard deviation of RQ at each period around first child birth in the main sample of parents. Panel (b) presents the mean and standard deviation of deviations from individual-specific RQ means, computed separately for pre- and post-birth periods.

In Table B.1, we examine which observable characteristics predict the timing of first child birth. The outcome is a dummy variable equal to one if the individual has their first child in the subsequent period. The specification includes RQ, dummies for college education, labour market activity and employment, log monthly labour income, weekly hours worked in the labour market, a marriage indicator and period dummies. We also include individual fixed effects, so the estimates reflect within-individual variation and indicate how changes in characteristics are associated with the timing of birth. The analysis is restricted to the period before first birth. The results show that changes in RQ prior to birth do not predict the timing of childbirth, which provides additional support for the no-anticipation assumption. Among the included covariates, only being married emerges as a significant predictor of first birth.

Table B.1: Observable characteristics and timing of birth.

	(1)
RQ	0.00877
	(0.025)
College educated	0.00905
	(0.096)
Active in labour mkt	0.106
	(0.139)
Employed	-0.185
	(0.136)
Log monthly labour income	0.0424
	(0.042)
Weekly work hours	-0.000765
	(0.002)
Married	0.125***
	(0.035)
Individual FE	✓
Period FE	✓
R-squared	0.557
Observations	1602

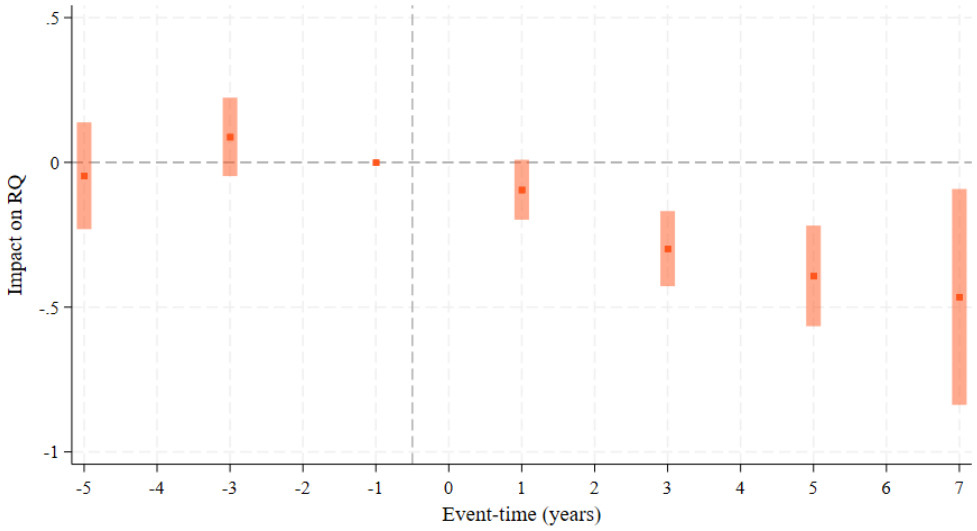
Notes: This table presents the results from regressing an indicator variable equal to one the individual's first child is born in the subsequent period (and 0 otherwise) on a set of observable characteristics. Regressors include RQ, dummies for college education, labour market activity and employment, log monthly labour income, weekly hours worked in the labour market, a marriage indicator, as well as period and individual fixed effects. The regression is estimated on pre-birth observations. Standard errors are reported in parentheses.

Conditional parallel trends cannot be directly tested, since it is an assumption about unrealized counterfactual scenarios. However, the flat pre-trends in Figure 2 provide suggestive evidence in its favour. We formally test whether the coefficients for all pre-treatment periods are statistically equal to zero and fail to reject this hypothesis. By default, Callaway and

Sant'Anna (2021) conducts "short" pre-birth comparisons, where consecutive periods are compared, whereas post-birth comparisons are "long", or relative to the baseline period. To address this asymmetry, Roth (2024) suggests using "long" comparisons for pre-birth periods as well, where all pre-birth periods are compared to a single baseline. While this does not affect post-birth coefficient estimates (as long as the baseline period is kept fixed), it alters pre-birth comparisons.

Implementing this is not straightforward in our context, given that RQ is never observed in two consecutive periods. To overcome this, we normalize the last observed period before birth as the baseline, pooling event-time periods $t - G_i = -2$ and $t - G_i = -1$. The remaining pre-birth periods are then redefined relative to this baseline. Figure B.2 presents the results of this approach. The slight differences in post-birth coefficients stem from this redefinition of the baseline period. Importantly, pre-birth estimates remain close to zero.

Figure B.2: Impact of first child birth on RQ using "long" pre-birth comparisons.



Notes: This figure presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The estimation conducts "long" comparisons for both pre- and post-birth periods, where all periods are compared to the baseline period. The baseline is defined as the last period when RQ is observed before first birth. Given the structure of the data, this may be one or two years prior to the event. All other periods are defined relative to this baseline, with gaps of one period due to RQ being observed only every other survey collection wave. We plot 95% confidence intervals.

A potential limitation of this approach is that we observe only realized fertility, not intended fertility. To address this, we separately examine individuals who experience a delay in the birth of their first child, identified as those who report a miscarriage or the use of fertility treatments.[36] While we cannot precisely determine when these individuals initially intended to have a child, we do observe that they faced a delay.[37] Figure B.3 (a) presents separate estimation

[36]Fertility treatment information is available only if the treatment was successful and led to a pregnancy since the previous interview. In our sample, approximately 9% of each treatment cohort experiences a delay in fertility.

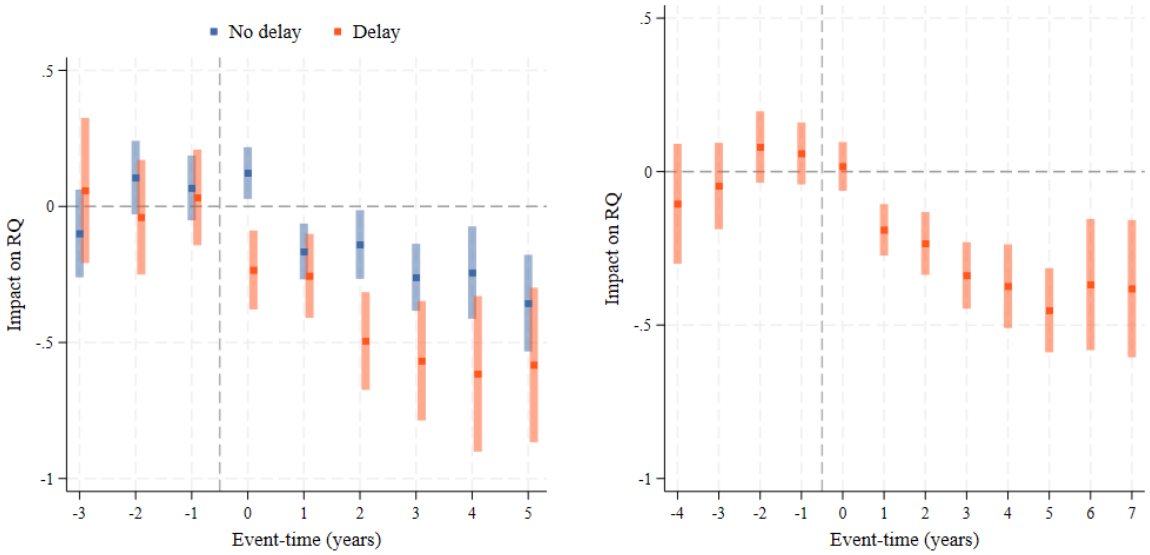
[37]On average, couples try to conceive for about a year before turning to assisted reproduction technologies

results for individuals who experience a delay in fertility and those who do not. The impact differs mainly on the birth period: parents without delays experience a slight increase in RQ, while those with delays show a decline. Although from event-time one onwards point estimates are similar, estimates for delayed parents are less precise. Importantly, before childbirth, there are no significant differences in RQ between those who experience delays, regardless of when they eventually have a child.

Figure B.3: Evidence on parallel trends assumption.

(a) Delayed pregnancy

(b) Similar baseline RQ



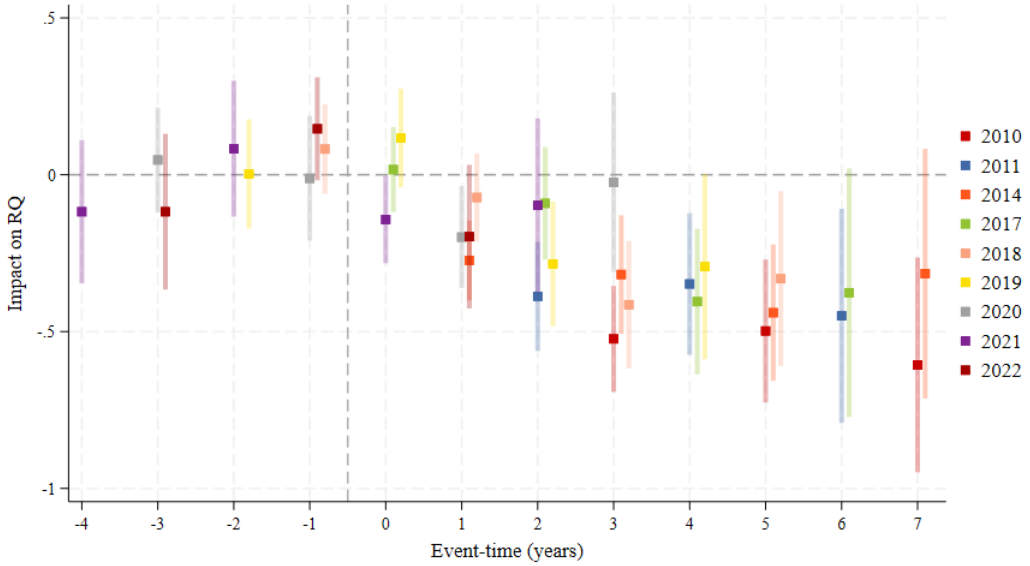
Notes: This graph provides evidence supporting the parallel trends assumption. Panel (a) presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ using the method proposed by Callaway and Sant'Anna (2021), separately for individuals who experienced a delay in childbirth and those who did not. We identify individuals who reported using fertility treatments or experiencing miscarriages before birth as subject to delayed fertility. Panel (b) replicates the main dynamic difference-in-differences estimation, restricting comparisons to individuals within the same RQ quintile before birth. We plot 95% confidence intervals.

Another way to assess the plausibility of parallel trends is to replicate our main analysis restricting comparisons to individuals with similar RQ levels before the birth of their first child but who had children in different periods. This specification makes the assumption more plausible. Specifically, we compare individuals within the same RQ quintile prior to birth. The results, shown in Figure B.3 (b), closely align with our main findings, further reinforcing the plausibility of the parallel trends assumption.

Although not an identifying assumption, a final condition required for the time aggregation in our main results is the **homogeneity of treatment effects across cohorts**. Figure B.4 presents the dynamic impact of first child birth separately for each treatment cohort. Due to sample size limitations, the estimates at this level of disaggregation are somewhat noisy. However, the point estimates remain similar across cohorts.

(Bögl et al., 2024).

Figure B.4: Impact by treatment cohort.



Notes: This graph presents the results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ using the method proposed by Callaway and Sant'Anna (2021), separately for each treatment cohort. A treatment cohort is defined by the calendar year when an individual's first child was born. We plot 95% confidence intervals.

B.3 Alternative Counterfactuals

Our main analysis sample is formed by individuals observed becoming parents at some point within our observation window. Coefficients are estimated by carrying out comparisons between individuals who become parents in a given period with individuals who have a child at a later date. Using this sample we focus on individuals who are exposed to the transition into parenthood and isolate its impact without conflating it with differences between parents and non-parents, who may differ systematically in unobservable ways. Moreover, restricting the sample to individuals who eventually become parents ensures that our estimates are directly relevant to study the consequences of family-oriented policies, which are designed to support individuals during and after the transition to parenthood.

In this section, we assess the robustness of the results obtained using the parent sample by considering two alternative counterfactuals or control groups: individuals who want to have children but do not manage to (*infertile individuals*), and individuals who do not want nor have children (*never parents*).

B.3.1 Infertile Individuals

First, we consider individuals who want to have children but do not manage to. This counterfactual could approximate how RQ would evolve in the case where individuals in our treatment sample did not manage to conceive. The causal interpretation of the estimates obtained us-

ing this control group relies on the assumption that the individuals identified as infertile are comparable to the treated sample.

Ideally, we would have data on fertility intentions, which could either be given by self-declared intentions or by reported usage of fertility treatments. Unfortunately, information on fertility is only available when a couple manages to conceive. We identify *infertile* individuals as those who manage to become pregnant but who do not have a live birth by the end of the observation window. While we are aware that this is only a subsample of the population of infertile individuals, since some couples may never conceive, they are the only cases which we are able to identify.[38]

The control sample of infertile individuals is formed by women having a pregnancy between ages 18 and 45 who never become mothers, and their partners. We observe around 126 such women, resulting in a sample of 1,362 observations, including their partners. Within this sample, the average number of failed pregnancies is 1.24, where about 80% of individuals have a single failed pregnancy. Out of the 2,015 individuals in our main analysis sample, only 112 report having had a failed pregnancy before first child birth, and the average number of failed pregnancies is 0.238 in the entire observation window. Thus, we still focus on first birth in the treatment group, and we use time of first pregnancy in the control group.

Table B.2 presents summary statistics for the main treatment sample in column (1) and the control sample of infertile individuals in column (2), during the period before first birth in the case of parents and before first pregnancy in the case of infertile individuals. The control group is formed by older and more educated individuals. They are more often active in the labour market and employed, and they also earn more and work more hours. Finally, infertile couples are in longer relationships and less likely to be married. This could indicate that women in this sample are more career-oriented and attempt motherhood at later ages, experiencing more difficulties.

We repeat the main analysis using the method proposed by Callaway and Sant'Anna (2021) and infertile individuals as the control group. We exploit the information on the year of individuals' first failed pregnancy as a proxy for fertility intentions. We acknowledge that this is an imperfect proxy, since individuals may have been trying to conceive for some time before their first recorded pregnancy.

The causal interpretation of the results relies on the usual assumptions of no anticipation and conditional parallel trends. The interpretation of no anticipation remains largely the same as in the main analysis: the time when individuals decide to get pregnant cannot be predicted by changes in RQ prior to pregnancy. To exploit fertility intentions, the parallel trends assumption

[38] According to Department of Health and Social Care (2023), one in five pregnancies ends in miscarriage, with most losses occurring within the first 12 to 14 weeks. This rate is substantially higher than the share captured in our data, which is expected because a pregnancy loss can only be reported if the pregnancy was known to the respondent at the time of the interview. Early losses that occur between survey waves, before the respondent reports a pregnancy, are therefore likely to be missed.

is now conditional on the year when the pregnancy occurs. In the treatment group pregnancy results in live birth, whereas in the control group it ends without child. Thus, the underlying assumption is that, if the pregnancy had not been successful for treated individuals, their RQ would have evolved like that of infertile individuals.

Figure B.5 displays the distribution of RQ for parents in the treatment group and infertile individuals in the control group separately (a) before and (b) after first birth/pregnancy. The distribution plotted in panel (a) indicates that the infertile sample has lower RQ ex ante. This could be due to prior difficulties to get pregnant, inducing RQ losses. Panel (b) indicates that the average RQ of infertile individuals also drops after the failed pregnancy. However, this drop is smaller than the drop documented for parents.

The results from estimating the effect of first child birth on RQ using the infertile sample as a control group and conditioning on the calendar year of the first pregnancy are presented in Figure B.6. The documented dynamics are similar to the main results presented in Figure 2. RQ decreases over the first four years after childbirth, and the magnitude of the coefficients is very similar using this control group.

B.3.2 Never Parents

The second control group we consider are individuals who do not wish to become parents and never become parents. We define *never parents* as those individuals never observed becoming pregnant or cohabiting with their children. We restrict the sample to respondents observed at least once during fertility ages, between 18 and 45. Since we do not observe fertility intentions until pregnancy, this group could contain individuals who want children but never manage to become pregnant, which we cannot distinguish.

We follow Kleven et al. (2019b) and use a design based on assigning placebo births to never parents. There are two types of individuals never observed becoming parents: those who have not had children by the end of their fertility cycle, and those who do not have children yet, but may have them before their fertile ages have passed, at age 40. We refer to cohorts 1951-1980 as old, non-truncated cohorts, and to cohorts 1981-2001 as young, truncated cohorts. This is a crucial distinction when assigning placebo births.

The first step before assigning placebo births to never parents is removing from the sample those individuals from young cohorts who are likely to have children in the future. We start by dividing individuals into ten-year cohorts, distinguishing between old (1951, 1961 and 1971) and young (1981, 1991 and 2001), and computing young cohort women's motherhood probability. This will then be extrapolated to their partners.

Using observations from old cohorts, we estimate a linear probability model of becoming a mother on a set of observable characteristics: cohort, within-cohort income quartile, within-cohort income quartile of the partner, educational level (college or not) and country (England,

Table B.2: Summary statistics of the alternative counterfactual samples.

	(1) Parents	(2) Infertile	(3) Never parents
(a) Individual characteristics			
Age	30.28	32.37	36.39
	(5.484)	(6.450)	(6.905)
Women (%)	54.34	60.39	59.94
	(49.82)	(49.07)	(49.03)
College educated (%)	41.45	58.87	54.26
	(49.25)	(49.38)	(49.85)
Active in labour mkt (%)	67.64	90.26	88.74
	(46.75)	(29.75)	(31.46)
Employed (%)	64.68	90.26	84.60
	(47.75)	(29.75)	(35.88)
Monthly labour income (£)	2181.4	2036.8	2201.9
	(1515.7)	(1323.1)	(1614.9)
Weekly work hours	23.10	32.94	30.59
	(18.82)	(14.41)	(15.56)
Weekly housework hours	7.177	8.424	7.732
	(5.362)	(6.709)	(5.676)
RQ	0.399	0.318	0.101
	(0.870)	(0.834)	(1.119)
Observations	5948	154	724
(b) Couple characteristics			
Tenure	4.991	5.451	8.379
	(4.850)	(4.552)	(6.875)
Married (%)	53.74	47.40	55.15
	(49.52)	(49.93)	(49.57)
Monthly household income (£)	4364.2	4932.7	4870.1
	(2621.1)	(2703.2)	(2607.5)
Female share of paid work	0.495	0.571	0.500
	(0.257)	(0.306)	(0.279)
Female share of housework	0.616	0.605	0.572
	(0.207)	(0.201)	(0.236)
Observations	3706	96	466

Notes: This table presents summary statistics for three different samples: (1) the main analysis sample of parents (*Parents*); (2) the sample of individuals becoming pregnant but never having a live birth (*Infertile*); (3) the sample of individuals never getting pregnant and never having children (*Never parents*). All individuals considered are above 18. We plot values for the period before (1) first birth, (2) first pregnancy and (3) placebo birth. Since RQ is biannual, not all individuals are observed on that period, resulting in a reduced sample. Income measures are given in gross terms. Standard errors are displayed in parentheses.

Wales, Scotland or Northern Ireland). We then use the obtained coefficient estimates, to compute the fitted values for the young cohorts. This gives us the probability that each woman becomes a mother before the end of their fertility period. We sort young cohort women by cohort based on that probability.

We next want to determine which women will have children before the end of their fertility cycle. To do that, we compute the share of mothers in old cohorts by cohort and impute births for young women based on their motherhood probability such that the share of women who become mothers is the same across old and young cohorts. The individuals that get a birth imputed stop forming part of our control sample of never parent individuals.

The final control group is formed by old cohort individuals who never had children and young cohort individuals who do not have children and are unlikely to have them in the future. This is the sample of individuals to which we assign placebo births. We do this by assigning placebo birth ages. We approximate the distribution of age at first birth with a cohort and education level specific log normal distribution. This is straightforward for old cohorts, since we observe the entire distribution of age at first birth. For young cohorts, we use the predicted average from estimating a linear trend on old cohorts as the average of the distribution. We then assign a random draw from the corresponding distribution to each individual and compute the placebo event-time periods.

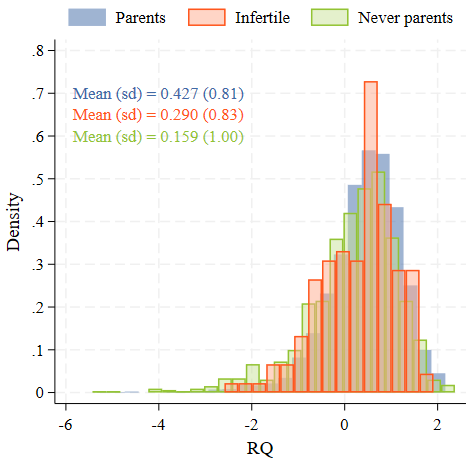
Table B.2 (3) presents summary statistics for the control sample of never parents, during the period before the placebo birth. Compared to the treatment group, in column (1), never parents are older, more often college educated, have higher earnings, have been together for longer and are more often married. Average RQ is also below the treatment sample mean before birth. However, as seen in Figure B.5 (b), average RQ becomes higher than parents' after the birth of their first child.

We repeat the main analysis using never parents as the control group and exploiting placebo births. However, the identification assumptions are somewhat less robust in this approach, particularly the parallel trends assumption. A key difference between the treatment and control groups is that never parents do not wish to have children. For the parallel trends assumption to hold, wanting children but being unable to conceive should have no impact on RQ. Infertile individuals experience a decline in RQ following a failed pregnancy. Arguably, if the treated group had been unable to become parents, a similar drop in RQ might have occurred.

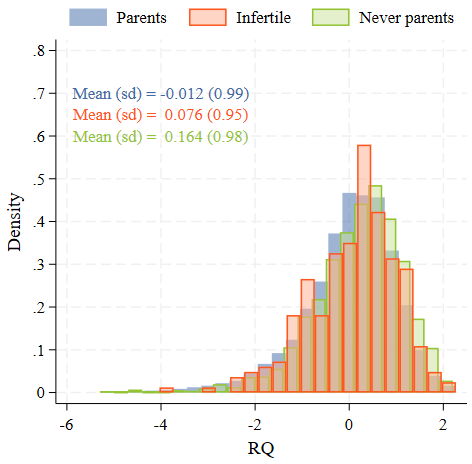
We plot the estimation results in Figure B.6. The dynamics and magnitudes of the estimated coefficients are very similar to the baseline estimates in Figure 2: they are virtually the same, but more precisely estimated.

Figure B.5: Distribution of RQ in different samples.

(a) Before

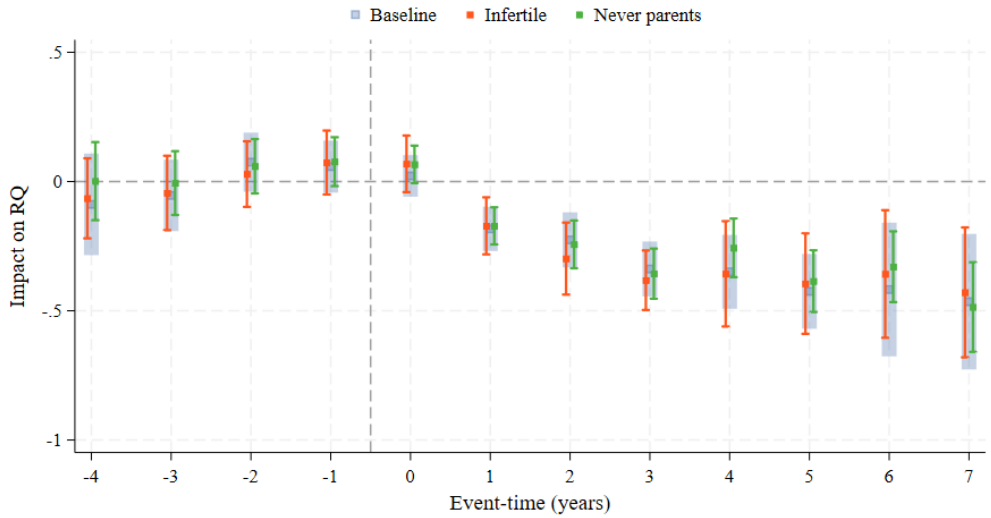


(b) After



Notes: This figure presents the distribution of RQ in three different samples: the main analysis sample of parents (*Parents*, in blue), the sample of individuals getting pregnant but never having a live birth (*Infertile*, orange) and the sample of individuals never having children (*Never parents*, green). Panel (a) contains the periods up to 4 years before birth for the treatment sample and up to 4 years before the first pregnancy attempt for the infertile sample. Panel (b) covers the period from birth up to 7 years after birth for the treatment sample and up to 7 years after the first pregnancy attempt for the infertile sample. Both panels display the full distribution for never parents.

Figure B.6: Impact of first child birth on RQ, using alternative control groups.



Notes: This figure presents the results from dynamic difference-in-differences estimations of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The main treatment sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. The baseline analysis (blue) uses individuals becoming parents at later dates as controls. The *infertile* analysis (orange) uses as the control group the set of individuals having a pregnancy between ages 18 and 45 who never become parents. We control for the period of first pregnancy. The *never parent* analysis (green) uses as controls individuals never becoming pregnant. We assign placebo births to the control group and control for placebo birth dates. We also plot 95% confidence intervals.

B.4 Surviving until First Birth

Our main analysis compares individuals who have already become parents with individuals who have not yet had a child but will do so later. Since RQ is observed only while couples remain together, our sample excludes couples whose relationships dissolve before a first birth. Thus, our main results concern couples who remain together until parenthood. However, if couples whose RQ deteriorates are more likely to separate before having a child, the sample of first-time parents may be selective in ways that affect the comparability of earlier- and later-treated parents. Flat pre-birth trends among observed eventual parents cannot rule out this form of selection.

Two complementary checks provide some reassurance. First, the placebo analysis of never-parents in Figure B.6 shows flat pre-trends in RQ, suggesting that the absence of pre-birth changes is not specific to our sample of eventual parents. Second, restricting the sample to couples whose first birth occurs after 3–7 years of relationship tenure makes earlier- and later-treated parents more comparable in relationship tenure. The results, in Figure D.8, remain similar, reducing concerns that differences in relationship duration drive our findings. However, neither check addresses selection through separation before parenthood directly. We therefore conduct a sensitivity analysis to assess how this selection could affect our estimates.

We proceed in two steps. First, using all cohabiting childless couples in the data, we estimate the probability of separation before the next survey wave and compute the associated change in RQ. We estimate these associations separately by gender, controlling for age, relationship tenure, both partners' education and a linear time trend. We use RQ in the wave preceding separation to benchmark the trajectory of couples who separate. Since RQ tends to deteriorate before dissolution, this provides a pessimistic benchmark for the counterfactual RQ trajectory of couples who separate before having a child.

Second, we use these estimates to assess how our main results would change if some couples who separate before having a child would otherwise have become parents. Let H denote the estimated cumulative probability of separation over a given event-time window, and let ΔS and ΔD denote the predicted changes in RQ for couples who remain together and couples who separate, respectively. We observe the RQ trajectory of the former but cannot observe how the latter's RQ would have evolved had they remained together. We introduce a sensitivity parameter s, which represents the share of separating childless couples that would have otherwise become parents. The adjusted group-time estimate is:

$$\text{ATT}^{\text{adj}}(g, t; s) = \text{ATT(g,t)} + sH(\Delta S - \Delta D)$$

When $s = 0$, none of the separating couples are assumed to have otherwise become parents, so the estimate is equal to our baseline result. When $s = 1$, separating childless couples

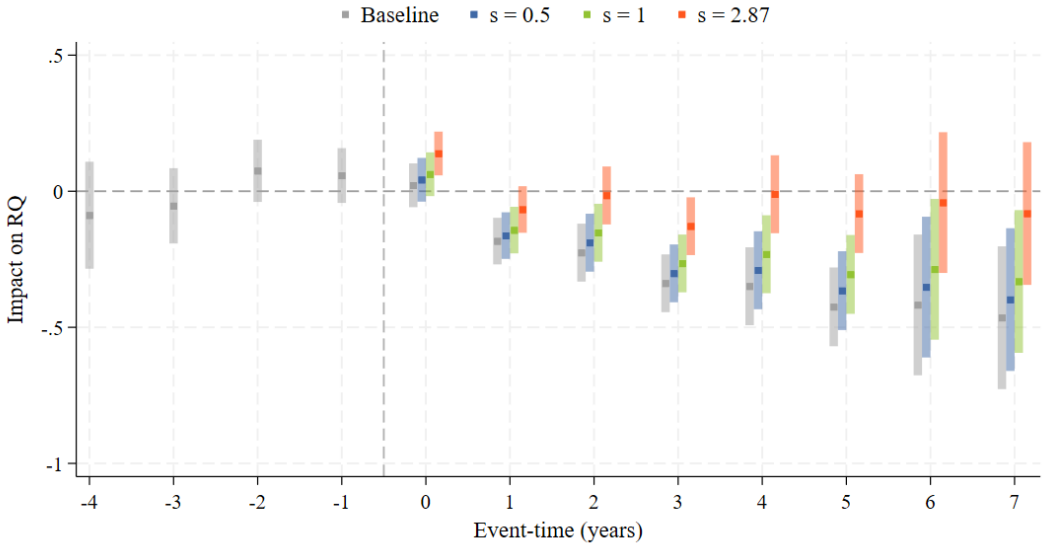
are assumed to have become parents at the same rate as comparable couples who remain together. Larger values allow for greater selection through separation. To bound the range of s, let π denote the share of couples who would become parents if they remained together. We approximate π using the observed share of couples who remain together and become parents within our observation window. In the most extreme scenario, all couples who separate before a hypothetical birth would otherwise have become parents, that is, $s_{\max} = 1/\pi \simeq 2.87$.

This exercise relies on the assumption that separation patterns among observed childless couples provide a relevant benchmark for couples who would otherwise have become parents. In particular, we use the predicted RQ trajectory before separation as a proxy for their unobserved counterfactual trajectory. The exercise is a sensitivity analysis, not an identified correction: we cannot observe the counterfactual fertility decisions or subsequent RQ of couples who separate.

We apply the adjustment to post-birth group-time estimates only and aggregate them into event-time coefficients using the same weights as in the main analysis. Before birth, both the treated and comparison groups consist of couples who remain together until the observation period. Applying the adjustment only to the comparison group would mechanically shift its estimated RQ trajectory relative to that of the treated group. Such a shift would reflect the adjustment itself rather than provide evidence on whether selection explains the observed pre-birth trends. We therefore retain the baseline pre-birth coefficients and use the sensitivity analysis to assess the robustness of the post-birth estimates.

We plot the resulting event-time coefficients in Figure B.7. The results suggest that the estimated decline in RQ is robust to moderate levels of selection through pre-birth separation. Under $s = 0.5$ and $s = 1$, the effect of first child birth remains negative and statistically significant between one and five years after birth. Six and seven years after birth, confidence intervals become large and include zero for all values of s, reflecting the small number of cohorts observed at these horizons. The average post-birth effect becomes statistically insignificant only for $s \geq 1.45$, while the three-year effect remains significant up to $s = 2.76$, close to the approximate upper limit.

Figure B.7: Impact on RQ adjusting for separations in the comparison group



Notes: This figure presents a sensitivity analysis of the estimated effect of first childbirth on RQ, based on a dynamic difference-in-differences design using the method proposed by Callaway and Sant'Anna (2021). Baseline estimates (gray) correspond to the main specification. Adjusted estimates show how the results change when post-birth group-time ATT(g,t) estimates are adjusted for predicted separations among childless couples, under alternative values of the sensitivity parameter s. The upper value, $s_{\max} = 1/\pi$, corresponds to the scenario in which all couples who separate before a hypothetical birth would otherwise have become parents, where π is the observed share of couples who remain together and become parents during the observation window. Pre-birth coefficients are unadjusted and coincide with the baseline estimates. We also plot 95% confidence intervals.

C Related Results

Figure C.1: Yearly couple dissolution rate.



Notes: This figure plots the number of couples that dissolve each year as a share of all observed couples in the Understanding Society dataset. Office for National Statistics (2022) reports that, on average, 1.44% of married couples of reproductive age (20–45 years old) in England and Wales divorced annually between 2009 and 2021.

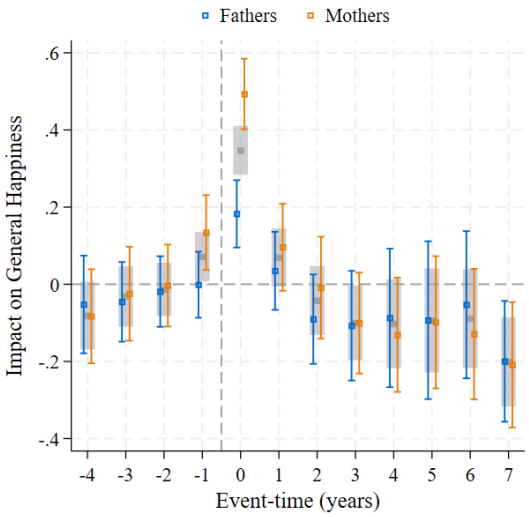
Table C.1: Regression of general happiness on RQ in the sample of never parents.

	General Happiness		
	(1)	(2)	(3)
RQ	0.253***	0.253***	0.191***
	(0.013)	(0.013)	(0.019)
Controls		✓	✓
Individual FE			✓
R-squared	0.058	0.058	0.017
Observations	14424	14424	14424

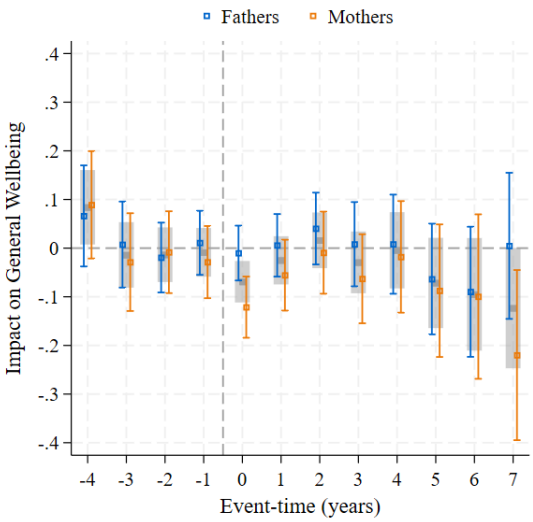
Notes: This table presents the results from regressing a standardized measure of general happiness on RQ in the sample of individuals who never become parents. The sample consists of individuals first observed between ages 18 and 45 who are never observed cohabiting with children. The general happiness measure is based on responses to the question "Have you recently been feeling reasonably happy, all things considered?". Responses are recorded on a four-point Likert scale. Controls include age, relationship tenure, period, sex, college education, employment status, marital status, and urban residence. Standard errors clustered at the couple level are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure C.2: Impact of children on general happiness and subjective well-being.

(a) General happiness

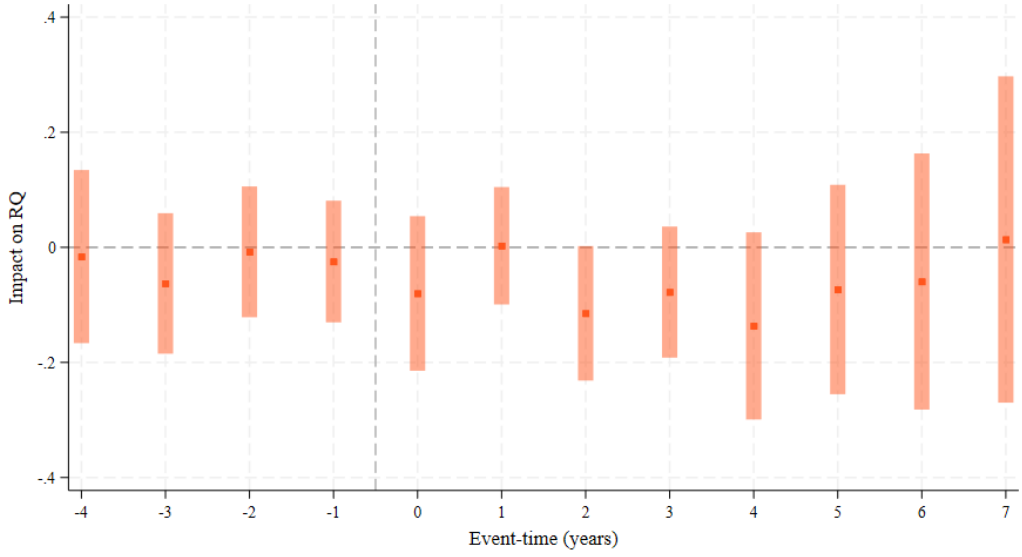


(b) Subjective well-being



Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on different outcomes, using the method proposed by Callaway and Sant'Anna (2021). Panel (a) reports results for a standardized measure of general happiness, based on responses to the question "Have you recently been feeling reasonably happy, all things considered?". Responses are recorded on a four-point Likert scale. Panel (b) displays results for a standardized measure of subjective well-being, constructed by Understanding Society using responses to the 12-item General Health Questionnaire (GHQ-12). We plot 95% confidence intervals.

Figure C.3: Impact of unemployment on RQ.



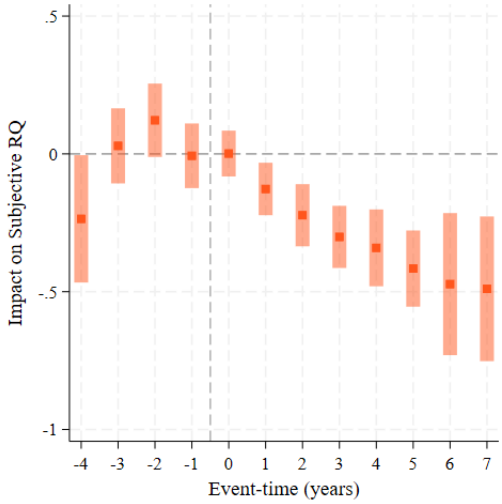
Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of unemployment events on RQ, using the method proposed by Callaway and Sant'Anna (2021). The sample consists of parents from the main analysis sample, restricted to individuals of working age (20-55). An unemployment event is defined as being unemployed at the time of the survey but not in the previous period. We only consider the first instance of unemployment events and the control group is formed only by individuals who experience unemployment at some point ("not yet unemployed"). We plot 95% confidence intervals.

D Robustness

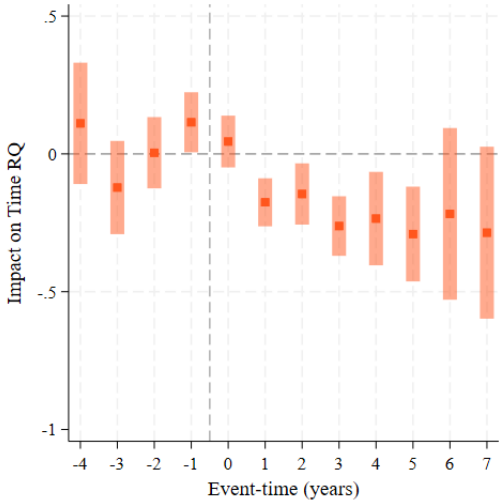
D.1 Time Invariance of RQ

Figure D.1: RQ by item block.

(a) Subjective assessment

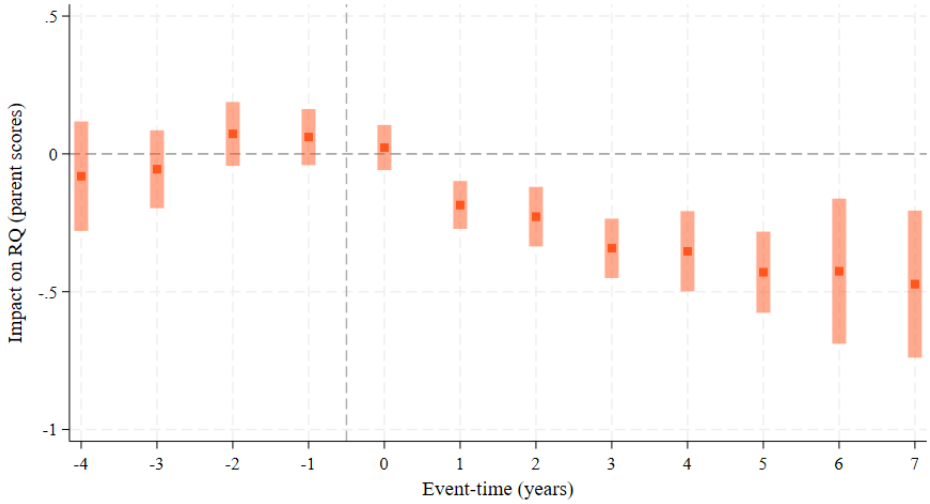


(b) Couple time use



Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on alternative RQ measures, using the method proposed by Callaway and Sant'Anna (2021). The outcome in panel (a) is constructed computing a factor analysis on the subjective assessment items in Table 1, retaining the first factor. The outcome in panel (b) is based on a factor analysis on the couple time use items in Table 1, also retaining the first factor. We plot 95% confidence intervals.

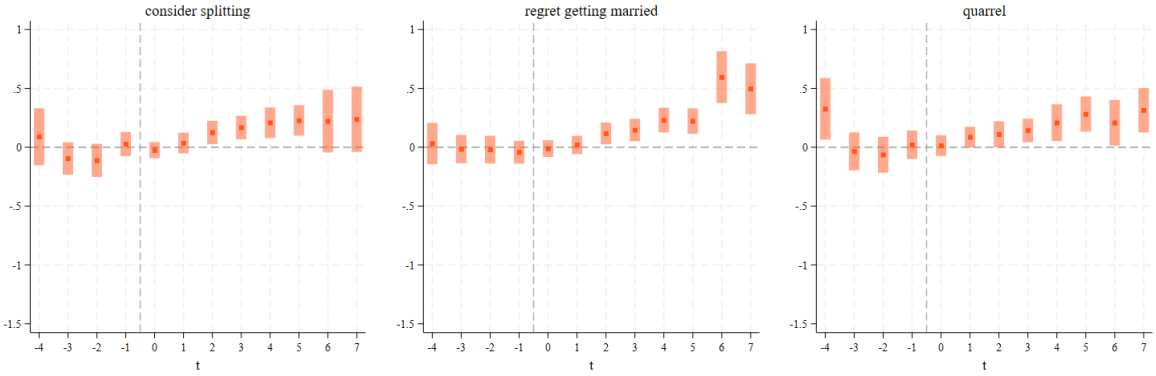
Figure D.2: Impact of children on RQ using factor scores after birth.



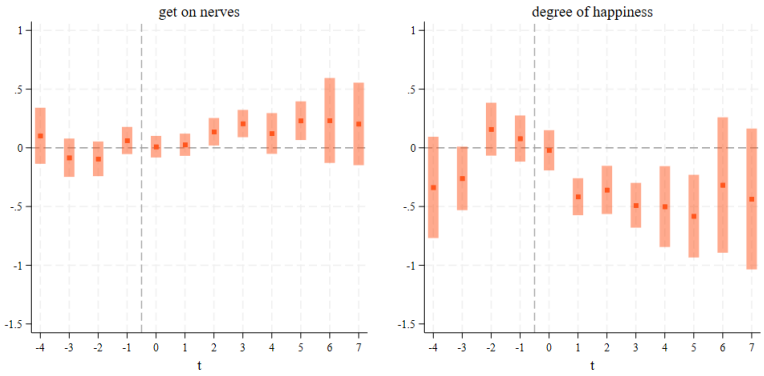
Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The outcome is derived from a factor analysis of the items in Table 1, retaining the first factor, and is constructed using only observations from individuals who have already become parents. We plot 95% confidence intervals.

Figure D.3: Impact on subjective assessment items.

(a) Consider splitting (b) Regret marriage (c) Quarrel



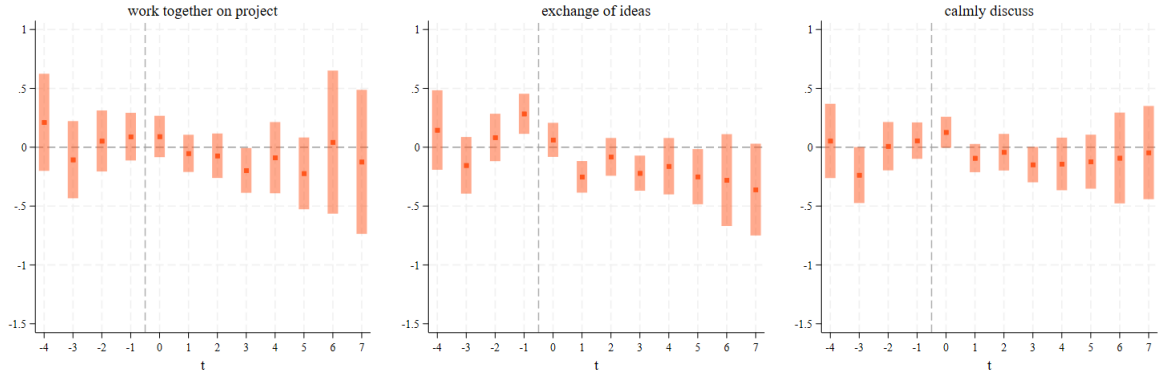
(d) Get on nerves (e) Degree of happiness



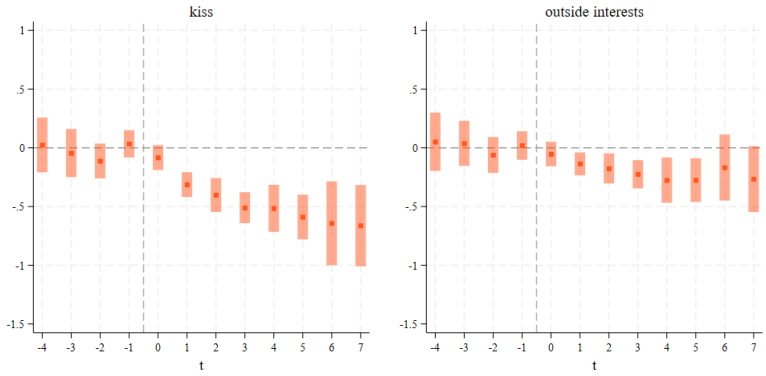
Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on different measures, using the method proposed by Callaway and Sant'Anna (2021). The outcomes are the standardized responses to the subjective assessment items in Table 1. We plot 95% confidence intervals.

Figure D.4: Impact on couple time use items.

(a) Work together (b) Exchange ideas (c) Calmly discuss



(d) Kiss (e) Outside interests

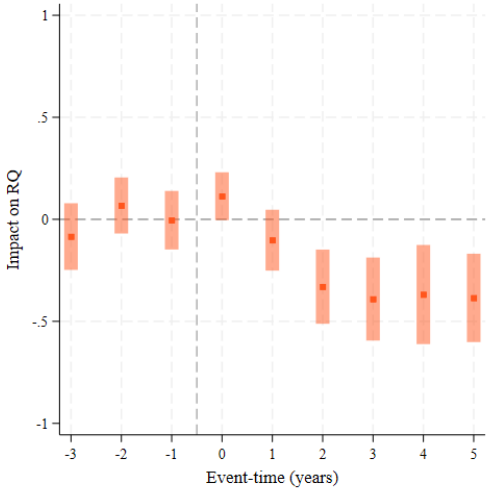


Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on different measures, using the method proposed by Callaway and Sant'Anna (2021). The outcomes are the standardized responses to the couple time use items in Table 1. We plot 95% confidence intervals.

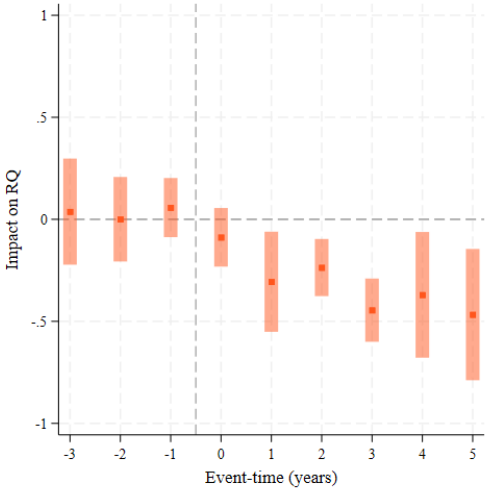
D.2 Subsequent Fertility

Figure D.5: Impact by final number of children.

(a) One child



(b) Two children



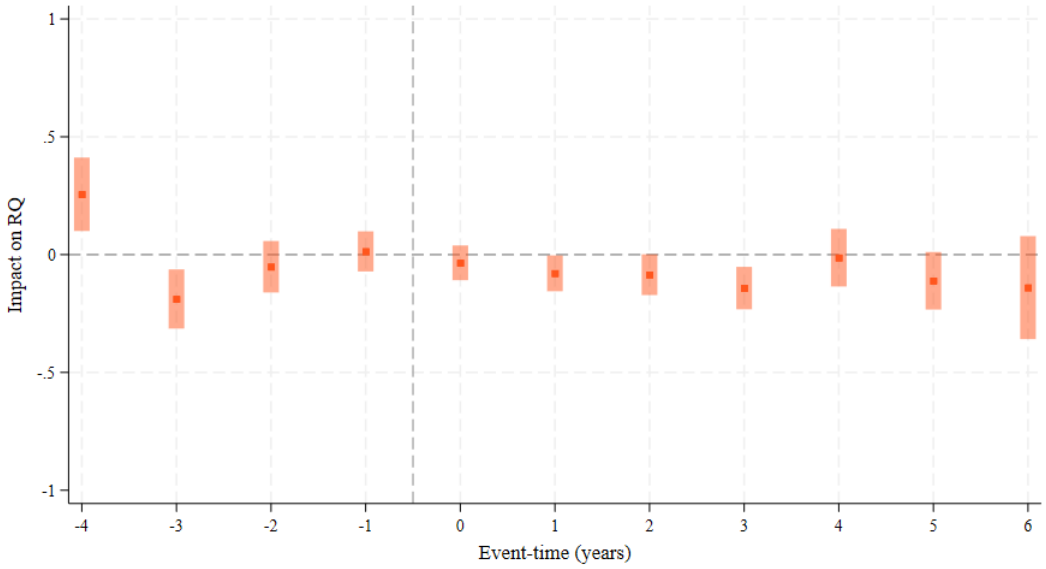
Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ in different samples, using the method proposed by Callaway and Sant'Anna (2021). Panel (a) shows results for individuals who have only one child, and panel (b) displays results for individuals with two or more children by the end of the observation period. We plot 95% confidence intervals.

Table D.1: Impact by total realized fertility.

	(1) One child	(2) More than one
ATT	-0.330*** (0.0782)	-0.323*** (0.0967)
Observations	914	1075

Notes: This table presents results from a static difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). Column (1) reports results for individuals who have only one child by the end of their fertility cycle (age 40), and column (2) presents results for those with more than one child. Standard errors clustered at the couple level are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure D.6: Impact of second child birth on RQ.

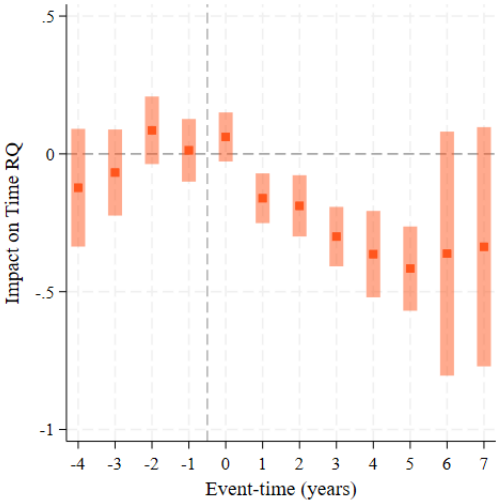


Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of second child birth on RQ in different samples, using the method proposed by Callaway and Sant'Anna (2021). The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, were not cohabiting with children from previous relationships, and became second-time parents. We plot 95% confidence intervals.

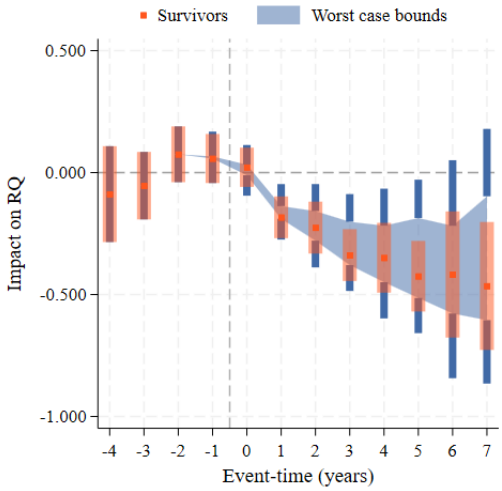
D.3 Attrition due to Post-birth Couple Dissolution

Figure D.7: Sample selection on separating couples.

(a) Impact on non-separating couples.



(b) Worst case sensitivity analysis.

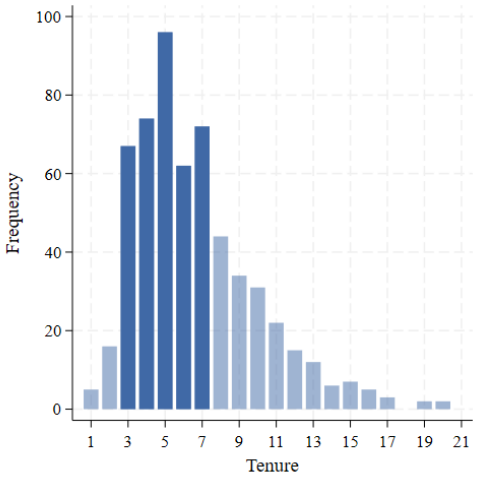


Notes: Panel (a) presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ in the sample of first-time parents who have not separated by the end of the observation period, using the method proposed by Callaway and Sant'Anna (2021). We plot 95% confidence intervals. Panel (b) reports the main estimates alongside a sensitivity analysis based on extreme imputations of unobserved RQ values following couple dissolution. Missing values are imputed at the 1st and 99th percentiles of the RQ distribution in the sample. The shaded area spans the point estimates from these two imputations. We report the lower 95% confidence limit for the 1st-percentile imputation and the upper 95% confidence limit for the 99th-percentile imputation.

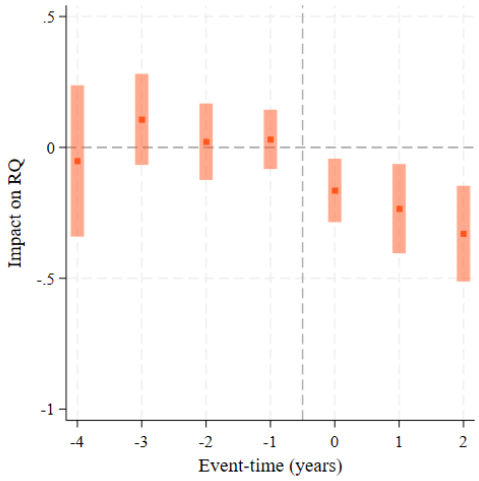
D.4 Age and Tenure at Birth

Figure D.8: Analysis by tenure at birth.

(a) First births at each tenure.



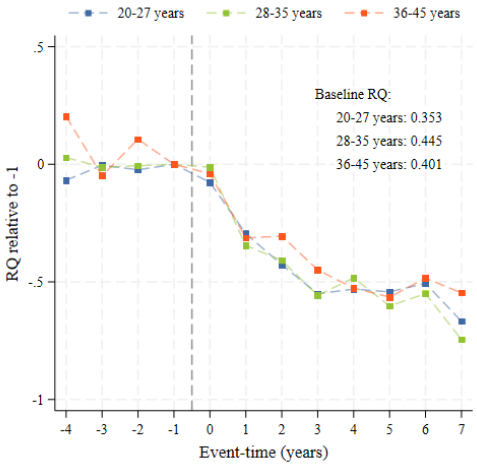
(b) Impact restricting to 3-5 tenure births.



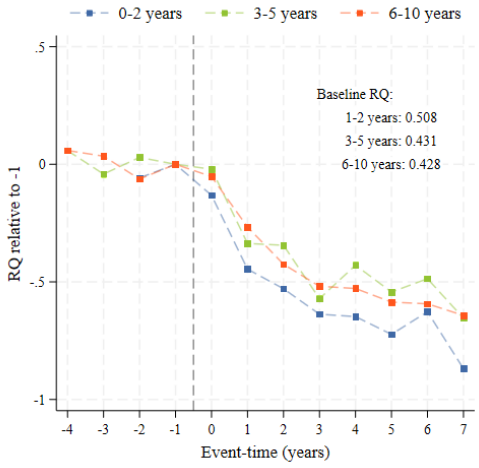
Notes: Panel (a) plots the distribution of relationship tenure at first birth in our sample of first-time parents. Panel (b) presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, restricting the sample to parents whose first child was born at relationship tenures 3-7, and using the method proposed by Callaway and Sant'Anna (2021).

Figure D.9: Average RQ per event-time period.

(a) Age bins

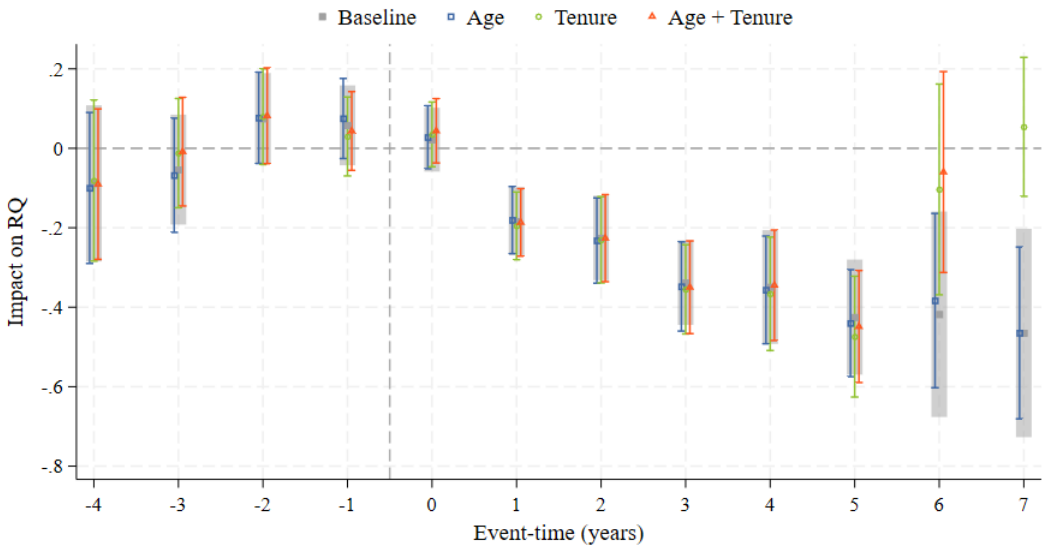


(b) Tenure bins



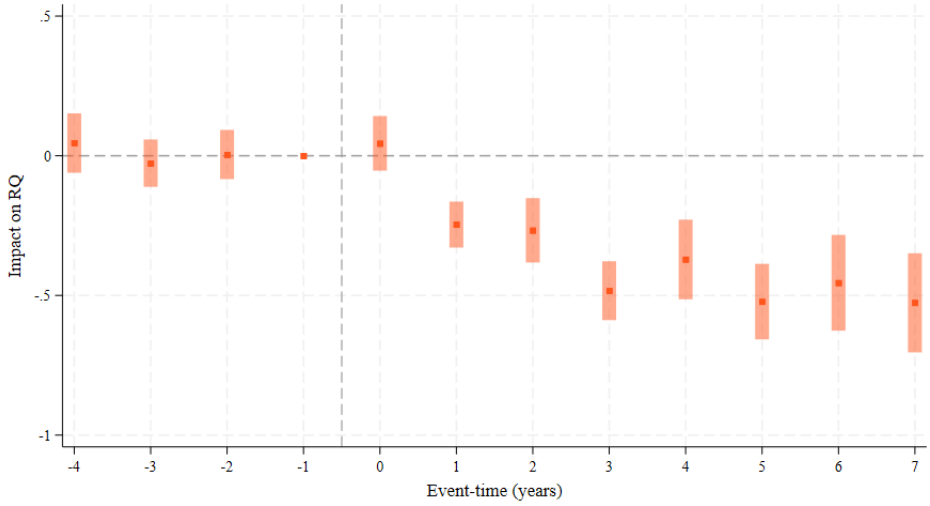
Notes: This figure plots the average RQ at each event-time period. The average in the period before birth ($t - G_i = -1$) is normalized to zero. Panel (a) displays these means separately for individuals who had their first child at different ages: 20-27 years, 28-35 years and 36-45 years. Panel (b) displays RQ means separately by relationship tenure at first birth: 1-2 years, 3-5 years and 6-10 years.

Figure D.10: Impact on RQ using different sets of controls.



Notes: This figure presents results from a dynamic difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021) and including different sets of controls. The method accounts for age and tenure by comparing individuals with the same value of the covariates in the period before birth. We plot 95% confidence intervals.

Figure D.11: Impact of first child birth on RQ using Kleven et al. (2019b).



Notes: This figure presents the result from estimating the following regression:

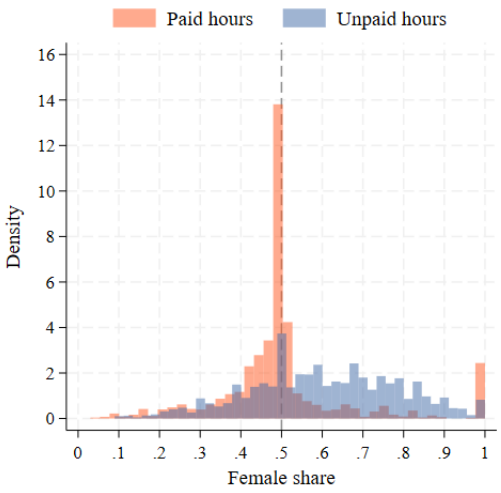
$$y_{i,t} = \mu_t + \sum_{j \neq -1} \mathbb{1}\{j = t - G_i\}\delta_j + \sum_a \mathbb{1}\{a = \text{age}_{i,t}\}\gamma_a + \sum_d \mathbb{1}\{d = \text{tenure}_{i,t}\}\lambda_d + u_{i,t}$$

This regression is the event-study specification popularized by Kleven et al. (2019b), and it accounts for the dynamic effects of age and tenure. The plotted coefficients $\hat{\delta}_j$ correspond to the event-time dummies. Standard errors clustered at the couple level are used to construct the plotted 95% confidence intervals.

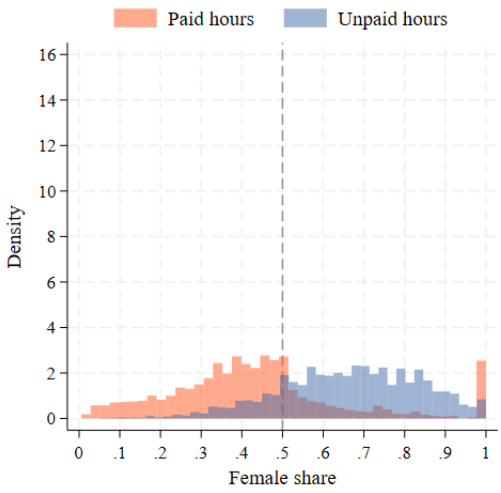
E Household Specialization

Figure E.1: Distribution of the female share of paid labour market and unpaid housework hours.

(a) Before birth

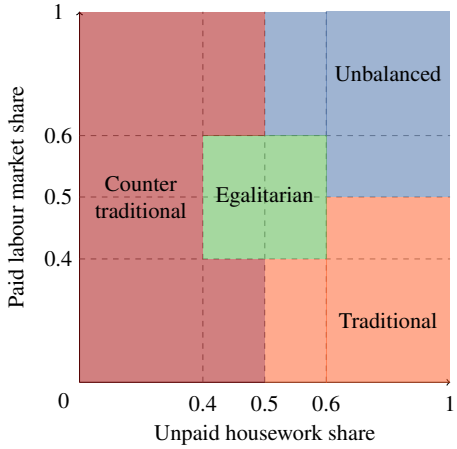


(b) After birth



Notes: This figure displays the distribution of the female shares of paid and unpaid work (defined as the share of total weekly household paid labour market hours and unpaid housework hours performed by women) in the sample of individuals who become parents. Panel (a) presents data for up to four years before the birth of the first child. Panel (b) covers the period from the year of birth to seven years after. These time frames correspond to the main analysis period.

Figure E.2: Classification of couples based on the female shares.



Notes: This figure presents the classification of couples based on the female share of paid labour market work and unpaid housework. These shares represent the proportion of total weekly household paid labour market hours and unpaid housework hours performed by women, each ranging from zero to one. Classification is based on average shares before the birth of the first child. We distinguish four couple types: (i) Traditional couples (orange), where women perform more than 50% of housework and men take a larger share of paid work; (ii) Unbalanced couples (blue), where women take on more than half of both types of work; (iii) Egalitarian couples (green), where men and women share both types of work roughly equally, with female shares between 40% and 60%; and (iv) Counter-traditional couples (red), where men contribute more than 50% to housework.

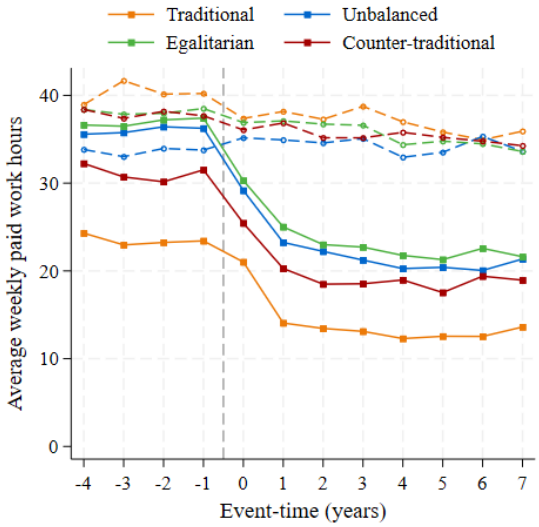
Table E.1: Summary statistics before the birth of the first child, by couple type.

	Traditional		Unbalanced		Egalitarian		Counter-tradit.	
	(1) Fathers	(2) Mothers	(3) Fathers	(4) Mothers	(5) Fathers	(6) Mothers	(7) Fathers	(8) Mothers
(a) Individual characteristics								
Age	31.31	28.88	32.51	30.02	32.22	30.34	32.60	29.97
	(5.520)	(4.820)	(5.922)	(4.825)	(5.198)	(4.181)	(5.920)	(4.686)
College educated (%)	43.94	50.82	45.05	60.29	46.58	63.29	50.89	67.54
	(49.70)	(50.06)	(49.88)	(49.05)	(49.97)	(48.29)	(50.10)	(46.92)
Active in labour mkt (%)	94.56	86.61	84.41	93.12	95	95.33	92.51	89.86
	(22.71)	(34.08)	(36.34)	(25.36)	(21.82)	(21.13)	(26.37)	(30.25)
Employed (%)	94.15	81.50	79.47	93.12	94.72	93.68	89.51	88.04
	(23.49)	(38.87)	(40.47)	(25.36)	(22.39)	(24.36)	(30.70)	(32.50)
Monthly labour income (£)	2710.2	1730.9	2499.5	2313.0	2826.4	2340.6	2635.5	2366.7
	(1528.8)	(1198.4)	(1790.5)	(1480.0)	(1443.4)	(1153.0)	(1792.6)	(1438.1)
Weekly work hours	37.55	27.26	28.87	33.90	37.38	34.90	32.66	30.96
	(11.58)	(14.96)	(16.58)	(11.30)	(9.722)	(9.571)	(14.77)	(13.20)
Weekly housework hours	3.667	10.78	4.617	9.677	6.893	6.976	9.055	5.114
	(2.501)	(6.298)	(3.890)	(5.269)	(3.563)	(3.619)	(4.250)	(3.446)
RQ	0.192	0.340	0.117	0.306	0.523	0.554	0.384	0.367
	(1.069)	(0.981)	(0.727)	(0.814)	(0.620)	(0.631)	(0.864)	(0.849)
Observations	496	508	263	276	360	364	267	276
(b) Couple characteristics								
Tenure	4.785		5.212		5.215		4.798	
	(2.909)		(3.154)		(3.103)		(3.125)	
Married (%)	68.73		63.57		66.93		62.32	
	(45.79)		(47.46)		(46.68)		(48.36)	
Monthly household income (£)	4193.7		4727.3		4924.2		4878.1	
	(2174.1)		(2745.2)		(2538.2)		(2712.7)	
Female share of paid work	0.384		0.603		0.484		0.509	
	(0.179)		(0.209)		(0.0687)		(0.220)	
Female share of housework	0.743		0.698		0.501		0.343	
	(0.144)		(0.153)		(0.112)		(0.138)	
Observations	534		280		375		284	

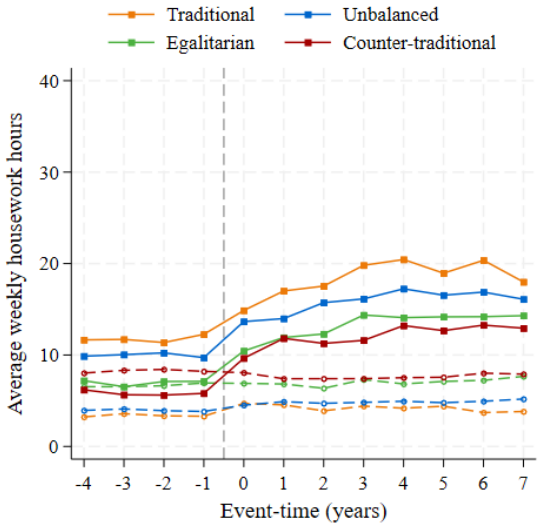
Notes: This table presents summary statistics for the main analysis sample of parents in the period before the birth of the first child, disaggregated by couple type based on their division of labour market and housework time prior to birth. The classification is illustrated in Figure E.2. Panel (a) reports mean values of individual characteristics separately for women and men, and panel (b) summarizes couple characteristics. Female shares of paid and unpaid work represent the share of total weekly household paid labour market hours and unpaid housework hours performed by women. Income measures are presented in gross terms. Standard deviations are reported in parentheses.

Figure E.3: Time use around first child birth by couple type.

(a) Paid work share

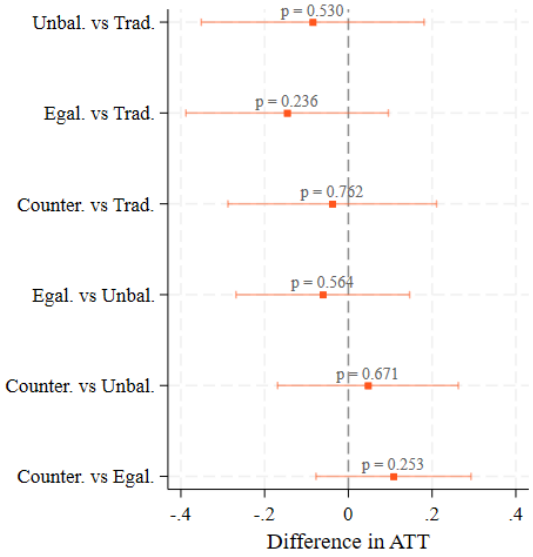


(b) Unpaid work share



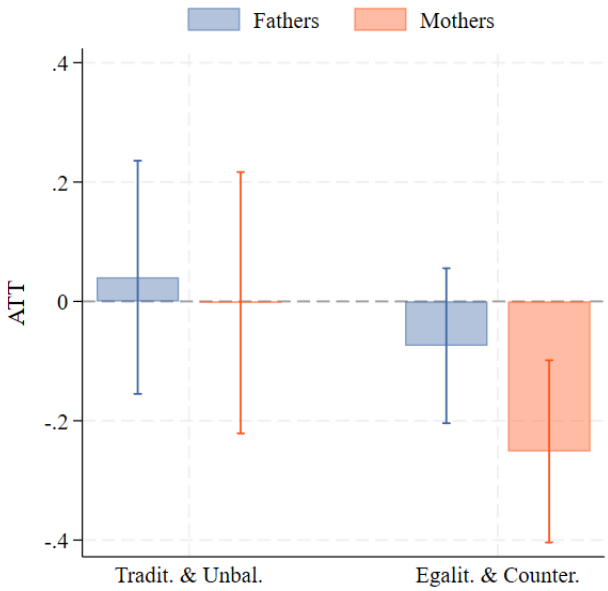
Notes: This figure displays the average weekly (a) paid labour market hours and (b) unpaid housework hours at each time around the birth of the first child. These averages are plotted separately by couple type, defined based on the distribution of tasks before first child birth, and by sex. The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships.

Figure E.4: ATT comparisons across couple types.



Notes: This figure reports p-values from pairwise tests of equality of the estimated effects of first childbirth on RQ across couple types. The estimates for each couple type are obtained from separate static difference-in-differences analyses, reported in Table 3. Each p-value tests whether the corresponding coefficients differ statistically across the two couple types being compared. The p-value of the joint test that all four ATTs are equal is 0.568.

Figure E.5: Impact of first birth by couple type and gender.



Notes: This figure presents the results from a static difference-in-differences estimation of the effect of first child birth on RQ, using the method proposed by Callaway and Sant'Anna (2021). The sample includes individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18–45 (women) or 18–50 (men) at the time of birth, and were not cohabiting with children from previous relationships. Estimates are obtained separately by sex and couple type, pooling traditional with unbalanced couples and egalitarian with counter-traditional couples, as defined in Figure E.2. For traditional and unbalanced couples, the estimation sample comprises 435 men and 473 women; for egalitarian and counter-traditional couples, it comprises 678 men and 719 women. The analysis covers event times from four years before to seven years after first childbirth, consistent with the main analysis. We also report 95% confidence intervals.

F Other Potential Channels

The main mechanism we study are changes in household specialization after the birth of the first child. In this section, we consider the possibility that other channels may could contribute to the RQ response. Specifically, we study changes in gender norms, household income and the within-couple earnings gap.

Gender norms. To measure gender norms, we use survey questions on gender role attitudes. We employ a subset of the standard questionnaire developed by the International Social Survey Programme (ISSP), widely used in the economics literature on gender norms (e.g., Bertrand et al., 2021; Farré et al., 2022). These questions are part of the Adult Gender Attitudes module of Understanding Society, which is available in waves 2, 4 and 10. Specifically, we use a set of items on a 5-point Likert scale on agreement with a set of statements related to the role of women and men at home and in the labour market, listed in Table F.1.[39]

Following Farré et al. (2022), we recode the items such that higher values indicate more gender-equal views: a score of 0 corresponds to the most conservative position (agreement with (a) and disagreement with (b) in Table F.1) and 4 corresponds to the most egalitarian views. We construct the measure of gender role attitudes by running a factor analysis on the 5 items and retaining the first factor. All items' loadings are positive and the factor explains 43.91% of the variation in the data. The resulting measure is standardized to zero mean and a standard deviation of one. Figure F.1 (a) presents the distribution of this measure before the birth of the first child. The distribution is skewed to the right, indicating relatively more progressive views on average, with considerable heterogeneity.

Table F.1: Items in the gender role attitude questionnaire.

Do you personally agree or disagree...	
(a) Conservative statements	(b) Progressive statements
A pre-school child is likely to suffer if his or her mother works.	Both the husband and wife should contribute to the household income.
All in all, family life suffers when the woman has a full-time job.	Employers should make special arrangements to help mothers combine jobs and childcare.
A husband's job is to earn money, a wife's job is to look after the home and family.	

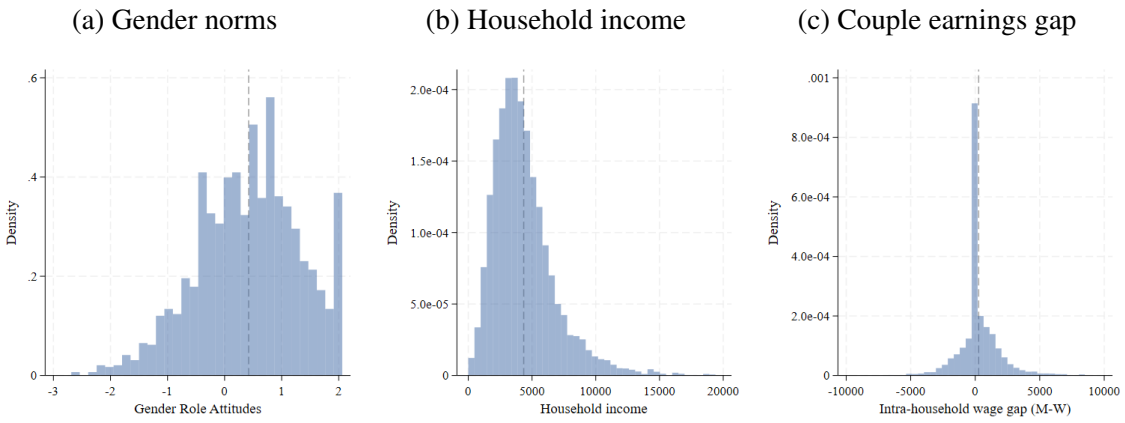
Notes: This table lists all the questions used to construct the gender role attitudes, drawn from the Understanding Society Adult Gender Attitudes module. This self-completion questionnaire is administered in waves 2, 4 and 10. We group these items in two blocks based on whether the statements are of (a) conservative or (b) progressive nature. The responses are given in a five-point Likert scale of agreement, where one means complete agreement and five complete disagreement. We recode the items such that higher values indicate more gender-equal views.

[39] It is missing the following items from the usual questionnaire: "A job is all right, but what most women really want is a home" and "Being a housewife is just as fulfilling as working for pay".

Household income. An alternative mechanism for the documented decrease in RQ could be financial strains generated by childbirth (Blanchflower and Clark, 2021). To assess this, we study the role of household resources. Our measure of household resources is gross monthly household income during the month prior to the interview, in pounds. Since household income is reported at the household level, both members of the couple have the same value. Figure F.1 (b) presents the pre-birth distribution of this variable, which averages around £4,000 per month but exhibits a long right tail.

Couple earnings gap. Finally, we construct the within-couple earnings difference using gross monthly individual labour income. We compute the gap as the male partner's monthly earnings minus the female partner's. Positive values indicate that men have higher earnings, while negative values capture cases where women are the primary earners. Figure F.1 (c) presents the pre-birth distribution. It is centred slightly above zero, consistent with an average gender gap in favour of men, but there is substantial variation on both sides of the distribution.

Figure F.1: Distribution of alternative mechanisms.



Notes: This figure plots the distribution, within the main analysis sample of parents and before childbirth, of (a) gender role attitudes, (b) household income and (c) the within-couple earnings gap. The sample consists of individuals in heterosexual cohabiting relationships who became first-time parents between 2009 and 2024, were aged 18-45 (women) or 18-50 (men) at the time of birth, and were not cohabiting with children from previous relationships.

Table F.2 reports the correlations between our main mechanism, female shares of paid and unpaid work as defined in Subsection 2.2, and the alternative channels discussed in this section. The strongest correlation arises between the couple earnings gap and the female share of paid work: as women contribute more hours in the labour market, the earnings gap shifts in favour of them being the primary earners. Conversely, a higher female share of unpaid work is associated with larger gaps in favour of men. The female share of paid work is positively correlated with more egalitarian gender role attitudes, suggesting an alignment between labour market participation and progressive views. The weakest correlation is between our main mechanism and household income.

Table F.2: Correlation between alternative mechanisms.

	Share of paid work	Share of unpaid work	Gender role attitudes	Household income	Couple earnings gap
Share of paid work	1				
Share of unpaid work	-0.334	1			
Gender role attitudes	0.244	-0.165	1		
Household income	0.118	-0.040	0.120	1	
Couple earnings gap	-0.568	0.283	-0.204	0.235	1

Notes: This table presents the correlations between our main mechanism (the female shares of paid and unpaid work) and the alternative mechanisms considered in this section (gender norms, household income and the intra-household earnings gap). The correlations are computed in the main analysis sample of first time parents across all available periods. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

F.1 Impact of First Child Birth

We start by examining how the different candidate channels change around the birth of the first child. Mechanisms that undergo larger shifts following childbirth have greater potential to contribute to the change in RQ. We consider both our main mechanism and the alternative channels discussed above. We standardize all measures to facilitate the comparison across outcomes. We estimate a static difference-in-differences specification in the main analysis sample of first time parents, using the method proposed by Callaway and Sant'Anna (2021), and use each channel as the outcome in separate regressions.

Table F.3 reports the results. The largest changes occur in the household specialization variables. After first birth, women's share of paid work falls by 0.43 standard deviations, while their share of housework rises by 0.39 standard deviations. We also find an increase in the within-couple earnings gap and a reduction in household income. Finally, the gender norms index declines, although the estimate is only marginally significant. Thus, household specialization exhibits the largest changes among the candidate channels considered.

Table F.3: Impact of first child birth on potential channels.

	(1) Paid share	(2) Unpaid share	(3) Gender norms	(4) HH income	(5) Couple gap
ATT	-0.427***	0.387***	-0.142*	-0.151***	0.364***
	(0.028)	(0.040)	(0.060)	(0.037)	(0.036)
Observations	28335	16035	1552	45030	38526

Notes: This table presents the results from a static difference-in-differences estimation of the effect of first child birth on alternative channels, using the method by Callaway and Sant'Anna (2021): (1) the female share of paid work, (2) the female share of unpaid work, (3) gender role attitudes, (4) gross monthly household income, (5) within-couple monthly labour earnings gap. All measures are standardized. We use the main analysis sample of first time parents, as defined in Subsection 2.3. Standard errors are reported in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

F.2 Children, RQ and Potential Channels

We next study the association between RQ and the different candidate channels, both overall and around childbirth. We first estimate a linear regression of RQ on all potential mechanisms, standardized to facilitate the comparison across coefficients.[40] As controls, we include second-order polynomials of age and tenure, dummies for employment status, labour market activity and college education, a marriage indicator, and the full set of time dummies. The sample is restricted to the usual observation window, covering four years before and seven years after first birth. Table F.4 (1) presents the results. RQ is significantly associated with gender norms and is marginally significantly associated with household income and the within-couple earnings gap. In Column (2), we add individual fixed effects to capture within-individual variation, which mechanically excludes individuals observed only once and thus reduces the sample size. Under this specification, only gender norms remain significant.

We then examine whether the impact of childbirth on RQ varies with pre-birth levels of the candidate channels. We compute the individual-specific average of each mechanism before birth and interact these measures with the post-birth indicator in a static difference-in-differences specification. We do not use the Callaway and Sant'Anna (2021) estimator, as it does not allow for interaction terms. Since the mechanisms are time-invariant in this specification and we include individual fixed effects, the coefficients on the pre-birth mechanism levels are absorbed, leaving the post-birth indicator and its interactions. The results, presented in Table F.4 (3), show no statistically significant heterogeneity in the effect of childbirth on RQ by pre-birth levels of any of the candidate mechanisms when considered jointly. The largest magnitude coefficients correspond to the female share of paid and unpaid hours, but these associations remain statistically indistinguishable from zero.

Taken together, these results provide suggestive evidence on the potential channels through which childbirth may affect RQ. Household specialization exhibits the largest changes around first birth, and changes in specialization are closely linked to changes in women's and men's time allocation between paid and unpaid work. The alternative channels considered also change around childbirth and are related to RQ to varying degrees, particularly gender norms. We therefore interpret the evidence as consistent with changes in household specialization being an important mechanism, while recognizing that the analyses do not separately identify the causal contribution of the different channel.

[40] Since housework time and gender norms are not collected in the same survey waves as RQ, we use their lagged values.

Table F.4: Association between RQ and alternative channels.

	(1)	(2)	(3)
Female share of paid hours	0.010	-0.025	
	(0.021)	(0.024)	
Female share of unpaid hours	-0.003	0.022	
	(0.013)	(0.016)	
Gender role attitudes	0.134***	0.070*	
	(0.013)	(0.034)	
Monthly gross household income	0.034*	-0.001	
	(0.016)	(0.023)	
Within-couple labour earnings gap	0.043*	0.001	
	(0.017)	(0.024)	
Post			0.006
			(0.044)
Post × Female share of paid hours			0.037
			(0.043)
Post × Female share of unpaid hours			0.040
			(0.026)
Post × Gender role attitudes			0.033
			(0.026)
Post × Monthly gross household income			-0.006
			(0.031)
Post × Within-couple labour earnings gap			-0.011
			(0.031)
Controls	✓	✓	✓
Individual FE		✓	✓
R-squared	0.064	0.744	0.671
Observations	5523	4824	3603

Notes: This table reports the associations between RQ and the alternative channels in the main analysis sample of first time parents, as defined in Subsection 2.3. All mechanism measures are standardized. Columns (1) and (2) present results from estimating a linear regression of RQ on all channels jointly. Column (3) reports results from a static difference-in-differences regression, where the treatment variable is interacted with individual-specific pre-birth averages of all channels. All specifications control for second-order polynomials of age and tenure, dummies for employment status, labour market activity and college education, a marriage indicator, and the full set of time dummies. Columns (2) and (3) additionally include individual fixed effects. Standard errors are reported in parentheses.